CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner:	County of Hawaii		
Contractor:	Nan, Inc.	**Project No.:**	WW-4705R
Project Name:	Hilo WWTP Phase 1	**Submittal Number:**	11313F - 001.0
Submittal Title:	Pump and Equipment		☐ For Information Only
TO:	County		☐ Designer of Record
From:	Nan Inc		☐ Government Approval

Specification No. and Subject of Submittal / Equipment Supplier			
Spec: 11313F	**Paragraph:** 1.05 & 1.06		
Authored By:	Optimal Water	**Date Submitted:**	06/09/2025

Submittal Certification

Check Either (A) or (B):

☑ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.

General Contractor's Reviewer's Signature: Jyun-Cheng Jhuo Digitally signed by Jyun-Cheng Jhuo DN: C=US, E=jcjhuo@nanhawaii.com, O=Nan Inc., CN=Jyun-Cheng Jhuo Date: 2025.06.09 16:03:45-10'00'

Printed Name and Title: Jyun-Cheng Jhuo

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:

Date Received PM/CM to Reviewer:

Date Received Reviewer to PM/CM:

Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 6/16/2025
SPECIFICATION SECTION # 11313
SPECIFICATION SUBMERSIBLE CHOPPER CENTRIFUGAL PUMPS
PARAGRAPH 1.05 1.06
DRAWING N/a
SUBCONTRACTOR N/a
SUPPLIER N/a
MANUFACTURER Vaughan

CERTIFIED BY CQCM or Designee : Makoa Ng Digitally signed by Makoa Ng DN: C=US, E=mng@nanhawaii.com, O="Nan, Inc.", CN=Makoa Ng Date: 2025.06.16 16:29:53-10'00'

TRANSMITTAL MEMO
ENGINEERING SUBMITTAL

DATE: April 28, 2025

TO: **OPTIMAL WATER**

Attn: Quintin Collier
Email: quintin@optimalwater.net

FROM: Hannah Hurd, Engineering Department

SUBJECT: PO#: 005625r3; Transmittal of Submittals; HILO WWTP; Request for Approval to Build; 2 EA, S3M2-460V-076.

Quintin:

This memo documents the e-mail transmittal of submittals for the above project.

We are holding production of this pump and related equipment until we receive written approval from you to proceed with manufacturing.

After we receive approval of this submittal, we will immediately submit your order to production, and supply applicable Installation, Operation and Maintenance instructions for these pumps.

If you need anything else concerning this submittal, please feel free to contact me.

Sincerely,

Hannah Hurd,
Submittal Specialist

S/N: 174838 A/B

VAUGHAN COMPANY, INC.

ENGINEERING SUBMITTAL, HILO WWTP, SUBMERSIBLE PUMP

SECTION 11312E

2 EACH VAUGHAN MODEL: S3M2-460V-076
WITH 10 HP, 1765 RPM ABB/BALDOR MOTOR

DATE: April 28, 2025
Prepared by: Hannah Hurd
Vaughan Engineering Department

VAUGHAN COMPANY, INC.
364 MONTE-ELMA ROAD
MONTESANO, WA 98563
(360) 249-4042
(360) 249-6155 FAX

TABLE OF CONTENTS
ENGINEERING SUBMITTAL
SUBMERSIBLE PUMP
HILO WWTP

LIST OF EQUIPMENT TO BE SUPPLIED
LOCATION: HILO WWTP
SUBMERSIBLE PUMP
SECTION 11312E

EQUIPMENT NOS: 03-PMP-1610 & 03-P,P-1620

2 EA VAUGHAN MODEL S3M2-460V-080 - Submersible Chopper Pumps
PUMP PERFORMANCE: 235 GPM @ 43 FT. TDH;
150 GPM @ 48 FT. TDH;
325 GPM @ 30 FT. TDH
400 GPM @ 15 FT. TDH

CONSISTING OF:
CASING, cast ductile iron with 3" ANSI CL 125 discharge flange.
IMPELLER, CUTTER BAR, CUTTER NUT UPPER CUTTER, cast steel, heat treated to minimum Rockwell C60. **IMPELLER BALANCE CERTIFICATE**
MECHANICAL SEAL, cartridge type with ductile iron gland, Viton O-rings, tungsten carbide faces, and integral stainless steel sleeve as manufactured by Vaughan.
ELASTOMERS, Buna N
BEARINGS, oil bath lubricated with minimum 100,000-hour L-10 bearing life.
BEARING HOUSING, cast ductile iron with piloted motor mount.
SHAFT, heat treated steel
FACTORY TESTING, Non-witnessed Performance tests and hydrostatic testing.
PUMP MONITOR RELAY for mounting in customer control panel to supply seal leakage and over temperature alarms for submersible motor.
DRIVE, 10 HP, 1750 RPM, 460V, 3 phase, 60 Hz, 1.15 SF, Explosion Proof (Class 1, Group C & D) 15 minute in air continuous duty submersible motor with tandem mechanical seals, moisture sensors, internal thermostats, and **50 FEET** of power and control cable, and **ROUTINE MOTOR TESTS** manufactured by ABB/BALDOR.
PREMIUM PUMP FINISH: Solvent wash, sandblast and coat with minimum 30 MDFT Tnemec Perma-Shield PL Series 431 epoxy. (Except Motor)

2 EA SPARK PROOF GUIDE RAIL SYSTEM, consisting of:
3" BASE ELBOW, cast ductile iron.
2" GUIDE BRACKET, cast non-sparking aluminum bronze.
(1) INTERMEDIATE STIFFENER BRACKETS, 316 stainless steel located every 10 feet.
TOP MOUNTING BRACKET AND CHAIN HOLDER BRACKET, 316 stainless steel.

NOTE: Customer to provide 2" Sch. 40 pipe rails.

1	**LOT SPECIAL TOOLS,** consisting of:		**VTK-200-10**
	SEAL CARTRIDGE CAP (VAUGHAN)		V104-748
	SEAL CAP TOOL (VAUGHAN)		V107-194
	GLAND TOOL		V104-361
	THRUST BEARING TOOL (STD - BEARINGS)		V3500-083
	UPPER CUTTER TOOL		V108-006
	CUTTER BAR PINS (SET)		V3500-101

1	**LOT SPARE PARTS,** consisting of:	
	1 EA SEAL	V801-301TC
	2 EA THRUST BEARING	V801-139
	1 EA IMPELLER	V102-545-080
	1 EA CUTTER BAR	V102-501
	1 EA CUTTER NUT	V104-611
	1 EA O-RING, BUNA-N	V850-239B
	1 EA CABLE KIT	802386201T

NOTE:

THE ITEMS CONTAINED IN THIS LIST OF EQUIPMENT TO BE SUPPLIED ARE THE ONLY ITEMS BEING SUPPLIED BY VAUGHAN COMPANY, THE PUMP MANUFACTURER. OTHER SPECIFIC ITEMS WHICH MAY BE REQUIRED BY THE SPECIFICATION, INCLUDING, BUT NOT LIMITED TO THE ITEMS LISTED BELOW, MUST BE FURNISHED BY OTHERS:

GAUGES, SWITCHES, VALVES AND OTHER SPECIALTIES NOT SPECIFICALLY CALLED OUT HEREIN.
SPECIAL COATINGS.
EQUIPMENT, LABOR, MATERIAL AND PERSONNEL REQUIRED TO PERFORM FIELD TESTING OF PUMPS.
SPECIAL MOTOR SPECIFICATIONS INCLUDING MILL AND CHEM DUTY, EXPLOSION PROOF, INTERNAL SPACE HEATERS, ETC.
FACTORY MOTOR TESTS.
INTRINSICALLY SAFE FEATURES.
LEVEL CONTROLS, VARIABLE FREQUENCY DRIVES, AND CONTROL PANELS.
ADDITIONAL LUBRICANTS OTHER THAN THOSE CONTAINED WITHIN THE PUMP.
ANCHOR BOLTS.

CLARIFICATIONS AND EXCEPTIONS
HILO WWTP

SECTION 11313F

1.04: Vaughan is furnishing pumps as described in the list of equipment to be supplied and designed to meet the specified performance requirements with the clarifications and exceptions herein. Anchorage and analysis are to be provided by others.
1.05.A: Vaughan is providing our standard submittal package.
1.05.B.1-2: BY others.
1.05.B.3: N/a per pump schedule.
1.05.C: Vaughan is furnishing our standard factory testing per our standard procedures, all field testing is to be provided by others. Tolerances for Vaughan equipment to be per pump specification included herein.
1.05.D.1: Cast steel is heat treated to minimum Rockwell C60.
1.05.D.3: Torsional analysis not purchased under this contact, as such not provided.
1.05.E: Vaughan is furnishing our standard, customized, I O&M manual package.
1.06: Vaughan is furnishing our standard warranty, terms and conditions herein.
2.02.A: Vaughan is only furnishing pumps and pump related equipment as described on the list of equipment to be supplied, all other equipment, guiderails, lifting devices, supports, electrical, controls, and appurtenances are to be furnished by others.
2.02.D.1.b:N/A
2.02.D.1.d: HI standards.
2.02.E: Vaughan is only furnishing equipment as specified in section 11313F and as described within the submittal.
2.05: Pump to be Vaughan standard and constructed as described on the pump specification sheet included within the submittal.
2.05.A.3 No back pull out option for this model.
2.05.C Pump motor is rated for maximum 15 minutes running dry.
2.06.D: Impeller will be dynamically balanced.
2.08.C :Motor inner/upper seal has carbon and ceramic faces. Motor lower/outer seal has silicon carbide seal faces. Pump mechanical seal (lowest of the three) is tungsten carbide.
2.08: Vaughan is furnishing our standard mechanical flushless seal.
2.09.A Motor shaft has permanently sealed grease lubricated bearings. Pump shaft has oil lubricated bearings. Bearings to be per pump specification herein.
2.10: Motor to be per list of equipment to be supplied.
2.10.A: VFD not indicated on the pump schedule in 3.04, Vaughan is furnishing pump intended for constant speed as specified.
2.10.B: VPMR provided.
2.10.D: Normally closed thermostats included.
2.12: Guide rails and lifting Devices Supplied by others.
2.13: All items by others.
2.14: Spare parts and special tools as described on the list of equipment to be supplied.
3.01: By onsite contractor.
3.02: Field testing by others. Vaughan authorized representative to provide startup and training assistance.
3.04: Pump Schedule – Required condition 3 – TDH to be approximately 35'.
Accessories – Hoist: By others.
Functional Testing – Field testing by others.

01600

1.02.A: Vaughan is not ISO 9001, Vaughan has included our QA MANUAL within the submittal in lieu of this requirement.
1.03.6: Samples N/A
1.03.7: Schedules by others.
1.03.8&12: Vaughan is furnishing our standard submittal package.
1.03.9-11: Vaughan is only furnishing spare parts and special tools as described on the list of equipment to be supplied.
2.02: Vaughan is furnishing pump and pump related equipment coated with our premium surface finish described within the submittal.
2.04: Vaughan is only furnishing spare parts and special tools as described on the list of equipment to be supplied.
3.01.A.3: Vaughan is furnishing our standard, customized, I O&M manual package following receipt of submittal approval.
3.01.B&C: Per Vaughans best crating practices.
3.02: By onsite contractor.
3.02.E: Spare parts will be packaged Vaughan standard. Tools box not purchased under this contract.
3.03: Storage by Onsite Contractor.
3.04: By onsite contractor.
3.05: By others.

15958

1.03: Standard submittals provided.
3.01: Vaughan is only furnishing factory testing as described on the list of equipment to be supplied. All field testing, instrumentation, and commissioning is to be furnished by others. Vaughan has included our factory testing procedures within the submittal.
3.02: N/A
3.04: Vaughan is furnishing testing with Job Motors.
3.05: N/A per pump schedule.
3.06: Vaughan is providing our standard hydro testing.
3.07: Vaughan will furnish impeller dynamically balanced. Vibrating testing not required per pump schedule.
3.08.A.1.a.1: Exception, Vaughan is furnishing 15 minute in air motor.
3.08.A.2&3: N/A per pump schedule.
3.08.B-D: N/A per pump schedule.
3.09: Vaughan will furnish factory test results for approval prior to shipment, all instrumentation and field testing to be provided by others.
3.10: Vaughan is only furnishing factory testing as described on the list of equipment to be supplied, no other testing is included under this contract.
3.11: By others.

09960

Vaughan is furnishing pump and pump related equipment coated in our premium surface finish.

01782

Vaughan is furnishing our standard, I O&M Manual package.

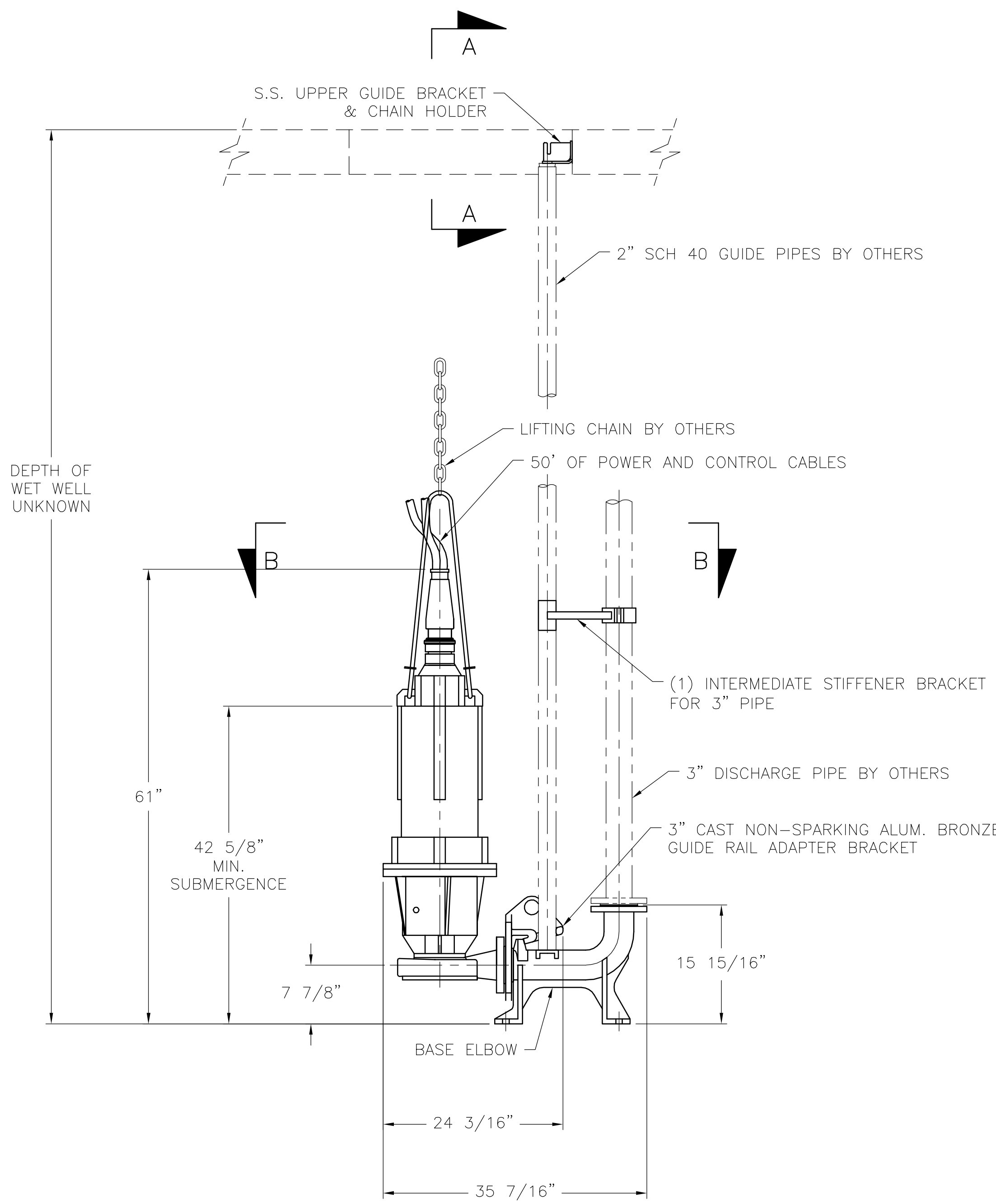
01330

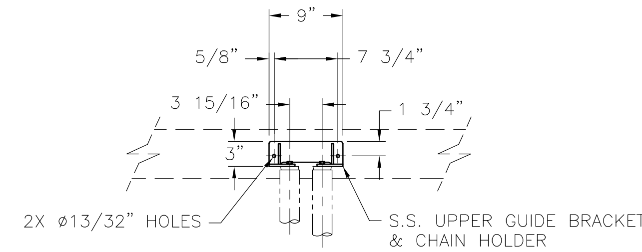
Vaughan is furnishing our standard submittal package.

NOTE:

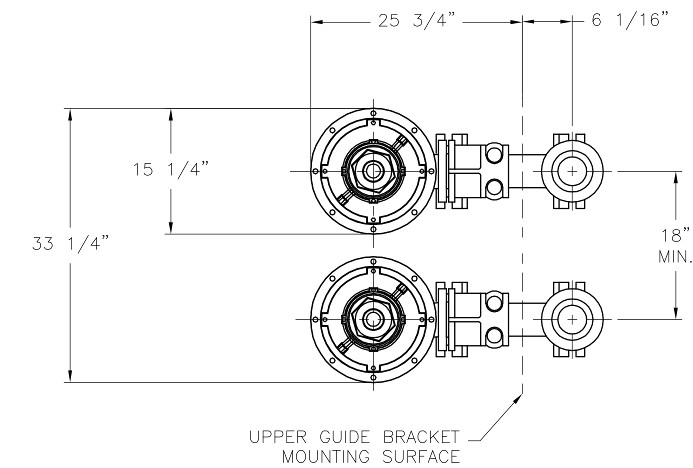
Warranty is limited to Vaughan standard warranty terms, as included herein.

Vaughan Co. will provide our standard, customized, Installation, Operation and Maintenance Manuals following submittal approval. All comments and resubmissions must be completed within one year of preliminary I,O&M submittals. No comments will be accepted after that time and the I,O&M will be marked approved. If no approval or comments have been received within that year I,O&M will be marked approved.

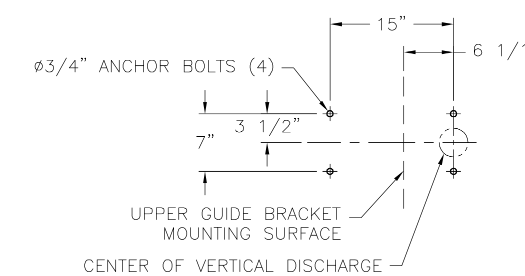




VIEW A–A



SECTION B–B



ANCHOR BOLT DETAIL

NOTES:

1. PUMP WEIGHT: 159 LBS
2. MOTOR WEIGHT: 335 LBS
3. TOTAL WEIGHT: 494 LBS
4. PERFORMANCE REQUIRED: 235 GPM @ 43' TDH
   PERFORMANCE REQUIRED: 150 GPM @ 48' TDH
   PERFORMANCE REQUIRED: 325 GPM @ 30' TDH
   PERFORMANCE REQUIRED: 400 GPM @ 15' TDH (STAMP ON TAG)
5. MOTOR: 10 HP, 1750 RPM, 460 V, 3 PH, 60 HZ, 1.15 SF, XP (CLASS 1, GROUP C & D), W/ TANDEM MECHANICAL SEALS, MOISTURE SENSORS, INTERNAL THERMOSTATS ROUTINE MOTOR TEST BY ABB
6. FINISH: SANDBLAST, SOLVENT WASHED & COATED W/ MIN. 30 MDFT TNEMEC SERIES 431 EPOXY (EXCEPT MOTOR)
7. (2) 3" SPARK PROOF GUIDE RAIL SYSTEMS REQUIRED.
8. BUNA N ELASTOMERS REQUIRED
9. FACTORY TESTING REQUIRED AS PER LIST OF EQUIP.
10. (2) VPMR MOISTURE & OVER TEMPERATURE RELAYS TO BE SUPPLIED.
11. OIL MONITOR NOT SUPPLIED. BEARING HOUSING TO BE PLUGGED.
12. TC SEAL FACES REQUIRED.

Vaughan
VAUGHAN CO., INC.
364 MONTE–ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249–4042
FAX: (360) 249–6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: OUTLINE DIMENSIONS
MODEL: S3M2–460V–076
(2) PRIMARY SCUM PUMPS
HILO WWTP, HI

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
PAB	AES	KCW	1"=1'	
JOB ORDER NUMBER:		DATE:	DRAWING NUMBER:	REVISION NUMBER:
174838A/B		4/21/25	124523	0

SPECIFICATION: S3M SUBMERSIBLE CHOPPER PUMPS

The submersible chopper pump shall be specifically designed to pump waste solids at heavy consistencies without plugging or dewatering of the solids. Materials shall be chopped and conditioned by the pump as an integral part of the pumping action. The pump must have demonstrated the ability to chop through and pump high concentrations of solids such as plastics, heavy rags, grease and hair balls, wood, paper products and stringy materials without plugging, both in tests and field applications. Pump shall be manufactured by Vaughan Co., Inc.

DETAILS OF CONSTRUCTION

A. **Casing:** Shall be of volute design, spiraling outward to the Class 125 flanged centerline discharge. Casing shall be cast ductile iron with all water passages to be smooth, and free of blowholes and imperfections for good flow characteristics. Casing shall include a replaceable Rockwell C60 alloy steel cutter to cut against the rotating impeller pump-out vanes for removing fiber and debris.

B. **Impeller:** Shall be a 5 blade (3M) semi-open type with pump out vanes to reduce seal area pressure. Chopping of materials shall be accomplished by the action of the cupped and sharpened leading edges of the impeller blades moving across the stationary cutter bar at the intake openings, with a set clearance between the impeller and cutter bar of 0.015-0.025” (40-.65 mm) cold. Impeller shall be cast alloy steel heat treated to minimum Rockwell C60 and dynamically balanced. The impeller shall be keyed to the shaft and shall have no axial adjustments and no set screws.

C. **Cutter Bar:** Shall be recessed into the pump casing and shall contain at least 2 shear bars extending diametrically across the intake opening to within 0.010-0.030” (0.25-0.75 mm) of the rotating cutter nut tooth, for the purpose of preventing intake opening blockage and wrapping of debris at the shaft area. Cutter bar shall be cast alloy steel heat-treated to minimum Rockwell C60.

D. **Cutter Nut**: The impeller shall be secured to the shaft using a cutter nut, designed to cut stringy materials and prevent binding using a raised, rotating cutter tooth. The cutter nut shall be cast alloy steel heat treated to minimum Rockwell C60.

E. **Upper Cutter:** Shall be threaded into the casing behind the impeller, designed to cut against the pump-out vanes and the impeller hub, reducing and removing stringy materials from the mechanical seal area. Upper cutter shall be cast alloy steel heat treated to minimum Rockwell C60. The upper cutter teeth are positioned as closely as possible to the center of shaft rotation to minimize cutting torque and nuisance motor tripping. The ratio of upper cutter cutting diameter to shaft diameter in the upper cutter area of the pump shall be 3.0 or less.

F. **Pump Shafting**: Shafting shall be heat treated alloy steel, with a minimum diameter of 1.5” (38 mm) in order to minimize deflection during solids chopping.

G. **Bearing Housing:** Shall be cast ductile iron, and machined with piloted bearing fits for concentricity of all components. Piloted motor mount shall securely align motor on top of bearing housing.

H. **Thrust Bearings:** Shaft thrust in both directions shall be taken up by two back-to-back mounted single-row angular contact ball bearings, with a minimum L-10 rated life of 100,000 hours. Overhang from the centerline of the lower thrust bearing to the seal faces shall be a maximum of 1.7" (43 mm). A third mechanical seal (two in motor) shall also be provided to isolate the bearings from the pumped media. The third seal, as well as the thrust bearings shall be oil bath lubricated in the bearing housing by ISO 46 hydraulic oil. Shaft overhang exceeding 1.7” (43 mm) from the center of the lowest thrust bearing to the seal faces shall be considered unacceptable.

I. **Pump Mechanical Seal:** The mechanical seal shall be located immediately behind the impeller hub to maximize the flushing available from the impeller pump-out vanes. The seal shall be a cartridge-type mechanical seal with Viton O-rings and tungsten carbide faces. This cartridge seal shall be pre-assembled and pre-tested so that no seal settings or adjustments are required from the installer. Any springs used to push the seal faces together must be shielded from the fluid to be pumped. The cartridge shall also include a 17-4PH, heat-treated seal sleeve and a cast ductile iron seal gland.

J. **Shaft Coupling:** The submersible motor shall be close coupled directly to the pump shaft using a solid sleeve coupling, which is keyed to both the pump and motor shafts. Slip clutches and shear pins between the shaft and the motor are considered unacceptable.

K. **Stainless Steel Nameplate:** The stainless steel nameplate giving the manufacturer's model and serial number, rated capacity, head, speed and all pertinent data shall be mounted to a larger stainless steel plaque. Warning tags and oil reservoir will be mounted to this same plaque. Plaque is to be fastened to wall or structure adjacent to pump.

L. **Submersible Motor:** The submersible motor shall be U/L or FM listed and suitable for Class I, Group C & D, Division I hazardous locations, rated at 10 HP, 1750 RPM, 460 Volts, 60 Hertz and 3 phase, 1.15 service factor with Class F insulation. Motor shall have tandem mechanical seals in oil bath and dual moisture sensing probes. Moisture probes must be connected to indicate water intrusion. The lower motor seal shall be exposed only to the lubricant in the pump bearing housing, with no exposure to the pumped media. Motor shall include two normally closed automatic resetting thermostats connected in series and embedded in adjoining phases. The thermostats must be connected per local, state, and/or the National Electric Code to maintain hazardous location rating and to disable motor starter if overheating occurs. Motor frame shall be cast iron, and all external hardware and shaft shall be stainless steel. Motor shall be sized for non-overloading conditions.

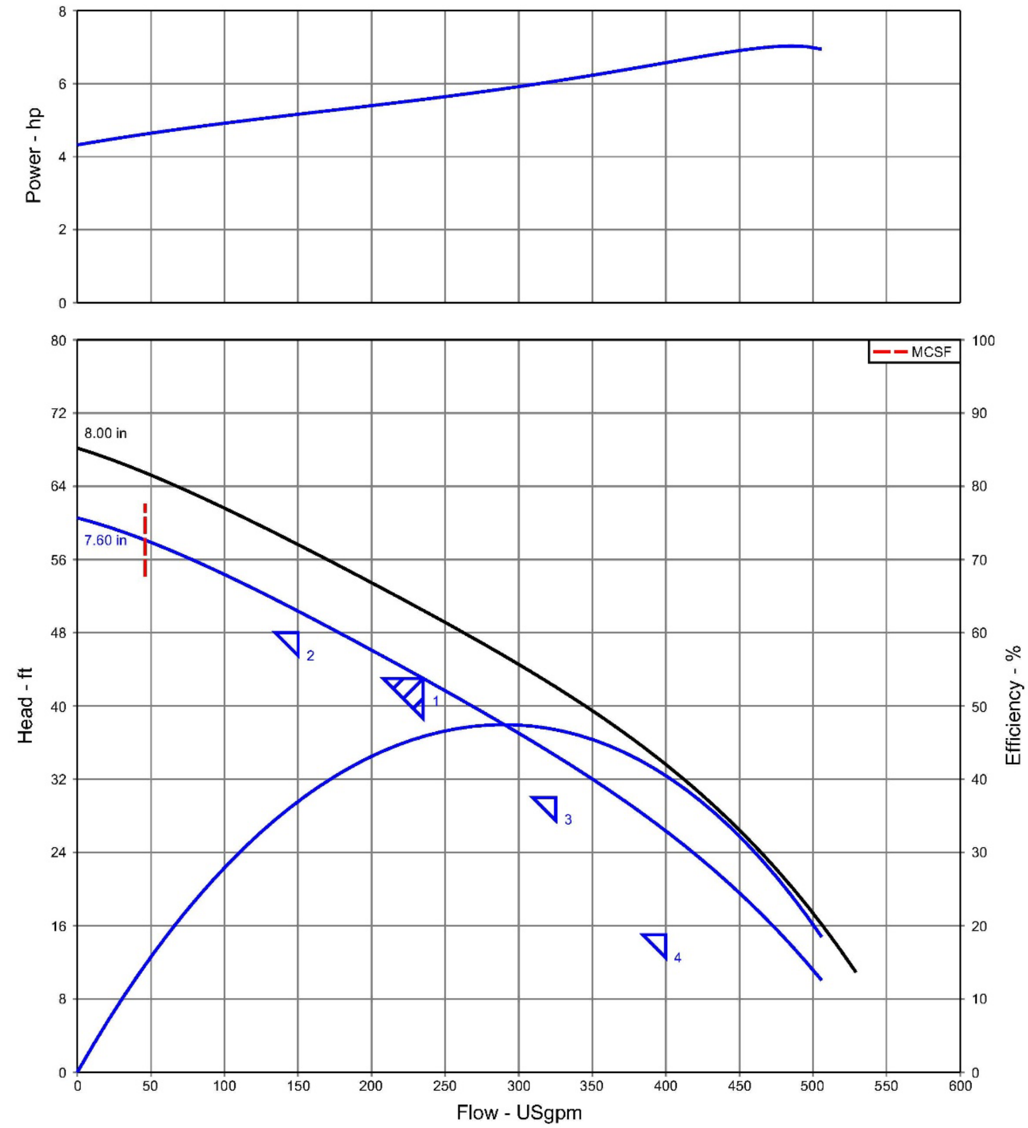
M. **Spark Proof Guide Rail System:** Provide a non-sparking guide rail system consisting of two galvanized or stainless steel guide rails (by others), cast bronze pump guide bracket, cast ductile iron discharge elbow with mounting feet and Class 125 flanges, 316 stainless steel upper guide rail mounting bracket, and 316 stainless steel intermediate guide rail stiffener bracket every 10 feet (3 m). System design shall prevent spark ignition of explosive gases during pump installation and removal.

N. **Premium Surface Preparation:** Solvent wash, sandblast, and coat with minimum 30 MDFT Tnemec Perma-Shield PL Series 431 epoxy (except motor).

PERFORMANCE CURVE - S3M2-460V-076

Customer :
End user :

Multiple Conditions Curve
Vaughan Quotation System 24.6.2



Item number	: 001	Size	: S3M - Trim Dia
Service	:	Stages	: 1
Quantity	: 1	Speed, rated	: 1750 rpm
Quote number	: 18642	Based on curve number	: S3M-1750-TD
Date last saved	: 28 Apr 2025 12:32 PM	Efficiency	: 45.84 %
Flow, rated	: 235.0 USgpm	Power, rated	: 5.57 hp
Differential head / pressure, rated	: 43.00 ft	NPSH required	: -
		Viscosity	: 1.00 cP
Fluid density, rated / max	: 1.000 / 1.000 SG	Cq/Ch/Ce/Cn [ANSI/HI 9.6.7-2010]	: 1.00 / 1.00 / 1.00 / 1.00

Vaughan Company, Inc. · 364 Monte-Elma Road · Montesano , WA 98563
phone: 1-888-249-CHOP · fax: 1-360-249-6155 · Chopperpumps.com

Customer :
End user :

Multiple Conditions Datasheet

Vaughan Quotation System 24.6.2

Item number	: 001	Quantity	: 1	Size	: S3M - Trim Dia
Service	:	Quote number	: 18642	Stages	: 1
		Date last saved	: 28 Apr 2025 12:32 PM	Speed, rated	: 1750

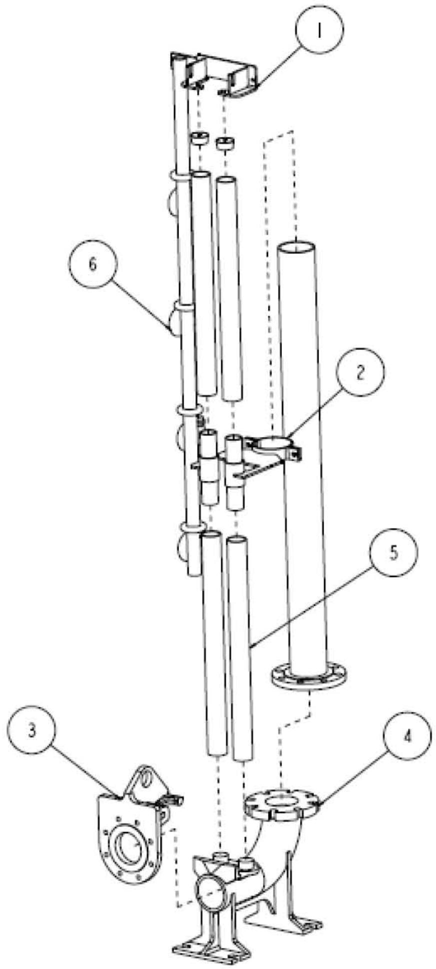
Condition #		1	2	3	4
Description		-	-	-	-
Temperature, max	deg F	68.00	68.00	68.00	68.00
Fluid density, rated / max	SG	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000
Viscosity, rated	cP	1.00	1.00	1.00	1.00
Primary condition		◉	○	○	○
Size		S3M - Trim Dia			
Stages		1			
Impeller diameter, rated	in	7.60			
Flow, rated	USgpm	235.0	150.0	325.0	400.0
Head, rated (requested)	ft	43.00	48.00	30.00	15.00
Head, rated (actual)	ft	43.01	50.36	34.59	26.35
Suction pressure, rated / max	psi.g	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
NPSH available, rated	ft	Ample	Ample	Ample	Ample
Speed, rated	rpm	1750	1750	1750	1750
Selection status		Acceptable	Acceptable	Acceptable	Acceptable
Cq/Ch/Ce/Cn [ANSI/HI 9.6.7-2010]		1.00 / 1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00 / 1.00	1.00 / 1.00 / 1.00 / 1.00
Efficiency	%	45.84	36.96	46.77	40.49
NPSH required	ft	-	-	-	-
Power, rated	hp	5.57	5.16	6.07	6.57

Vaughan Company, Inc. · 364 Monte-Elma Road · Montesano , WA 98563
phone: 1-888-249-CHOP · fax: 1-360-249-6155 · Chopperpumps.com

EXPLODED ASSEMBLIES - S3M2-460V-076

Unmatched Reliability Vaughan®

2" Guiderail System
Materials of Construction
All 3"– 8" Pumps



ITEM	DESCRIPTION	STANDARD	SPARK PROOF
1	Upper Guide Bracket	Stainless Steel	Stainless Steel
2	Intermediate Stiffener Bracket	Stainless Steel	Stainless Steel
3	Guide Bracket	Ductile Iron	Aluminum Bronze
4	Base Elbow	Ductile Iron	Ductile Iron
5	Guide Pipes 2" Sch. 40	Galvanized or Stainless Steel	Galvanize or Stainless Steel
6	Retractable Float Switch Assy.	3/4" Stainless Steel Pipe/ABS Floats	3/4" Stainless Steel Pipe/ABS Floats

Refer to Bill of Materials for part numbers.

VAUGHAN CO., INC.
364 Monte Elma Road, Montesano, WA 98563
Phone: 1-360-249-4042 / Fax: 1-360-249-6155
Toll Free Phone (US only): 1-888-249-CHOP (2467)
Web Site: www.chopperpumps.com
Company E-mail: info@chopperpumps.com

For all current patents, see http://www.chopperpumps.com/patents.htm

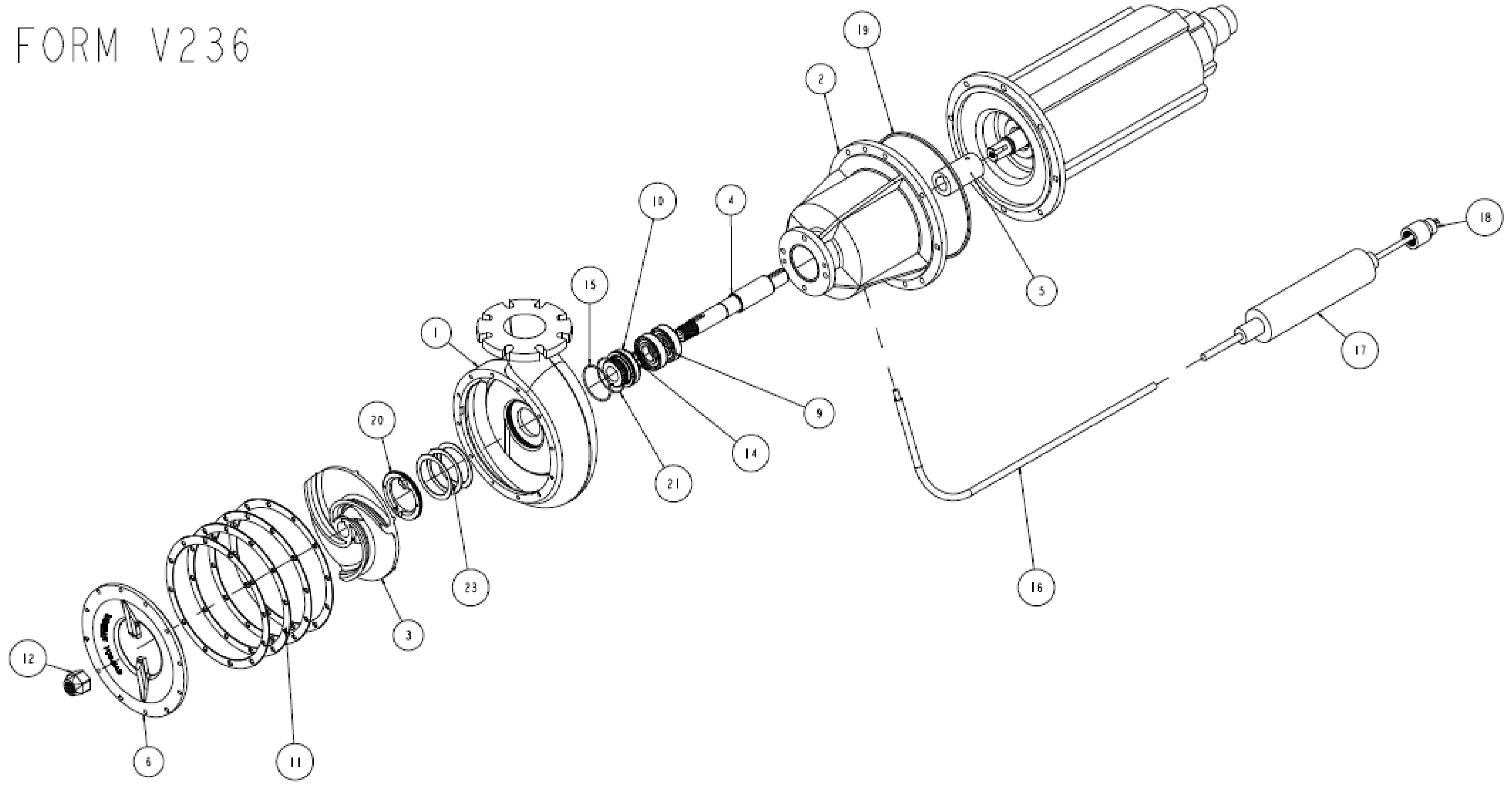
GUARANTEED SINCE Vaughan 1960 PERFORMANCE

MADE IN THE USA

V275, Rev. 1

10/19, ECN4543

FORM V236



NO.	DESCRIPTION	MATERIAL	NOTES
1	CASING	DUCTILE IRON	
2	BEARING HOUSING	DUCTILE IRON	
3	IMPELLER	CAST STEEL, HT TO RC 60+	
4	SHAFT	SHAFT STEEL	
5	COUPLING SLEEVE	STEEL	
6	CUTTER BAR	CAST STEEL, HT TO RC 60+	
9	THRUST BEARING	ANGULAR CONTACT BALL BEARING	(2) REQUIRED
10	MECHANICAL SEAL	VAUGHAN CARTRIDGE SEAL	
11	CUTTER BAR SHIM	FIBER GASKET MATERIAL	
12	CUTTER NUT	CAST STEEL, HT TO RC 60+	
14	O-RING	BUNA-N OR VITON	SEAL SLEEVE O-RING, (2) REQUIRED
15	O-RING	BUNA-N OR VITON	CASING/BRG HSG O-RING
16	OIL HOSE	1/2" ID RUBBER	
17	OIL RESERVIOR	URETHANE	
18	OIL LEVEL MONITOR	URETHANE CAP /W S.S. INTERNAL SENSOR	
19	GORETEX SEALANT	GORETEX	
20	UPPER CUTTER	CAST STEEL, HT TO RC 60+	
21	O-RING	BUNA-N OR VITON	SEAL/BRG HSG O-RING
23	UPPER CUTTER SHIMS	FIBER GASKET MATERIAL	

REV	DATE	DESCRIPTION	BY
3	9/24/13	ADDED ITEMS #14 & #21, ECN 2932	EPL
2	6/26/13	CHANGED ADJUSTER STYLE, ECN2896	EPL
1	7/17/12	CHANGED ADJUSTER STYLE, ECN2720	AJC

MODEL NAME: S4PR_ASSY

Vaughan

VAUGHAN CO., INC.

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TITLE: ASSEMBLY DRAWING
S SERIES SUBMERSIBLE
WITHOUT BACKPLATE
SHEET 1 OF 2

CHK'D BY	DRAWN BY	SCALE	DATE	DRAWING NUMBER	REV
RAB	AJC	0.188	9/22/10	114239	5

AC induction motor performance data

Record #86077 - Typical performance - not guaranteed values

Winding	07WGA0017
Type	0738M
Enclosure	TENV

Nameplate data

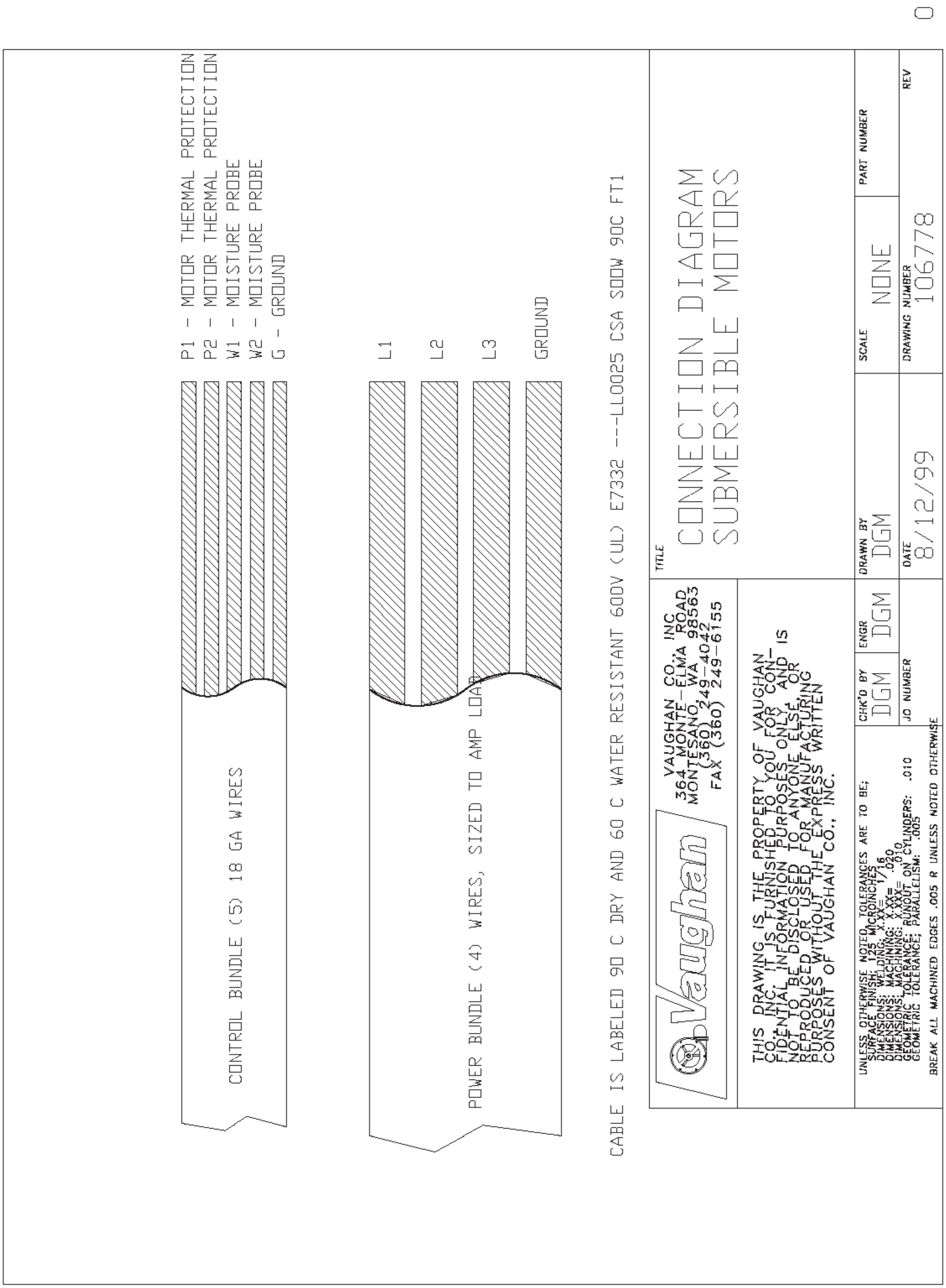
Rated Output			10
Volts			230/460
Full Load Amps			25/12.5
R.P.M.			1765
Hz	60	Phase	3
KVA Code			H
S.F.			1.15
NEMA Nom. Eff.	90.2	Power Factor	82
Duty			15 MIN
S.F. Amps			

460 V, 60 Hz:
High Voltage Connection

Full Load Torque	29.74 LB-FT
Start Configuration	direct on line
Breakdown Torque	83.88 LB-FT
Pull-up Torque	29.9 LB-FT
Locked-rotor Torque	45.11 LB-FT
Starting Current	87.77 A
No-load Current	4.92
Line-line Res. @ 25°C	1.25 Ω
Temp. Rise @ Rated Load	15°C
Temp. Rise @ S.F. Load	18°C
Locked-rotor Power Factor	37.1
Rotor inertia	0.934 lb-ft²

Load Characteristics 460 V, 60 Hz, 10 HP

% of Rated Load	NL	25	50	75	100	125	150	SF
Power Factor	6	47	68	78	82	84	84	83
Efficiency	0	88.1	90.8	91.9	90.9	90	88.6	90.4
Speed	1799	1792	1785	1777	1768	1758	1747	1762
Line amperes	4.92	5.73	7.53	9.86	12.49	15.44	18.79	14.3


P1 - MOTOR THERMAL PROTECTION
P2 - MOTOR THERMAL PROTECTION
W1 - MOISTURE PROBE
W2 - MOISTURE PROBE
G - GROUND
CONTROL BUNDLE (5) 18 GA WIRES
L1
L2
L3
GROUND
POWER BUNDLE (4) WIRES, SIZED TO AMP LOAD
CABLE IS LABELED 90 C DRY AND 60 C WATER RESISTANT 600V (UL) E7332 ---LL0025 CSA SOOW 90C FT1
Vaughan
VAUGHAN CO., INC
364 MONTE-ELMA ROAD
MONTESANO, WA 98563
(360) 249-4042
FAX (360) 249-6155
TITLE
CONNECTION DIAGRAM
SUBMERSIBLE MOTORS
THIS DRAWING IS THE PROPERTY OF VAUGHAN CO., INC. IT IS FURNISHED TO YOU FOR CONFIDENTIAL INFORMATION PURPOSES ONLY AND IS NOT TO BE DISCLOSED TO ANYONE ELSE, OR REPRODUCED OR USED FOR MANUFACTURING PURPOSES WITHOUT THE EXPRESS WRITTEN CONSENT OF VAUGHAN CO., INC.
UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE;
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS: WELDING: X.XX= 1/16
DIMENSIONS: MACHINING: X.XX= .020
DIMENSIONS: MACHINING: X.XXX= .010
GEOMETRIC TOLERANCE: RUNOUT ON CYLINDERS: .010
GEOMETRIC TOLERANCE: PARALLELISM: .005
BREAK ALL MACHINED EDGES .005 R UNLESS NOTED OTHERWISE
CHK'D BY DGM
ENGR DGM
DRAWN BY DGM
SCALE NONE
PART NUMBER
JO NUMBER
DATE 8/12/99
DRAWING NUMBER 106778
REV
0

SPECIFICATIONS

Submersible motors

Water and sewage U/L listed, Class 1, Groups C and D

Offering a solution for both wet and dry pit applications, Baldor-Reliance® submersible motors provide a proven solution for municipal and industrial wastewater operations as well as slurry pumps, aerators and mixers.

1-450 Hp submersible wet pit 15 minute in air

1.0 Scope
This specification details the mechanical and electrical requirements for squirrel-cage induction motors, both single and three phase, designed for wet well and dry well submersible applications in water and sewage.

It is the intent of this specification to define submersible premium quality motors which will provide efficient operation with high mechanical integrity under adverse operating conditions for maximum life and minimum life cycle costs. This specification covers sewage wet well and dry well applications defined by the National Electric Code as Class 1: Division 1, hazardous locations section 501-8 (a) requiring explosion proof construction.

2.0 General

2.1 All motors covered by this specification shall conform to the latest applicable requirements of NEMA, IEEE, ANSI, and NEC standards.

1-150 Hp submersible dry pit continuous in air

2.2 Motors shall be designed for continuous submerged duty in water and sewage, and minimum 15 minute duty continuous in air under full load operating conditions. Motors used in dry well operation shall be designed for continuous in air under full load operating conditions.

2.3 Three (3) phase shall be rated 200/400, 230/460 or 575 volt. Single voltage motors will also be available. Single phase motors shall be rated 115/230 volts. Multi-voltage motors shall be final connected for the highest voltage unless specified by customer and designed for easy field reconnection.

2.4 Ratings will be based on 40 C ambient conditions.

2.5 Motor construction shall be designed to withstand 100 psi* water pressure at all seal locations. Maximum submerged depth is 160 feet.
*Total sum of operating depth and reflected pump pressure

2.6 All motors shall be furnished with Class F rated insulation materials or better. Insulation materials rated lower than Class F (Class B or A) are specifically prohibited.

2.7 Motors 1 horsepower up to and including 135 horsepower, shall be rated as Class F, 1.15 service factor, Class 1, Groups C& D. Motors under 1 Hp shall be rated as Class B, 1.0 service factor, Class 1, Group D.

2.8 All motors will be CSA (Canadian Standard Association) and U.L (Underwriters Laboratories) approved and nameplated accordingly.

2.9 All motors shall be manufactured in the United States of America. The ability to provide any and all replacement parts, engineering design support, complete dynamometer testing, and U/L rerate capability shall be provided domestically.

3.0 **Mechanical**

3.1.0 Bearings and Lubrication

3.1.1 Bearings shall be ball, single row, deep groove, Conrad type, and shall have a Class 3 internal fit conforming to AFBMA Std. 20. Standard bearing size for 440TY frame is 7222 duplex bearing. Other arrangements are available depending on shaft loading. Please contact your local Baldor representative for assistance.

3.1.2 Bearing shall be selected to provide a minimum L_{10} rating life of 17,500 hours.

3.1.3 The motor shall be designed to limit the bearing temperature rise to a maximum of 60 degree C under full load conditions.

3.1.4 Motors shall be greased by the manufacturer with a premium moisture resistant polyurea thickened grease containing rust inhibitors and suitable for operation over a temperature range of -25 degree C to + 120 degree C.

3.2.0 Shaft seal

3.2.1 Two independently-mounted mechanical face type seals shall be provided. The inner and outer seals shall be separated by an oil filled chamber. The oil chamber shall act as a barrier to trap moisture and provide sufficient time for a planned shutdown. The oil shall also provide lubrication to the internal seal.

Seal	Description	Application
Type 21 carbon ceramic faces	Standard seal offering. A general purpose seal of stainless steel construction with carbon ceramic seal faces.	Used in food processing, petrochemical and wastewater applications with relatively clean effluent.
Type 21 tungsten carbide faces	Same type of seal except with tungsten carbide faces. Harder seal faces.	For more demanding applications, more abrasive fluids.
Type 21 silicon carbide faces	Same type of seal except with silicon carbide faces.	For the most demanding, slurry type applications.
Slurry type seal *	Seal is completely contained in the oil chamber. Non-explosion proof design only.	Used in non-explosion proof slurry applications. Can be used continuous in air.
Hydropad seal **	Special seal design with hydropad scallops to run cooler.	Recommended for applications with low specific gravity or lighter film between the faces.

* Slurry design is not UL approved, and is limited to 15 psi.
** Recommended not to operate in dry-run condition

3.2.2a Buna-N O-rings and elastomers are standard. Viton elastomers are recommended for ambients of 61 C and higher.

3.2.3 The outer seal construction shall be designed for easy replacement.

3.2.4 The outer seal assembly is selected to prevent the entrance of moisture into the motor oil chamber. The OEM is responsible for protecting the outer seal from exposure to solids and foreign materials (such as banding, ropes, or strings). It is the responsibility of the pump OEM to insure the outer seal is lubricated by the effluent for both 15 minute in air and continuous in air applications.

3.2.5 In compliance with U/L standards for explosion proof motors, a flamepath shall be proved by a labyrinth slinger in the bottom flange in order to prevent the ignition of ambient gases. Under such conditions the seal design shall allow for pressure relief across either seal face.

3.3.0 Moisture detector system

3.3.1 Warrick type, dual (2) moisture sensing probes are to be provided that extend into the oil chamber to detect the presence of moisture should the outer seal fail.

3.3.2 The moisture detection system shall be selected by the customer to utilize the Warrick sensing probes. A suitable relay such as a Warrick Type 2810, an Ametek APT Model 8040, or equivalent must be provided by others. For additional information on the moisture probes and Warrick Type 2810 relay, go to www.gemssensors.com. For additional information on the Ametek APT Model 8040 relay, go to www.bwallen.com/alarm-leak-detection.html.

3.4.0 Cap/cable assembly

3.4.1 The power cable and cap assembly shall be designed to prevent moisture from wicking through the cable assembly even when the cable jacket has been punctured.

3.4.2 Power and control cable entry into the lead connection chamber shall be epoxy encapsulated for positive moisture sealing.

3.4.3 A Buna-N power and control cable grommet shall be provided in addition to the epoxy filled leads.

3.5.0 Enclosure and shaft

3.5.1 The motor enclosure including frame, end brackets, flange and cap assembly shall be cast iron ASTM type A-48, Class 25 or better.

3.5.2 Motor 180 through 360 frame's construction will not have fins and will be a smooth surface to prevent the clogging of solids and provide for easy cleaning. 440 frames are constructed with finned stator frames and smooth end brackets.

3.5.3 The top end bracket will include integrally cast provisions for vertical lifting capability.

3.5.4 All mating frame fits to have rabbet joints with large overlap as well as O-ring shall be Buna-N (nitrile). Viton O-rings may be supplied as an option and are required for ambient conditions of 61 degrees C and higher.

3.5.5 Motor shaft material is 416 stainless steel. Other materials available as options are 303, 304, 410 stainless steel, 17-4PH, Carpenter 20, Monel, or Nitronic 50.

3.5.6 All external hardware including the motor nameplate shall be made of stainless steel.

3.5.7 Motor rotor construction shall be die cast aluminum. Rotors shall be dynamically balanced to NEMA limits per MG1-7.8.2 Table 7-1. Balance weights if required shall be secured to the rotor fan blades by rivets. Machine screws and nuts are prohibited.

3.5.8 The frame, opposite drive end bracket, and cable cap assembly shall receive an alkyd primer and epoxy ester finish coat of high grade paint to resist rust and corrosion.

3.5.9 The drive end bracket will have oil fill and drain holes.

4.0 Electrical

4.1 All motors shall successfully operate under power supply variations per NEMA MG1-14.30.

4.2 Motors shall be designed to limit the maximum surface temperature to NEC specifications for Division 1, Class 1, Group D or Class 1, Group C&D for hazardous locations.

4.3 All motors shall be NEMA Design B or A with torque and starting current in accordance with NEMA MG-1.

4.4 Motors shall have copper windings.

4.5 Motor insulation system shall be Class F minimum, utilizing material and insulation systems evaluated in accordance with IEEE 117 classification tests.

4.6 Motor leads shall be nonwicking type, Class F temperature rating or better and permanently numbered for identification.

4.7 The stator insulation system shall be treated with a non hydroscopic epoxy varnish with a dip and bake process.

4.8 All motors to include two (2) normally closed thermostats connected in series and embedded in adjoining phases as required by Underwriters Laboratories for motors of 1 Hp or higher.

4.9 All motors operated on inverter power will be rated as Class 1, Group D with a T2A temperature code.

4.10 Each completed and assembled motor shall receive a routine factory test per NEMA MG-1. Tests are in accordance with IEEE Std. 112.

—
ABB Motors and Mechanical Inc.
5711 R.S. Boreham, Jr. Street
Fort Smith, AR 72901
Ph: 1.479.646.4711

new.abb.com/motors-generators

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9AKK107106581 07.2018

Vaughan Pump Monitor Relay

MADE IN THE U.S.A.

UL FILE #E101681

OPERATION

The VPMR provides Motor Over Temperature and Seal Leakage alarms for Vaughan Submersible Pumps.

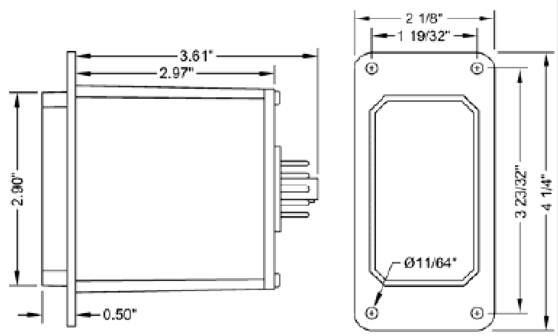
Motor Over Temperature Alarm - The unit applies a low voltage DC signal to the Motor Thermal Sensor to check its status. If the unit detects that the Motor Thermal Sensor contacts are closed (normal condition), the Overtemp indication remains off, and the Overtemp Relay is energized closing the contacts between terminals 2 and 11.

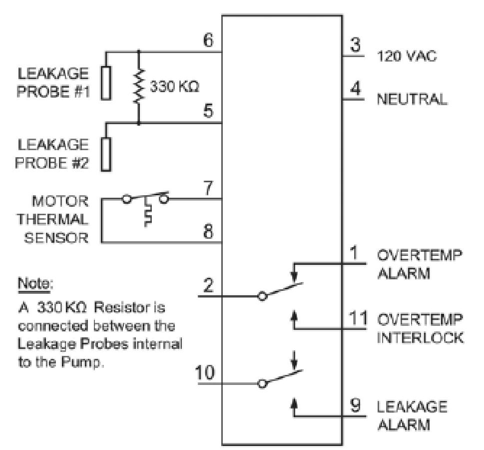
If the Motor Thermal Sensor contacts open (Over Temperture condition), the Overtemp Indication is turned on and the Overtemp Alarm Relay is de-energized opening the contacts between terminals 2 and 11 and closing the contacts between terminals 2 and 1.

When the High Motor Temperature condition has cleared, the unit will reset based on the position of Alarm Reset Mode Select Switch (Auto or Manual). When in the Auto position, the Overtemp Alarm resets automatically. If the switch is in the Manual position, the Overtemp Reset Push-button must be pushed for approximately 1.5 seconds to clear the alarm.

Seal Leakage Alarm - The unit applies a low voltage AC signal across the two Leakage Probes to detect moisture in the pump motor. A Seal Leakage condition is considered present when the amount of moisture in the motor causes the resistance between the two probes to drop below the setting on the potentiometer. When this occurs the unit turns on the Leakage Indication and energizes the Leakage Alarm Relay closing the contacts between terminals 9 and 10.

The alarm trip point may be set by the following procedure: Isolate the Leakage Probes from terminals 5 and 6. Connect a resistor, with the desired trip value, across terminals 5 and 6. Slowly adjust the potentiometer to the point where the alarm turns on. Remove the resistor and reconnect the Leakage Probes to terminals 5 and 6.





SPECIFICATIONS

Input Power:	120 VAC ±10%, 7.0 VA max
Output Rating:	8A Resistive @ 120VAC
Operating Temp:	-20°C to +65 °C
Storage Temp:	-45°C to +85 °C
Temp Sensor Voltage:	6.6 VDC ±10%
Leak Sensor Voltage:	4.7 VAC ±10%
Enclosure:	White Lexan
Base:	Phenolic

ORDERING INFORMATION

Part Number: V800-497

Vaughan Co., Inc. 364 Monte-Elma Road, Montesano, WA 98563 (360) 249-4042

TNEMEC

PRODUCT DATA SHEET

PERMA-SHIELD® PL SERIES 431

PRODUCT PROFILE

GENERIC DESCRIPTION Modified Polyamine Ceramic Epoxy

COMMON USAGE A 100% solids, abrasion-resistant lining specifically designed for wastewater immersion and fume environments and exposure to corrosive soils. Provides low permeation to H_2S gas, protects against MIC and provides chemical resistance to steel, ductile iron pipe and fittings for severe wastewater or buried exposures. A coal-tar free, resin-rich formulation with low pigment volume concentration (PVC) for maximum performance.

COLORS 5024 Sewer Pipe Green. **Note:** Epoxies chalk with extended exposure to sunlight.

FINISH Gloss

SPECIAL QUALIFICATIONS Contains 20% ceramic microspheres for increased abrasion resistance. Compatible with high-velocity jet sewer cleaning (hydrocleaning) with 0-degree tips (Reference Technical Bulletin No. 11-86).
Meets the performance requirements of AWWA C 210 (not for potable water contact).

COATING SYSTEM

PRIMERS Self-priming, Series 61 or N140. **Note:** Self-priming or Series 61 is recommended for use in mesophilic anaerobic digesters and other severe exposures. **Note:** Series 431 must be applied to the Series 61 or N140 within 7 days. Scarify the surface with fine abrasive before topcoating if this maximum recoat window is exceeded. Contact your Tnemec representative for more information.

SURFACE PREPARATION

Prepare surfaces by method suitable for exposure and service.

STEEL **Wastewater Service:** SSPC-SP5/NACE 1 White Metal Blast Cleaning or ISO Sa 3 Blast Cleaning to Visually Clean Steel with a minimum angular anchor profile of 3.0 mil (76.2 microns).
Raw water or Buried Service: SSPC-SP10/NACE 2 Near-White Blast Cleaning or ISO Sa 2 1/2 Very Thorough Blast Cleaning with a minimum angular anchor profile of 3.0 mils (76.2 microns).

DUCTILE IRON All surfaces of ductile iron pipe and fittings shall be delivered to the application facility free of asphalt or any other protective lining on the interior surface. All oils, small deposits of asphalt paint, grease, and soluble deposits shall be removed in accordance with NAPF 500-03-01 Solvent Cleaning prior to abrasive blasting.
Pipe Interior: Uniformly rotary-abrasive blast using angular abrasive to a NAPF 500-03-04: Internal Pipe Surface Condition. When viewed without magnification, the interior surfaces shall be free of visible dirt, dust, annealing layer, rust, mold coating and other foreign matter. Random staining and tightly adhered annealing oxide residue shall be limited to no more than 5 percent and may consist of random shadows, streaks or discolorations that can be clearly distinguished from the substrate and is trapped in the pits, pores, indentations, profile or other capillary structures on the surface. Any area where rust appears before application shall be re-blasted. The surface shall contain a minimum angular anchor profile of 3.0 mils (76.2 microns) (Reference NACE RP0287 or ASTM D 4417, Method C).
Pipe Exterior: Uniformly abrasive blast the entire surface using angular abrasive to an NAPF 500-03-04: External Pipe Surface Condition. When viewed without magnification, the exterior surfaces shall be free of all visible dirt, dust, loose annealing oxide, mold coating, rust and other foreign matter. Tightly adherent annealing oxide and rust staining may remain on the surface provided they cannot be removed by lifting with a dully putty knife. Any area where rust reappears before application shall be reblasted. The surface shall contain a minimum angular anchor profile of 3.0 mils (76.2 microns). The exterior surfaces shall be primed with recommended epoxy primer at 3-5 mils (76.2 to 127 microns) dry film thickness.
Fittings: Uniformly abrasive blast using angular abrasive to a NAPF 500-03-05: Fitting Blast Clean #1 condition, no staining. When viewed without magnification, the interior surfaces shall be free of all visible dirt, dust, annealing oxide, rust, mold coating and other foreign matter. Any area where rust reappears before application shall be reblasted. The surface shall contain a minimum angular anchor profile of 3.0 mils (76.2 microns) (Reference NACE RP0287 or ASTM D 4417, Method C).

ALL SURFACES Must be clean, dry and free of oil, grease and other contaminants.

TECHNICAL DATA

VOLUME SOLIDS 100% (mixed)

RECOMMENDED DFT **Carbon Steel:** 30.0 to 50.0 mils (760 to 1270 microns) in one or more coats.
Ductile Iron: 40 mils (1015 microns) (nominal) in one or more coats.
Note: Number of coats and thickness requirements will vary with substrate, application method and exposure. Use Series 44-721 to increase film build. Amounts vary by kit size, refer to the Series 44-721 product data sheet. Contact your Tnemec representative for more information.

CURING TIME

Temperature	Set to Touch	Max. Recoat	To Place in Service
90°F (32°C)	1-2 hours	7 days	24 hours
75°F (24°C)	2-3 hours	7 days	2 days
55°F (13°C)	8-9 hours	7 days	3 days

Note: If more than 7 days have elapsed between coats, the Series 431 coated surface must be mechanically abraded (scarified) before topcoating. Curing time will vary with surface temperature, air movement, humidity and film thickness.

VOLATILE ORGANIC COMPOUNDS EPA Method 24: 0.19 lbs/gallon (23 grams/litre)

HAPS 0.00 lbs/gal solids

THEORETICAL COVERAGE 1,604 mil sq ft/gal (39.4 m²/L at 25 microns). See APPLICATION section for coverage rates.

NUMBER OF COMPONENTS Two: Part A (amine) and Part B (epoxy)

MIXING RATIO By volume: One (Part A) to one (Part B)

PERMA-SHIELD® PL | SERIES 431

PACKAGING

	PART A (partially filled)	PART B (partially filled)	When Mixed
Drum Sets †	55 gallon drum	55 gallon drum	100 gallons
Large Kit †	5 gallon pail	5 gallon pail	8 gallons (30.28 L)
Small Kit	1 gallon pail	1 gallon pail	1 gallon (3.78L)
Extra Small Kit	750 ml cartridge	750 ml cartridge	1,500 mL
Touch-Up Kit †† (1 tube)	4 ounces	4 ounces	8 ounces (236 mL)

† Plural Component application only.
†† Touch-Up Kit consists of one (1) tube along with two (2) disposable static mixers.

NET WEIGHT PER GALLON 9.48 ± 0.25 lbs (4.3 ± 0.11 kg) (mixed)

STORAGE TEMPERATURE Minimum 25°F (-4°C) Maximum 110°F (43°C)
For optimal handling and application characteristics both material components should be conditioned to a minimum of 80°F (27°C) or higher 48 hours prior to use.

TEMPERATURE RESISTANCE (Dry) Continuous 275°F (135°C) Intermittent 300°F (149°C) (Wet) Intermittent 150°F (65°C)

SHELF LIFE 24 months at recommended storage temperature.

FLASH POINT - SETA Part A: N/A Part B: 200°F (93°C)

HEALTH & SAFETY This product contains chemical ingredients which are considered hazardous. Read container label warning and Material Safety Data Sheet for important health and safety information prior to the use of this product.
Keep out of the reach of children.

APPLICATION

COVERAGE RATES Before commencing, obtain and thoroughly read the Series 431 Surface Preparation and Application Guide.

Dry Mils (Microns)	Wet Mils (Microns)	Sq Ft/Gal (m²/Gal)
30.0 (760)	30.0 (760)	53 (4.9)
40.0 (1015)	40.0 (1015)	40 (3.7)
50.0 (1270)	50.0 (1270)	32 (3.0)

Note: Recommended DFT will depend on substrate condition and system design. Allow for overspray and surface irregularities. Film thickness is rounded to the nearest 0.5 mil or 5 microns. Add Series 44-721 to increase film build. Application of coating below minimum or above maximum recommended dry film thicknesses may adversely affect coating performance. **Note:** At 20.0 mils (510 microns), the Touch Up kit covers approximately 500 sq inches.

MIXING **Drum Set:** For Plural Component Application. Place band heaters on drums. Remove the lid and insert the mixing blade shaft through the center two-inch bung; reinstall the lid. Mixing blade should be adequately sized to fully agitate material. The material should be 80°F-90°F (27°C-32°C) before the mixing blade is turned on. Insert 5:1 feed pumps into the outside two-inch bung. Place the recirculation line in the 3/4 inch outside bung. Recirculate the material through the primary heaters and heated hose bundle back into the containers. Continue recirculation under agitation until Component A reaches 110°F-120°F (43°C-49°C) and Component B reaches 100°F-110°F (38°C-43°C). Do not exceed 120°F (49°C) for either component. Consult Technical Services for specific details.

Large Kit: For Plural Component Application. Agitate Parts A & B separately making sure no pigment or solids remain on the bottom of the can. **DO NOT MIX PART A WITH PART B.** Use a 1 (Part A amine) to 1 (Part B epoxy) mix ratio heated plural component airless spray unit. **Note:** Product component A (amine) must be heated to 110°F to 120°F (43°C to 49°C) and component B (epoxy) must be heated to 100°F to 110°F (38°C to 43°C) prior to and during plural component application. Do not exceed 120°F (49°C) for either component. Keep containers tightly sealed prior to use. Consult Technical Services for specific details.

Small Kit: Agitate Parts A & B separately ensuring no pigment or solids remain on the bottom of the can. Scrape all of the Part B into Part A can using a flexible spatula. Use a variable speed drill with a PS Jiffy blade ad mix the blended components for a minimum of two minutes. During the mixing process, scrape the sides and bottom of the container to ensure complete blending of materials. Apply the mixed material within 15 to 20 minutes, or before the material reaches 100°F following agitation.

Extra Small Kit: Contains one (1) 2K-cartridge and one (1) 32 element helical spray mixer and is used in conjunction with a pneumatic spray dispenser (supplied by Sulzer). Please reference Series 431 Extra Small Packaging Mixing & Application Procedures for additional information.

Touch-Up Kit: Equipment: A dispensing gun with a thrust ratio of 26:1 is required (F100-TKAP). Material tube must be used in conjunction with provided disposable static mixer in order to ensure proper mixing. **Usage:** Unscrew retaining ring and remove plug. Save plug in case entire tube is not used. Install static mixing element, replace retaining screw ring, and install tube in gun. Point assembly up and slowly pull the trigger to de-air the mixer. Dispense approximately 1 fluid ounce (29.8 mL) of material to waste and continue to pump until material is of uniform color with the Part A completely blended with the Part B. Use a putty knife, brush or spatula to ensure adequate coverage and repair.

THINNING **DO NOT THIN**

PURGE TIME 2 minutes (PC application)

POT LIFE 15 minutes at 75°F (24°C) (for single leg - airless application)
Material temperatures above 80°F (27°C) will significantly reduce the spray and pot life.

SPRAY LIFE 15 minutes at 75°F (24°C) (for single leg - airless application)
Flush the pump and lines immediately after spraying.

APPLICATION EQUIPMENT

PLURAL COMPONENT AIRLESS EQUIPMENT. The preferred application method for Series 431 Perma-Shield PL is using plural component equipment. Plural component equipment reduces material waste, solvent consumption and reduces material viscosity. Contact Tnemec Technical Service for complete Series 431 Plural Component Recommendations.

Airless Spray:

Pump Size	Rotary Gun †	Mat'l Hose ID	Manifold Filter
45:1 or 56:1	Model 712-216	3/8" (9.5mm)	NR

† **Rotary Spray Gun:** Series 431 shall be applied to the interior surfaces of pipe or fittings using a rotary coater pistol spray gun. Spray-Quip (Houston, TX), Model 712-216, or similar rotary lance, to produce a monolithic and level film.
Note: Pump assembly should include a moisture trap and oiler, air regulator with gauge and fluid outlet drain valve and outfitted with a gravity fed material hopper (material will not feed through a suction tube).

For other airless spray configurations, contact Tnemec Technical Services for additional information.

Roller: Use high quality 3/8" to 1/2" woven nap roller covers.
Brush: Recommended for bell sockets, spigot ends, and small repairs.

SURFACE TEMPERATURE

Minimum of 50°F (10°C) Maximum of 130°F (54°C).
The surface temperature should be dry and at least 5°F (3°C) above the dew point. The coating will not cure properly below minimum surface temperature.

HOLIDAY TESTING

High Voltage Discontinuity (spark) testing shall be performed in accordance with ASTM D 5162 or NACE SP0274 with a minimum voltage setting of 100 to 125 V/mil.

CLEANUP

Flush and clean all equipment immediately after use with Tnemec's No. 4 Thinner or MEK.

VAUGHAN COMPANY OFF LOADING AND LONG-TERM STORAGE INSTRUCTIONS
ELECTRICAL SUBMERSIBLE PUMPS

OFF LOADING AND INSPECTION INSTRUCTIONS:

Prior to shipment Vaughan pumps are carefully crated and inspected to ensure arrival at your site in good condition. On receiving your pump, examine it carefully to assure that no damaged crating or broken parts have resulted from mishandling during shipping. Look for signs that the pump has been dropped, such as missing paint, dented flanges, cracked housings, or leaking oil. Turn the pump shaft by hand and verify that it turns over smoothly. If the shaft binds, look for debris between impeller and cutter bar. Otherwise, shaft binding could indicate damage. If damage has occurred, report to your carrier immediately, and consult your local Vaughan representative or call Vaughan Co. for advice. Be sure to remove the flange cover at time of installation.

STORAGE REQUIREMENTS TO BE UNDERTAKEN BY CONTRACTOR:

If equipment is to be stored for longer than two weeks, take the following action:

1. Coat exposed steel with a light layer of grease or protective spray-on lubricant to protect the equipment from corrosion.
2. Rotate the motor 1-¼ turn once each week to keep the bearings from sitting in one position for extended periods of time.
3. Avoid storing rotating equipment near other vibrating equipment. The vibrations can cause damage to the ball bearings and cause premature failure once the equipment is started up.
4. Store rotating equipment in a clean, dry, heated area away from areas where it could be damaged from impact, smoke, dirt, vibration, corrosive fumes or liquids, or from condensation inside the motor or pump. It is helpful to cover equipment with plastic.
5. The bearing housing located below the submersible pump motor is about 85% oil filled. (An air bubble needs to be kept in this housing to avoid ruining the seal from high pressure if outside temperature increases during shipping or storage.) These housings should be kept 85% filled with ISO Grade 46 hydraulic oil during storage to be sure the bearings are kept covered to avoid corrosion.

Unmatched Reliability

RECORD OF PUMP AND MOTOR SHAFT ROTATION ONCE EVERY WEEK.

PUMP AND MOTOR SHAFT ARE TO BE ROTATED 1 ¼ REVOLUTIONS

- SHEET __ OF __ SHEETS –
-

PROJECT: ____________________

LOCATION: ______________________

PUMP MODEL: ____________________

PUMP SERIAL NUMBER: _________________

DATE	SHAFT ROTATED 1.25 TURNS? (CHECK BOX BELOW)	SIGNATURE OF PERSON ROTATING SHAFT	ANY COMMENTS?

VAUGHAN CO., INC. PRODUCT TWO YEAR WARRANTY

HILO WWTP

SN: 174838 A/B

Vaughan Company, Inc. (Vaughan Co.) warrants to the original purchaser/end user (Purchaser) all pumps and pump parts manufactured by Vaughan Co. to be free from defects in workmanship or material for a period of twenty-four (24) months from date of startup, not to exceed thirty (30) months from the date of shipment from Vaughan Co. Startup data must be submitted to Vaughan Co. within 30 days of startup. If Purchaser fails to submit startup data within 30 days of startup, then Vaughan, in its sole discretion, may elect to void this warranty at any time. Purchaser must contact Vaughan Co. prior to commencing any repair attempts, or removing pump or parts from service. If Purchaser fails to contact Vaughan Co. prior to commencing any repair attempts or removing pumps or parts from service, then Vaughan, in its sole discretion, may elect to void this warranty at any time.

If during said warranty period, any pump or pump parts manufactured by Vaughan Co. prove to be defective in workmanship or material under normal use and service, and if such pump or pump parts are returned to Vaughan Co.'s factory at Montesano, WA, or to a Vaughan authorized Service Facility, as directed by Vaughan Co., transportation charges prepaid, and if the pump or pump parts are found to be defective in workmanship or material, they will be replaced or repaired by Vaughan Co. free of charge. Products repaired or replaced from the Vaughan Co. factory or a Vaughan authorized Service Facility under this warranty will be returned freight prepaid. Vaughan Co. shall not be responsible for the cost of pump or part removal and/or re-installation.

All warranty claims must be submitted in writing to Vaughan Co. not later than thirty (30) days after warranty breach occurrence. The original warranty length shall not be extended with respect to pumps or parts repaired or replaced by Vaughan Co. under this Warranty. This Warranty is voided as to pumps or parts repaired/replaced by other than Vaughan Co. or its duly authorized representatives.

Vaughan Co. shall not be liable for consequential damages of any kind, including, but not limited to, claims for property damage, personal injury, attorneys' fees, lost profits, loss of use, liability of Purchaser to customers, loss of goodwill, interest on money withheld by customers, damages related to third party claims, travel expenses, rented equipment, third party contractor's fees, or unauthorized repair service or parts. The Purchaser, by acceptance of delivery, assumes all liability for the consequences of the use or misuse of Vaughan Co. products by the Purchaser, its employees or others.

Equipment and accessories purchased by Vaughan Co. from outside sources which are incorporated into any Vaughan pump or any pump part are warranted only to the extent of and by the original manufacturer's warranty or guarantee, if any, which warranty, if appropriate, will be assigned by Vaughan Co. to the Purchaser. It is Purchaser's responsibility to consult the applicable product documentation for specific warranty information. Specific product documentation is available upon request.
Any warranty shall be void if the total contract amount is not paid in full.

Vaughan Co. neither assumes, nor authorizes any person or company to assume for it, any other obligation in connection with the sale of its equipment with the exception of a valid Vaughan "Performance Guarantee" or "Extended Warranty," if applicable. Any other enlargement or modification of this warranty by a representative or other selling agent shall not be legally binding on Vaughan Co.

Warranty eligibility determination is at Vaughan Co.'s sole discretion.

<u>Warranty Limitations</u>:
This warranty shall not apply to any pump or pump part which has been subjected to or been damaged by any of the following non-exclusive list of causes:

- Misuse
- Abuse
- Accident
- Negligence
- Operated in the dashed portion of the published pump curves
- Used in a manner contrary to Vaughan's printed instructions
- Defective power supply
- Improper electrical protection
- Improper storage
- Faulty installation, maintenance, or repair
- Wear caused by pumping abrasive or corrosive fluids or by cavitation
- Dissatisfaction due to buyer's remorse
- Damages incurred during transportation
- Damages incurred during installation or maintenance

THIS IS VAUGHAN CO.'S SOLE WARRANTY AND IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, WHICH ARE HEREBY EXCLUDED INCLUDING IN PARTICULAR ALL WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOS

VAUGHAN FACTORY PERFORMANCE TEST PROCEDURE

for Horizontal, Vertical and Submersible pumps tested in horizontal configuration
(For 50-60 Degree F Clear Water)
Consult Vaughan for Test Pricing
revised November 12, 2001
revised June 20, 2003
revised January 31, 2005
revised January 13, 2006

I: PURPOSE

The purpose of this procedure is to describe the equipment and methods required to accurately test and evaluate Vaughan pump performance in accordance with the accepted standards of the latest edition of the Hydraulic institute Standards test Code.

II: TEST TOLERANCES

In performing tests under the Hydraulic institute test code, no minus tolerance or margin shall be allowed with respect to capacity, total head, or efficiency at the rated or specified conditions.

The pump must be within the following tolerances for acceptance:

At rated head : from +0% to +10% of rated capacity,

or

At rated capacity : from +0% to +8% of rated head.

Conformity with only one of the above tolerances is required. It should be noted, that there may be an increase in horsepower at the rated condition when complying to the plus tolerance for head or capacity.

Pump speed shall be maintained within 0.3% of rated speed. If pump speed cannot be maintained within 0.3%, speed shall be recorded and all data corrected to rated speed using the affinity laws.

III: EQUIPMENT

1. Pressure test gauge(s), various ranges as necessary, with 4-20 mA output signal and current NIST certification.

2. Pump torque, speed, and horsepower measuring equipment as supplied by S. Himmelstein & Co., consisting of:

    A. MCRT 9-02T(4-3) 4000 inch-pound Non-contact torquemeter, serial # 9T-855 with speed pickup properly mounted and aligned between the pump and drive motor shafts for measuring the torque applied to the pump shaft and the rotational speed (RPM) of the shaft.

    B. 28004TN(4-3) 4000 inch-pound Non-contact torquemeter, serial # 28004T-101 with speed pickup properly mounted and aligned between the pump and drive motor shafts for measuring the torque applied to the pump shaft and the rotational speed (RPM) of the shaft.

C. MCRT 2804TAN(1-4) 10,000 inch-pound Non-contact torquemeter, serial # 2804TA-419 with speed pickup properly mounted and aligned between the pump and drive motor shafts for measuring the torque applied to the pump shaft and the rotational speed (RPM) of the shaft.

D. MCRT 28003TC(1-3)NZ 1,000 inch-pound Non-contact torquemeter, serial # 28003T-127 with speed pickup properly mounted and aligned between the pump and drive motor shafts for measuring the torque applied to the pump shaft and the rotational speed (RPM) of the shaft.

E. System 7 (backup test equipment) data acquisition system, serial # 7-147, with input ports for three torquemeters above, multiple pressure gauges, 3 phase voltage monitors (Ohio Semitronics model VT8-009E, serial numbers 9120230, 9120231 & 9120232) and 3 phase current monitors (Ohio Semitronics, 0-300 Amp AC $\pm$ model CTA215H/CTL-401S/300Y04, serial numbers 9120212/8102927, 9120213/8102928 & 9120214/8102929).

F. National Instruments LabView 7.1 Full Dev System for Windows (primary test equipment) data acquisition system, with SCXI-1000 4 slot chassis, NI SCXI-1600 USB data acquisition and control module, SCXI-1102B 32-Channel Amplifier, SCXI-1308 32-Channel Current Input Terminal Block, and Himmelstein Model 721 Signal Conditioner with input ports for three torquemeters above, multiple pressure gauges, 3 phase voltage monitors (Ohio Semitronics model VT8-009E, serial numbers 9120230, 9120231 & 9120232) and 3 phase current monitors (Ohio Semitronics, 0-300 Amp AC $\pm$ model CTA215H/CTL-401S/300Y04, serial numbers 9120212/8102927, 9120213/8102928 & 9120214/8102929).

3. Two adjacent test pits, one for suction and one for discharge, 11,000 gallons each. Suction tank is provided with a 18" suction manifold with connections for 6", 8", 10", 12" and 18" 150 pound ANSI pump flange suctions. Suction tank is also provided with clear PVC sight glass marked off in 100 gallon increments to be used for level indication in calculating volumetric change in the test tank and provided with port for connection of a level transmitter. This system is arranged so that repeatable flow rate measurements (i.e., change in volume per unit time) within 0.5% accuracy may be taken as required to meet the requirements of the Hydraulic Institute Standards Test Code.

4. Mag Flow meters for measurement of flow stream in 6" or 16" piping loops.

5. An accurate, calibrated stop watch for measuring the time increments between float level points for calculating volumetric flow rate (used only with System 6 test).

6. A 125 HP, 200 HP, or other suitable electric motor shall be direct coupled to the appropriate Himmelstein Torquemeter which shall in turn be direct coupled to the pump to be tested.

6A. Alternatively, a field type operational test is available. If required by the applicable specification, the assembled pump (except for vertical recirculator pumps and wet-well pumps longer than 8 feet) may be run with the customer's motor. Torquemeter outputs of speed and torque may not be available in this configuration, so during these operational type tests, motor nameplate speed or strobe light tachometer is used in place of torquemeter speed, and power calculations are based on observed amperage readings and published motor data. These tests may not meet the Hydraulic Institute accuracy requirements for speed and power measurements, so these tests are only done when we are specifically directed to do so by the customer or the engineer.

7. A 150 HP Constant Torque / 200 HP Variable Torque VFD (Variable Frequency Drive) by Square D, model ATV-6615N4U, provides 460 Volt 60 Hz power or 380 Volt 50 Hz power to the appropriate electric drive motor.

8. A fresh water supply for filling and topping off the test tanks.

9. 1 Ton bridge crane to allow lifting of pumps, motors, valves, and piping in the test facility.

10. Thermometer for measuring water temperature in the test pits.

11. Tachometer for measuring shaft speed during tests where the Himmelstein equipment cannot be used, such as when the pump must be tested with the Customer's electric motor. (This method does not meet the Hydraulic Institute requirements for accuracy, but may occasionally be required by the customer.)

IV: TEST SETUP

1. Pump testing for horizontal end-suction, vertical wet well, vertical pedestal, vertical recirculator, submersible, and cantilever pumps are accomplished by testing the wet end components (impeller, casing, and cutterbar) in the horizontal end suction arrangement, as shown in Figure 1, Testing Layout for Vaughan Chopper Pumps. Upon request, submersible pumps may be run fully assembled with the submersible motor submerged as an operational type test, **not** meeting all Hydraulic Institute Standards.

2. Verify that test motor starters are off, and that VFD is off, and then connect power cord for pump test motor, pressure gauge(s) and torquemeter cables. Open pump Discharge valve partially.

3. Turn on power for VFD and the LabView computer. Double click on the LabView icon on the computer screen. Select "Open" and the appropriate Virtual Instrument name for the pump being tested.

4. Enter the Model, Serial #, Date, Speed and Test # in the appropriate fields in the VI header. Click the "Run" button in the upper left corner of the screen, navigate to the folder My Documents\Test Data\LabView Data at the prompt, and assign a test file name of xxxx (where xxxx is test # from test log). LabView will now begin displaying real-time data from all the instrumentation, but has not yet begun to record any data.

5. Transfer water into the suction pit, bleed out air from suction header, pressure lines and high point vent on pump casing (if provided).

6. Use the jog function on the VFD, (or run @ 5 hz) verify that pump is turning the proper direction.

7. Bring pump up to proper operating speed (60 hz or 50 hz for customer motors). Close discharge valve. If VFD coarse speed adjustment arrow buttons do not have sufficient resolution to achieve correct operating speed (within $\pm$0.3%), push "Enter" twice to reach fine speed adjustment screen. Arrow down to high speed limit, push "Enter" to allow adjustment, adjust to high speed limit to desired operating frequency, push "Enter" to lock adjustment. Verify that Ramp Up and Ramp Down Times are at least 6 seconds – if not, adjust as needed – minimum 10 seconds is recommended for 8" – 12" pumps. Push "ESC" twice to return to main screen. Push "Speed Increase" button if required to increase output frequency to match high speed limit. Check observed operating speed on LabView screen and repeat speed adjustment procedure if required.

8. Push VFD "Stop" button, top off suction pit water, vent test header and casing vent again, and then push "Start" button.

9. Watch LabView RPM readout or listen for pump to reach full operating speed. When full speed is achieved, click the “Save” button to begin saving data at pre-determined time intervals. A green indicator light will flash on the computer screen every time a set of data is saved.

10. Allow a few sets of data to be recorded at shutoff, and then crack open the Discharge Valve. Allow a few sets of data to be recorded at this low flow point. Then, open the discharge valve approximately 1 turn immediately after the red light flashes. It is acceptable to allow multiple sets of data to be recorded at any flow rate during the test.

11. If the Suction pit is emptied during before the test is completed, push the “Stop” button on the VFD, transfer additional water back into the Suction pit. Vent air from suction header, casing vent, and pressure lines as needed. Push the “Start” button on VFD, and allow LabView to save at least 2 sets of data. Then proceed with testing by opening discharge valve further.

12. Data shall be collected for a minimum of 10 operating points. Test may be concluded when discharge valve is fully open if data for 10 operating points has been collected. The “Save” button to stop recording data. Click the “Stop” button or press the “End” key to end the test.

13. Minimize or exit LabView, and double click the Microsoft Excel Icon. Click “File”, “Open”, and then in the bottom left box of the screen window, select files of type “All Files”, select file name xxxx from My Documents\Test Data\LabView Data, and click on the “Open” button. Choose “Finish” to open the data file. Select “File”, “Save”, if necessary re-type file name without parenthesis, dot or extension, and save as type “Microsoft Excel Workbook (*.xls) in the folder My Documents\Test Data\(appropriate sub-directory).

14. Select “File”, “Save As”, and save the file in S:\engr\perfdata\testdata\(appropriate sub-directory)\xxxx.xls, and record test information in test log.

15. Copy cells for first row of calculated flow through last row of calculated efficiency into EasyPlot graphing program. Compare resulting performance curve with required design point(s) to determine whether pump passed or failed. Add annotations as required.

16. If pump passed, paste curve onto test certificate in Microsoft Excel and save on Network.

17. If LabView 7.1 and/or related components are not available, proceed with testing using Himmelstein System 7 as follows:

18. Verify that test motor starters are off, and that VFD is off, and then connect power cord for pump test motor, pressure gauge(s) and torquemeter cables. Open pump Discharge valve partially.

19. Turn on power for VFD and the System 7 computer. Double click on the System 7 icon on the computer screen. Select “Configure”, “Load”, and the appropriate program name from for the pump being tested.

20. If pump being tested is an Assemble Submersible Pump or any other pump being tested without benefit of a Himmelstein torquemeter, select “Configure”, “User” and input the rated full load RPM for the motor.

21. Select “Execute”, “Test”, and assign a test file name of C:\data\xxxx (where xxxx is test # from test log) and fill in the Model, Serial #, Date, Speed and Test # on the screen and push “F10” twice per on-screen instructions. The System 7 will now begin displaying real-time data from all the instrumentation, but has not yet begun to record any data.

22. Transfer water into the suction pit, bleed out air from suction header, pressure lines and high point vent on pump casing (if provided).

23. Use the jog function on the VFD, (or run @ 5 hz) verify that pump is turning the proper direction.

24. Bring pump up to proper operating speed (60 hz or 50 hz for customer motors). Close discharge valve. If VFD coarse speed adjustment arrow buttons do not have sufficient resolution to achieve correct operating speed (within $\pm$0.3%), push "Enter" twice to reach fine speed adjustment screen. Arrow down to high speed limit, push "Enter" to allow adjustment, adjust to high speed limit to desired operating frequency, push "Enter" to lock adjustment. Verify that Ramp Up and Ramp Down Times are at least 6 seconds – if not, adjust as needed – minimum 10 seconds is recommended for 8" – 12" pumps. Push "ESC" twice to return to main screen. Push "Speed Increase" button if required to increase output frequency to match high speed limit. Check observed operating speed on System 7 screen and repeat speed adjustment procedure if required.

25. Push VFD "Stop" button, top off suction pit water, vent test header and casing vent again, and then push "Start" button.

26. Watch System 7 RPM readout or listen for pump to reach full operating speed. When full speed is achieved, push "V" to begin Saving data at pre-determined time intervals. A red indicator light will flash on the computer screen every time a set of data is saved.

27. Allow a few sets of data to be recorded at shutoff, and then crack open the Discharge Valve. Allow a few sets of data to be recorded at this low flow point. Then, open the discharge valve approximately 1 turn immediately after the red light flashes. It is acceptable to allow multiple sets of data to be recorded at any flow rate during the test.

28. If the Suction pit is emptied during before the test is completed, push the "Stop" button on the VFD, transfer additional water back into the Suction pit. Vent air from suction header, casing vent, and pressure lines as needed. Push the "Start" button on VFD, and allow System 7 to save at least 2 sets of data. Then proceed with testing by opening discharge valve further.

29. Data shall be collected for a minimum of 10 operating points. Test may be concluded when discharge valve is fully open if data for 10 operating points has been collected. Push "V" to stop recording data. Push "E" to return to main screen. Push "E" and then "A" to write ASCII data file, and "F10" to return to main menu.

30. "Alt Tab" or exit System 7, and double click the Microsoft Excel Icon. Click "File", "Open", and then in the bottom left box of the screen window, select files of type "Text (*.prn; *.txt; *.csv), select file name c:\data\xxxx, and click on the "Open" button. Choose "Text Delimited", "Next", "Comma Delimited" and "Finish" to open the data file. Select "File", "Save", re-type file name without parenthesis, dot or extension, and save as type "Microsoft Excel Workbook (*.xls) in the c:\data\excel sub-directory.

31. Import the appropriate calculation block, and verify that pipe diameter cell matches diameter of pipe at discharge gauge location. Fill formulas down as needed. Save. Save as s:\engr\perfdata\testdata\(appropriate sub-directory)\xxxx.xls, and record test information in test log.

32. Copy cells for first row of calculated flow through last row of calculated efficiency into EasyPlot graphing program. Compare resulting performance curve with required design point(s) to determine whether pump passed or failed. Add annotations as required.

33. If pump passed, paste curve onto test certificate in Microsoft Excel and save on Network.

34. If Customer's electric motor is to be used, and the Himmelstein torque meter equipment cannot be used, turn on the strobe tachometer and allow it to warm up for 15 minutes

prior to testing. Position the strobe light so that it flashes on a suitable reference mark. Note : this method is not available for submersible pumps.

VI: TEST DATA REDUCTION

1. Basic equation used to determine Total Dynamic Head (TDH).

   TDH (ft)= $(TDH)_{discharge} - (TDH)_{suction}$

   = $(\text{Velocity Head} + \text{Pressure Head} + \text{Elevation Head})_{discharge}$

   - $(\text{Velocity Head} + \text{Pressure Head} + \text{Elevation Head})_{suction}$

   Velocity Head (ft) = $(\text{Velocity})^2 / 2g$, where g=32.17

   Pressure Head (ft) = (Pressure Gauge Reading) * 2.31
   (for 60 deg. water @ sea level)

   Elevation Head (ft) = the average distance from the center of the pressure gauge to the water level during a particular test run.

2. Flow rate is measured by calibrated mag flow meters.

3. Horsepower is read directly from the Himmelstein power meter. Torque may be read from Himmelstein and horsepower calculated using the formula:

   HP = Torque (IN*LBS) * Speed (RPM) / 63025 (constant)

   Note this method may not be available when running pumps with job motors. In that case, pump input power shall be calculated using motor manufacturer's published motor data and motor input current and/or kw measurements.

4. Efficiency is calculated as follows:

   Efficiency (%) = TDH (ft) * Q (GPM) / (3960 * HP req'd)

   (Note: this equation is for 60 deg water at sea level)

VII: CURVE PLOTTING

1. Three performance curves, TDH vs Capacity, Horsepower vs Capacity, and Efficiency vs Capacity are to be plotted on a single sheet of paper, with Capacity on the horizontal scale and TDH, HP, and Efficiency on the vertical scale. Additional curves showing motor current or input kilowatts may also be provided if pump is tested with job motor.

2. After data points are plotted, any data points which deviate more than 5% from the curve drawn through the average of the points may to be discarded and the curve redrawn throughout the average of the remaining points.

3. An example of a completed curve is shown in Figure 2.

VIII: SECURING THE TEST FACILITY

After testing is completed, the following items must be completed to secure the test facility:

A. Push the "Stop" button on the VFD to stop the pump.
B. Push the "Stop" button of the appropriate motor starter.
C. Turn off the circuit breaker to the VFD.
D. Exit the LabView, Excel & EasyPlot programs on the computer
E. Push “Start”, “Shutdown”, “Shutdown the Computer” and then when prompted, turn off power to the computer.
F. Close and lock the outside test facility door.
G. Turn off the lights in the test facility.

IX: NOTES:

Vaughan Company, Inc. reserves the right to make upgrades to test methods and test equipment without notice.

TEST PROCEDURE FOR HYDROSTATIC PRESSURE TESTING OF ASSEMBLED VAUGHAN CHOPPER PUMPS

revised November 12, 2001
revised July 14, 2003

I: PURPOSE

The purpose of this procedure is to describe the equipment and methods required to perform hydrostatic pressure tests for Vaughan Chopper Pumps.

II: TEST TOLERANCES

In performing hydrostatic pressure tests under the Hydraulic Institute test code, the liquid containing components of the pump must be subjected to no less than the greater of 150% of design point pressure or 125% of shutoff point pressure, for a minimum of 3 minutes (10 minutes for pumps greater than 100 HP). Care must be taken to prevent subjecting the mechanical seal to pressures in excess of those specified above.

III: EQUIPMENT

A. Blocking plates shall be constructed as needed to isolate the liquid containing components to be pressurized. Suitable pressure taps for pressurizing and monitoring pressure shall be provided.

B. A fresh water supply for filling and topping off the liquid containing components to be pressurized.

C. Pressure source shall be plant air pressure or any low volume high pressure water pump.

D. Pressure test gauge(s).

E. An accurate, calibrated stop watch for timing the duration of the test shall be provided.

IV: PRESSURIZED COMPONENTS

A) Pressure containing components for horizontal pumps shall consist of the pump suction manifold and casing, and testing shall be done after final assembly.

B) Pressure containing components for vertical wet-well and cantilever pumps may be considered as just the casing and testing shall be done prior to final assembly and painting. However, if the customer has specified that the discharge pipe is to be included, testing must be done after final assembly.

C) Pressure containing components for vertical recirculating pumps shall consist of the pump casing only, as the non-leak tight recirculator valve prevents pressurizing of the discharge pipe. Testing shall be done on the casing prior to final assembly.

D) Pressure containing components for submersible pumps shall consist of the pump casing only. Testing shall be done prior to final assembly.

V: TEST SEQUENCE

A) Isolate the pressure containing components with blocking flanges as needed.

B) Fill the pressure containing components with water.

C) Pressurize the pressure containing components to the required pressure.

D) Remove or isolate the pressure source.

E) Using the stop watch, wait the required amount of time and observe the current pressure.

G) Inspect the pressure containing components for signs of leakage.

H) If the pressure did not drop, and there are no signs of leakage, the pressure containing components have passed the test.

I) Fill out the hydrostatic pressure test certificate, sign it, and place it to the job order packet.

J) Carefully bleed the pressure off, remove the blocking plates, drain the water, dry off the pump/components, and return the pump to production personnel for final assembly, painting, crating, other testing, etc.

Quality Assurance Manual

VAUGHAN COMPANY, INC.
364 MONTE-ELMA ROAD, MONTESANO, WASHINGTON 98563
PHONE: (360) 249-4042 FAX (360) 249-6155
WWW.CHOPPERPUMPS.COM

VAUGHAN COMPANY., INC. APPROVALS

President: Derek Vaughan Date: 8/20/24

Vice- President: Stacie Tobey Date: 8/24/2024

Vice- President of Production: Tom Vaughan Date: 8/20/24

Operations Manager: Taylor Smith Date: 8/20/24

Chief Engineer: Ric E Swenson Date: 8/20/2024

TABLE OF CONTENTS

1.0 INTRODUCTION

This manual describes Vaughan Company's quality assurance system, the way we want work accomplished and verified at this facility. This manual applies both to the items we produce and to the items we buy from our suppliers.

More specific written procedures may be required to implement the policies set forth in this manual.

This manual is a commitment from the president, vice president, and engineering and production managers of Vaughan Co. to our customers and to the future of this company to adhere to the letter and spirit of this manual. Our further commitment is to work with all Vaughan employees to emphasize the importance of adhering to this manual.

No changes may be made to this manual or to any supplementary quality control procedures unless approved by the company president, vice president, and the engineering and production managers.

This quality and inspection system promulgated herein will have the following basic elements:

1. Organization - setting and assigning specific authority and responsibility for each phase of the quality system.
2. Quality Planning - writing work instructions with realistic "defect prevention" rules, looking at manufacturing processes for possible quality trouble spots, setting acceptance/rejection standards, controlling accepted/rejected products, and setting up a means of using suppliers' and customers' failure information to improve product quality.
3. Product Specification Control - Making sure everyone always has the latest technical data for properly producing, inspecting, and shipping the product.
4. Supplier Product Quality Control - Watching purchases to make sure that the people we buy from know and observe our quality requirements as well as our technical specifications.
5. Measurement and Test Equipment Control - Setting up a system to ensure that such equipment is properly and regularly calibrated to established standards.
6. Nonconforming Material Control - Spotting defects in production as early as possible and keeping faulty items from reaching our customers.
7. Records and Reports - Setting up a system that tracks all steps of the production, inspection, and shipping cycle to identify existing and potential problem areas.

2.0 SCOPE

2.1 This quality control system includes:

- Receiving, identifying, stocking, and issuing material, parts, and complete assemblies.
- All manufacturing processes.
- Packing, storing, and shipping of parts and complete assemblies.

2.2 This system is designed to ensure customer satisfaction through quality control management of parts and equipment made and service performed at Vaughan Co. and at our suppliers' facilities. It is designed to spot processing problems early so we can correct them before we've produced faulty items.

2.3 Written inspection and test procedures may be prepared to supplement drawings and other specifications, as necessary.

3.0 RESPONSIBILITIES

3.1 Quality assurance and quality audits will be administered by the Vaughan engineering department. Quality inspections and tool control will be the responsibility of the Vaughan production department. Final inspections may require inspections by both Vaughan production and engineering organizations. See the organization chart, Appendix A. The heads of the engineering and production departments will report directly to the company president.

3.2 ENGINEERING MANAGER (OR DESIGNATED REPRESENTATIVE)

3.2.1 The engineering department manager will be responsible for:

1. Research, design, development and testing of all new products.
2. Preparation of all drawings, specifications, and other guidelines needed for production, quality control and testing.
3. Evaluate all changes requested by other groups and issue new revisions to drawings, specifications, guidelines, test procedures, etc.
4. Maintain records of engineering design, analysis, and testing.
5. Provide technical assistance for all other groups with the approval of the company president.
6. Staffing and supervising all engineering positions.
7. Provide supervision for all technical services supplied by vendors.
8. Provide supervision for field services as required based on job criteria.
9. Provide resolution to all customer warranty problems. Inform sales and production of any company corrective action needed to avoid recurrence of any particular warranty problems.
10. Provide all technical manuals and videos for equipment installation, startup, operation, maintenance, troubleshooting, and repair.
11. Prepare all warning and information labels, decals and plates for installation on equipment. Also engineering is to provide drawings or instructions documenting where these items are to be mounted on equipment. The wording for safety warnings is to be reviewed by Vaughan Co.'s legal consultant.
12. Provide documentation (engineering submittals and Installation, Operation and Maintenance Manuals) for special projects.
13. Provide procedures for testing of equipment.
14. Perform hydraulic testing, analysis, and test reports on all pumping equipment as required, and maintain yearly calibrations on all test equipment.

3.2.2 Quality assurance under the engineering department will include responsibility for:

1. Planning how to meet a particular customer's quality requirements as well as how to meet general product quality requirements.
2. Reviewing specific customer drawings and specifications.
3. Determining, in consultation with production, the appropriate inspection points for various products.
4. Writing, in consultation with production, inspection and test instructions.
5. Auditing quality assurance records on at least a twice yearly basis (February and August) and overseeing corrective action.

6. Coordinating correction of deficiencies found by customers, and determining and explaining corrective action to be taken. This will be done in consultation with the production manager.

3.3 PRODUCTION MANAGER (OR DESIGNATED REPRESENATIVE)

3.3.1 The production department manager will be responsible for:

1. Making sure parts and equipment on order are produced:
   A. in accordance with established engineering drawings and specifications;
   B. within the time frame agreed to with the customer;
   C. within costs and times consistent with those used in pricing the equipment.
2. Preparing any and all necessary production process sheets, flow charts and instructions.
3. Ordering raw materials, castings, tools, and fixtures.
4. Supervising tool and equipment handling.
5. Staffing and supervising the purchasing department.
6. Maintaining production records, sales order files, etc.
7. Preparing estimates of costs for new products.
8. Staffing and supervising all production personnel.
9. Supervising pilot or prototype production runs on all new products.

3.3.2 Quality assurance under the production department will include responsibility for:

1. Reviewing Vaughan engineering drawings and specifications to determine how to meet requirements.
2. Determining the appropriate inspection points.
3. Reviewing inspection and test instructions for parts or assemblies prepared by engineering.
4. Establishing the percentage of parts and sequence of all parts and materials to be checked.
5. Establishing (and making sure employees follow) the most effective and efficient quality assurance procedures possible. This will involve daily reviews of all quality records.
6. Correcting of deficiencies found and determining corrective action to be taken. This may be done in consultation with engineering.
7. Keeping quality assurance records.
8. Establishing procedures and providing for receiving inspection of all incoming parts, raw material, and castings. This will also include assuring that suppliers' take corrective action for quality problems.
9. Inspecting all special and standard gages, test equipment, and tooling used to manufacture products when we acquire them and calibrating them on a regularly scheduled, annual basis.
10. Making sure those performing inspections make unbiased decisions to accept or reject items.
11. Establishing a procedure for incorporating all engineering changes to drawings, specifications, or tests into the production system.

3.4 SALES MANAGER (OR DESIGNATED REPRESENTATIVE)

3.4.1 The sales manager will be responsible for:

1. Organizing and supervising the sales force.
2. Preparing sales literature, brochures and videos. This material will be reviewed by the legal consultant.
3. Preparing all sales quotations and bids.
4. Analysis of customer complaints and referral to engineering or production manager, as appropriate.
5. Administering all sales and service agreements in consultation with the company legal consultant.
6. Organizing the company's participation in trade shows and on-site demonstrations of equipment.
7. Staffing and supervising the sales department.

4.0 ORDER ENTRY CONTROL

4.1 All orders for pumps and assembled equipment will be presented to Vaughan Order Entry by Vaughan Sales only after a technical review by the cognizant sales engineer. Sales will present a completed Order Entry Form (see Appendix G) to the production department. No equipment is to be entered into production without this technical review and a completed Order Entry Form.

4.2 Any orders which are not clearly defined by existing drawings must be sent to engineering.

4.3 The completed Order Entry Form shall be made a permanent part of the sales order file.

4.4 Sales Orders (see Appendix H) and associated Job Orders shall:

1. Completely describe the work to be done, including referencing appropriate drawings, shop instructions, testing requirements, etc., complete with the correct revision of these documents, as required.
2. Job Orders shall have attached copies of the drawings, shop instructions, etc. unless the assemblies to be built are standard items for which the plant already has detailed drawings, specifications, test instructions, etc.
3. Be revised and reissued, as needed, to make sure that the plant has correct information and that the customer obtains the correct equipment. All revised sales orders shall be marked with the current revision.
4. Be accompanied by the correct Quality Assurance checklist for the product to be manufactured. This completed QA sheet shall be filed with the completed sales order file. All blanks on the QA sheet shall be initialed or marked as N/A, as appropriate.

Note: If a QA checklist does not exist for a custom product, the engineering department shall make one before the product is entered into production.

4.5 The following documents, specific to the product, shall be included in the shipping crate:

1. IO&M Manual
    a. The QA form shall be signed by staff indicating this step was completed.
2. Copy of the Sales Order
3. Other documents (if applicable), for example:
    a. International Orders
        CE Document Package
        CE/ATEX Document Package

4.6 Sales order status shall be listed on the Production Report and updated on a weekly basis.

5.0 PURCHASE ORDER CONTROL

5.1 All purchase orders to suppliers must be approved by the company president, or by an authorized representative of the company president, such as the company vice president. (see Appendix B).

5.2 When the purchase order is released, our buyer will send the supplier all required drawings, specifications, and other customer requirements (such as requirements for material or process certifications, physical or chemical analysis, and criteria for acceptance at Vaughan Co. receiving inspection, if any) with the purchase order. It is understood that routinely purchased material, supplies, tools, etc. will be purchased simply by referencing the supplier's product name and part number.

5.3 If there is a drawing or specification change after our order is placed with the supplier, our buyer will send the supplier a revised purchase order (marked as Revision 1, or 2, etc. as appropriate) and the revised drawing or specification

5.4 Copies of all purchase orders will be kept on file for at least 7 years.

6.0 DRAWING AND SPECIFICATION CHANGE CONTROL

6.1 All requests for changes to specifications, drawings, test procedures, operating or overhaul manuals, etc. must be made to the engineering department by means of a completed Engineering Change Notice (ECN) form (see Appendix C).

6.2 Before changes can be made to specifications, drawings, test procedures, etc. the change must be approved on the ECN by the engineering and production department managers.

6.3 Changes shall be incorporated in engineering documents in a timely manner. The document will be revised, listing the ECN number, the date of the change, and the initials of the person making the change. The current revision of engineering drawings shall also be listed in a computer database "Drawing Log" file. All shop orders for plant parts will be prepared listing the current revision of the appropriate drawing by referring to the Drawing Log.

6.4 An ECN log shall be used to keep track of the status of ECNs. An ECN log entry will be complete when the change has been incorporated into a drawing and the Drawing Log has been updated to show the latest revision.

6.5 The engineering department will provide the production manager with the latest revisions of all drawings, specifications, in-plant test procedures and operation and maintenance manuals, as needed or as requested.

6.6 The production manager shall establish a system to make sure drawing prints, specifications, and test procedure copies filed in the plant are updated to the latest revision.

6.7 Accountability: When creating a new drawing, the person creating the drawing shall place his name in the "drawn by" box. This applies, even if the new drawing differs from the parent drawing in only subtle ways (i.e. creating a new outline dimension drawing from a drawing of an identical pump).

7.0 RECEIVING INSPECTION

7.1 All materials will be received and logged in as having been received on the plant copy of the Purchase Order by Receiving. The receiving person will enter his initials, date received and no. of items received. The plant copy of the PO will be filed in the PO file. For motor receiving specifically, utilize applicable Motor Receiving Inspection Sheet (Appendix E).

7.2 All parts and materials will be inspected for adherence to the purchase order and attached documents (drawings, specifications, etc.) after being logged in on the purchase order. For most routinely purchased parts this inspection will involve verifying that the item(s) ordered were, in fact, received. All items described by engineering drawings or specifications will be inspected as described in 7.4.

7.3 Receiving inspection will assure that proper certification, physical and chemical test data, special process certifications, or source inspection certifications are with the items to be inspected, if and as required.

7.4 When engineering drawings or specifications describe the product ordered or when "INSPECTION" is noted under "routing" on the PO, the receiving inspector must document the complete results of all inspections and tests on the Inspection Data Form (Appendix E). This form is to be attached and filed with the purchase order. The percentage of items inspected (generally 10%) will be determined by the production manager in consultation with engineering. If any items are found to be out of specification, 100% of the parts are to be inspected.

7.5 An inspection acceptance or rejection tag or stamp shall be placed on material or parts received to indicate that they have been inspected. For boxes of items, such as bearings, the entire box may be stamped or marked as accepted.

7.6 Accepted material will be identified and entered into plant stock.

7.7 Rejected material will be tagged, stamped or otherwise marked as rejected and set aside until purchasing, production control, and engineering decide on disposition. Any decision to accept non-conforming material must be unanimous. Reworked or repaired material must be re-inspected after rework or repair.

7.8 The purchasing department has the responsibility of:
1. Assuring that a pattern of receiving continually faulty items from any supplier doesn't develop.
2. Assuring supplier corrective action.

7.9 The production department will follow-up to see that a supplier, that has sent us items we reject, has effectively corrected the problem.

7.10 Special receiving inspection instructions may be written depending on the complexity of the parts, material received, and customer (or engineering) requirements. Follow customer (or engineering) instructions (if any) for inspection.

7.11 The production quality department will review receiving inspection records in the PO file at least once per year to see if any suppliers are consistently failing to meet acceptance standards.

7.12 Inspection records will show the number inspected, the number rejected, and the name of the inspector.

7.13 Inspection records will document that supplier-provided records and data have been filed with the appropriate PO.

8.0 RAW MATERIAL CONTROL

8.1 For raw materials, bar stock, and sheet stock which require material traceability, they will be marked so they can be traced to their certification and stored in an area apart from the normal flow of in-process material.

8.2 Where traceability is required, copies of all material certifications will be filed in the sales order file with the appropriate sales order, so they are available for customer review. If certified material is to be used on more than one sales order. then copies of certifications will also be kept with the applicable purchase order and receiving inspection form.

8.3 Only raw material accepted by receiving inspection will be released for production. It is not anticipated that raw material will generally require material certification, and that receiving inspection will involve verifying that the material is, in fact, what was ordered.

8.4 Certified, traceable stock will be issued from a temporary storage area in the plant to fulfill specific sales order requirements. Traceable stock will not be a normally stocked item.

8.5 Verification of suppliers' certifications will be ordered from independent testing laboratories, when deemed necessary, by the quality department or to meet our customers' requirements.

8.6 All certifications will be traceable to purchase order, date of receipt of the material, and the inspector of the material.

8.7 If shop operations, such as cutting, should remove identifying marks from raw materials, the remnants must be clearly re-marked with indelible paint before being returned to storage.

8.8 Additional material control and inspection criteria are found in the following appendices:

1. Appendix J; Mechanical Seal Face Inspection Criteria
2. Appendix K; Casting Inspection and Testing Procedures
3. Appendix L; Rotamix Nozzle Inspection Procedures

9.0 IN PROCESS INSPECTION

9.1 No production runs on computer numerically controlled (CNC) machines or manual machines will be made until first piece inspection is accepted as meeting the requirements of the engineering drawing(s). This first piece inspection will be documented on the Production Floor Traveler, in the area provided (Appendix F). First piece inspection will be performed by an independent machinist (i.e. by another machinist, not the machinist running the part).

9.2 Finished parts from production runs (more than 1 piece, see 9.6 below) on manually operated machines will be spot checked as meeting the requirements of the engineering drawing(s) by an independent machinist. This inspection will be documented on the IDF in the area provided for this inspection.

9.3 After first piece inspection on CNC production runs, in-process spot inspections will be made by the operating machinist on a percentage basis as determined by the production manager (e.g., 10% of the production run, or as determined). Any problems found during spot inspections will require stopping the run, and inspecting 100% of the parts made. The Production Manager or Lead shall be notified prior to restarting the production run. Verification of percentage checks shall be documented by the machinist on the IDF in the area provided for this inspection. If all 10% of parts do not pass dimensional checks, 100% of parts will be dimensionally inspected and the results noted on the IDF. Any parts which do not pass dimensional checks will be marked with fluorescent red paint and either tagged with a Discrepancy, Repairable or Scrap tag (Appendix D).

9.4 All production parts from an accepted production run will be marked with the work order number to show conformity with requirements. If parts are too small for marking, then the packaging shall be marked with the work order number. All non-conforming items will be marked with fluorescent red paint and either a Discrepancy, Repairable or Scrap tag and stored in a separate "SCRAP" or "REWORK" area, out of the normal production flow, for disposition by the production and engineering managers.

9.5 The first production part made from a jig, tool, or fixture will be independently inspected by another machinist to ensure that the part made from the jig, tool, or fixture matches the engineering drawing. This inspection will be documented on the IDF in the location provided. The jig, tool or fixture will be given a tool number (see 13.1) and the tool number will be noted on the engineering drawing of the part for which this jig, tool or fixture was made.

9.6 Runs of 1 part made on manual machine tools will be inspected by the machinist as they are made. All results from these inspections will be listed on the IDF.

9.7 All non-conforming items accepted by both production and engineering will be tagged with a Discrepancy Tag with the restrictions for use clearly stated.

9.8 Appendix I shows the locations of stations within the plant where inspections typically will take place by production personnel. The QA Verification Forms (Appendix N) detail the inspections to be done on each type of equipment being assembled.

10.0 ASSEMBLY INSPECTION AND FUNCTIONAL TESTING

10.1 Production personnel will make assembly inspections and do functional testing, as required by the QA verification forms. Assemblies (e.g., oil monitor probes, bearing/seal assemblies, etc.) will be built, inspected, and tagged per sections 9.4 and 9.6.

10.2 Sales orders for equipment requiring special tests or certifications will be accompanied by a Pump Test & Certification Checklist (Appendix M) printed on yellow stationary and filled out by Vaughan Engineering. Assembly of this equipment will begin only after performance tests, hydro's, and special balancing have been completed. Shipping of equipment will be as described in section 15.2.

10.3 Recurring rejections of assemblies, or subassemblies, will be addressed by the production department in consultation with the engineering department.

11.0 FINAL INSPECTION AND TESTING

11.1 Final inspection will be done on 100% of all products prior to crating and prior to shipping. QA verification forms(s) will be completed by production. If products, or assemblies, are described by a specific engineering drawing, engineering will be required to verify conformance to the drawing and sign off that item on the QA verification form.

11.2 The completed QA verification form will be returned to the office for filing in the permanent sales order file.

11.3 Any product rejected at final inspection will be marked with fluorescent red paint and either a Discrepancy, Repairable or Scrap tag and set aside so it cannot be inadvertently shipped. The product cannot be shipped until the problem is fixed and the unit is re-inspected and accepted on the QA verification form.

12.0 FAULTY (DISCREPANT) MATERIAL CONTROL

12.1 All faulty (non-conforming) material, supplies, parts, or assemblies will be placed in a "DO NOT USE" area and marked with fluorescent red paint and either a Discrepancy, Repairable, or Scrap tag. The items will be clearly marked with any information necessary.

12.2 Material may be removed from the "DO NOT USE" area or the Discrepancy, Repairable, or Scrap tags may be removed only after a disposition is determined by agreement between the production manager and the engineering manager. A discrepancy tag shall be prepared and signed by the production and engineering managers clearly stating any limitations on how or where the discrepant part or assembly may be used. If this part or assembly is unacceptable for use under any condition, the part or assembly will be scrapped or reworked, as appropriate.

12.3 The production quality department will control all lots of parts until acceptance inspection is accomplished. Each lot will be kept as a unit, apart from other lots, and out of the normal flow of material.

12.4 The production quality department will set up a system so the stage of inspection each item is in can easily be identified.

12.5 Unidentified material will be taken out of the normal flow of production until it is inspected to insure it meets all specifications.

12.6 Reworked material will be segregated from other material or marked with a Repairable tag until the quality department determines its status.

13.0 TOOL AND GAGE CONTROL

13.1 All special tools, jigs, fixtures, gages, and measuring equipment must be properly identified by tool number assigned by production.

13.2 Each new or reworked tool, jig, fixture, gage, and item of measuring equipment will be inspected before issued for use.

13.3 All gages, measuring and test equipment will be calibrated to standards set by the National Bureau of Standards.

13.4 A written schedule for calibrating gages, measuring, and testing equipment will be set and strictly followed. Frequency of calibration will be based on the type and purpose of the equipment and severity of usage.

13.5 A restricted area for storing, calibrating gages, measuring standards, and testing equipment will be set up. A strict system of issue, control, and return will be set and followed for this equipment.

13.6 If a customer supplies special gages, they will be checked at the intervals the customer sets. If the customer supplies no inspection schedule, the equipment will be checked according to a schedule that takes into account type, purpose, and severity of use.

13.7 Calibration will follow any written procedures kept in the calibration area.

13.8 Obsolete or out-of-service tools and gages will be tagged or discarded.

13.9 Decals or stickers will be put on tools and gages, or their containers, to show the last date of calibration and the due date of the next calibration.

13.10 Personal, as well as company-owned production and inspection tools, must be properly and regularly calibrated.

14.0 PARTS STOCK CONTROL

14.1 The production quality department will monitor parts inventory to make sure all parts are marked as inspected and accepted.

14.2 The part number and work order number will be shown on each part if built in-house. Parts built out-of-house will show part number and PO number.

14.3 The production quality department will monitor storage of parts to be sure deterioration is not taking place.

14.4 If a revision to a part or drawing number takes place, it will be the responsibility of the production quality department to inspect the part stock to determine whether any parts are affected. Parts affected will be either reworked to the latest revision, scrapped, or they will be accepted as discrepant items and marked with an appropriate discrepancy tag signed by the production and engineering managers. See para. 12.2.

15.0 PACKING AND SHIPPING

15.1 No order will be shipped to a customer until all items have been verified to have inspection decals or marks on them indicating they are accepted. (See also 14.2 and 16.1.)

15.2 No order will be shipped until all required certifications, test reports, special samples, manuals, etc., have been sent to and approved by the customer by the engineering department. The item of equipment will be marked with "HOLD" on the sales order until approval for shipment is obtained from Vaughan Engineering.

15.3 All material will be packed to prevent damage, deterioration, and substitution.

15.4 The customer will be identified on the packaging or parts, or as otherwise necessary to prevent lost and misdirected shipments.

16.0 IDENTIFICATION

16.1 All materials and articles will be identified by a part number.

16.2 Large assemblies, such as pumps, will also carry a serial number which is the same as the sales order number with the addition of the month and year of the order entry.

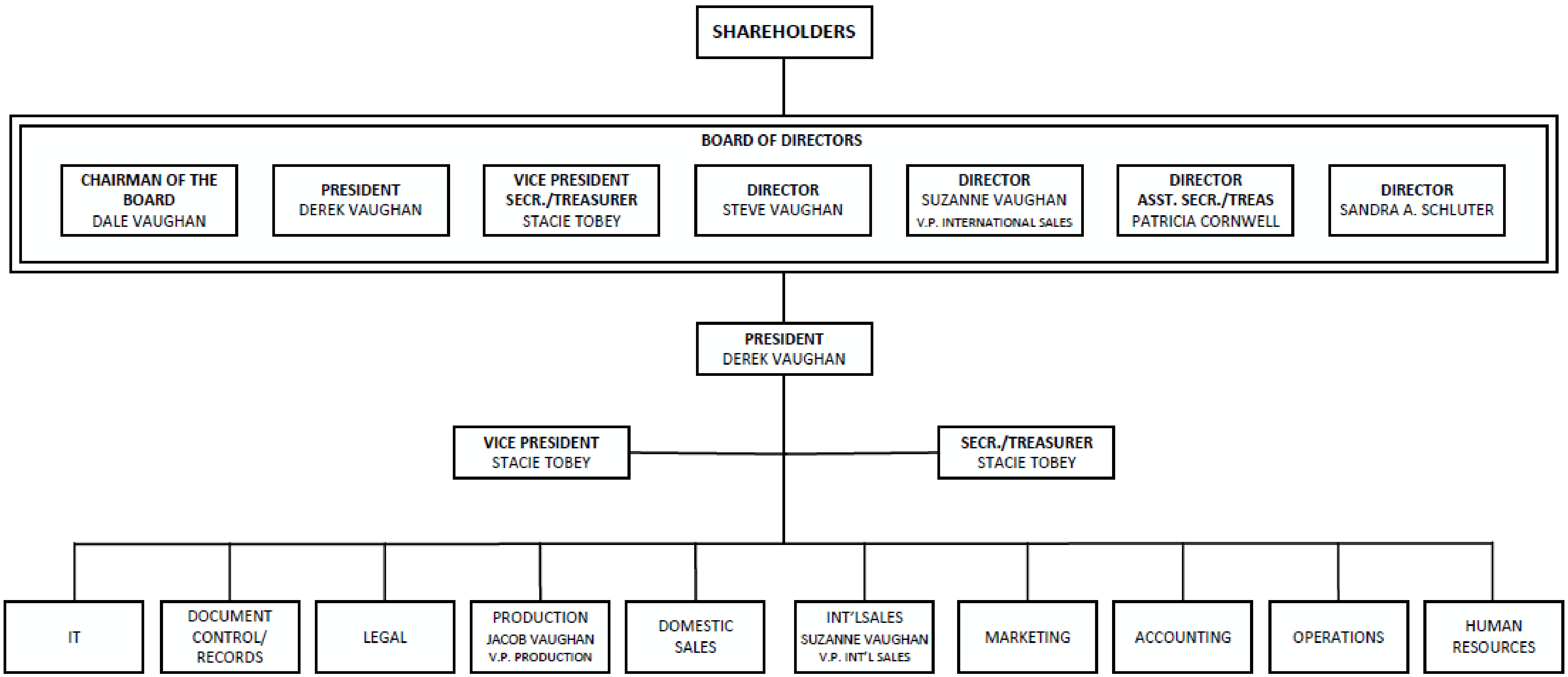
16.3 Pumps will be further identified by a Vaughan Co. nameplate listing the model number, impeller diameter, serial number, motor speed, horsepower, pump speed, and head and flow performance. The nameplate will also list Vaughan Co.'s address and phone number.

16.4 Prior to shipping all pumps will be checked for proper mounting of all decals, labels, and warning and nameplates in accordance with the information contained on the appropriate QA verification form.

APPENDIX A

ORGANIZATIONAL CHART

Vaughan Co., Inc. Corporate Structure
April 26, 2024



APPENDIX B

PURCHASE ORDER FORM

Chopper Rotamix Triton

364 Monte Elma Rd. Montesano, WA 98563
Phone: 360-249-4042 Fax: 360-249-0532

Purchase Order 032570-00
Vendor 000357

Purchasing Phone: 360-249-0725 Fax: 360-249-4071

To :

VAUGHAN CO., INC.
364 MONTE-ELMA RD.
MONTESANO, WA 98563

Ship to :

VAUGHAN CO., INC.
364 MONTE ELMA RD
MONTESANO WA 98563
United States

Phone (360)249-4042 Fax (360)249-0532

PO Date	Ship Via	FOB	Planner	Confirming to	Terms
07/17/2017	BEST WAY	OUR PLANT	02	TAYLOR SMITH	NET 5

Item	Facility / Part / Rev / Description / Details	Quantity	Promised Delivery	Unit Cost	Extended Cost
1	SAMPLE ITEM QA DOC SAMPLE ITEM QA DOC Rev NS U/M EA Purchasing Category : OUTSIDE JOB SHOP (N	1.00	07/18/2017	0.00	0.00

Please confirm this PO via either Fax or Email

Please reference our purchase order number on all correspondence. Notification of changes regarding quantities to be shipped and changes in the delivery schedule are required.
ALL PARTS MUST BE PART MARKED WITH VAUGHAN SUPPLIED PART NUMBER
DELIVERY TIMES ARE FROM 7:00AM TO 3:00PM MON-FRI

Total Items Price	0.00
Sales Tax	0.00
Fixed Cost	0.00
Total PO Price	**US$ 0.00**

PURCHASING DEPARTMENT **Page # 1**

Authorized Signature

APPENDIX C

ENGINEERING CHANGE NOTICE (ECN) FORM

5530 ECN	DATE	REVISED BY	RELEASED BY	CAR #	REASON FOR ECN

DRAWING NO.	REV	PART NO.	PENDING	DESCRIPTION OF CHANGES
		V-	0	
		V-	0	
		V-	0	
		V-	0	
		V-	0	
		V-	0	
		V-	0	
		V-	0	
		V-	0	
		V-	0	

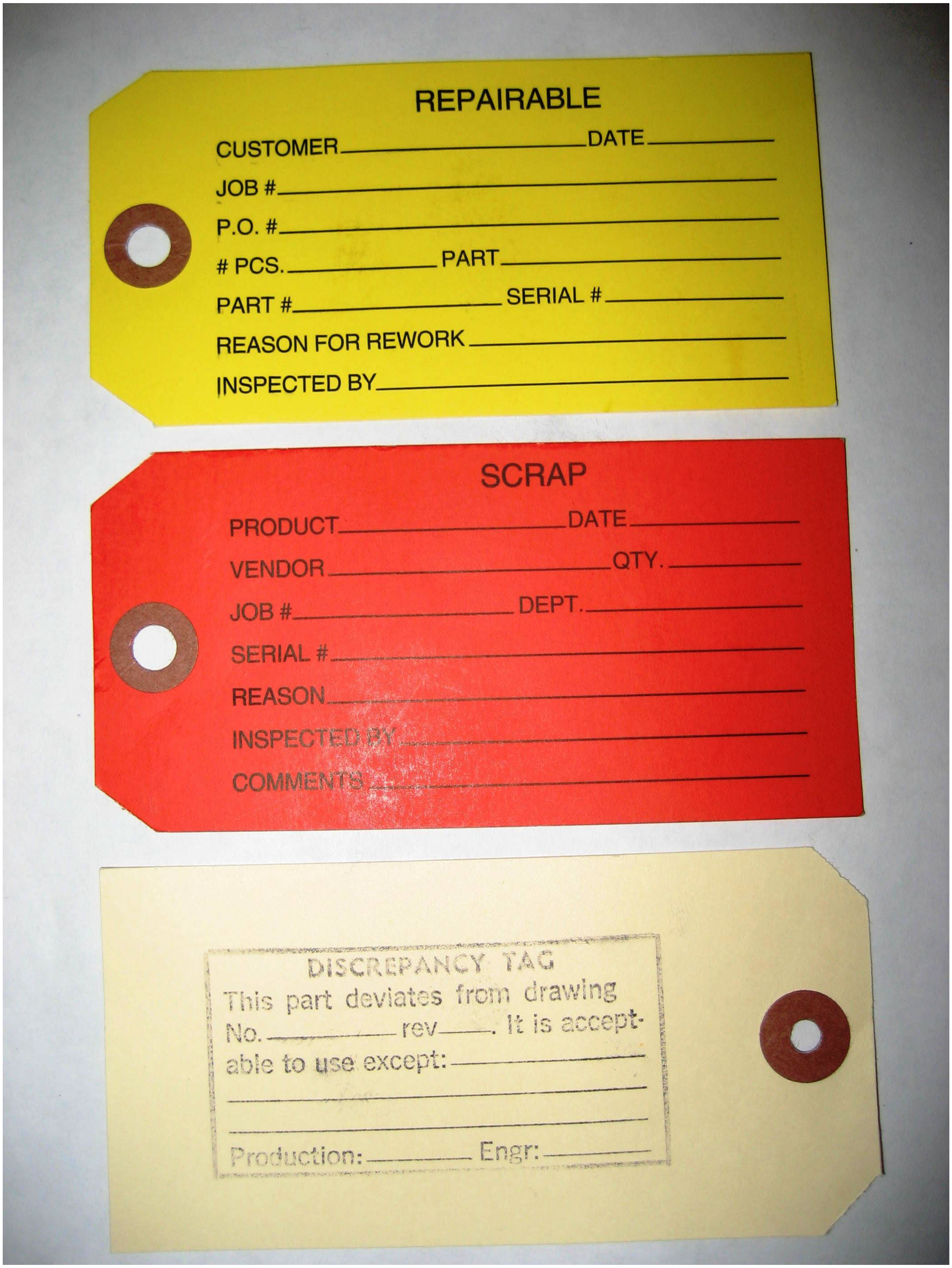
DISPOSITION							
INITIAL RELEASE	REWORK INVENTORY	REWORK TOOLING	USE AS-IS & PULL NONSTARTED JOBS	RETURN TO VENDOR	SCRAP	DOCUMENTATION ONLY	NOTIFY OUTSIDE VENDOR

IMPACT OF ECN															
EXPLODED ASSEMBLY		BURNOUT FILES		I, O & M		OVERHAUL MANUAL		BOMS		SPEC SHEETS		ATEX		OTHER:	
YES:		YES:		YES:		YES:		YES:		YES:		YES:		YES:	
NO:		NO:		NO:		NO:		NO:		NO:		NO:		NO:	

APPROVAL										
SALES		ENGINEERING		ACCOUNTING		PRODUCTION		DALE		REASON FOR NO
YES:		YES:		YES:		YES:		YES:		
NO:		NO:		NO:		NO:		NO:		

APPENDIX D

NON-CONFORMANCE TAGS


REPAIRABLE
CUSTOMER DATE
JOB #
P.O. #
PCS. PART
PART # SERIAL #
REASON FOR REWORK
INSPECTED BY
SCRAP
PRODUCT DATE
VENDOR QTY.
JOB # DEPT.
SERIAL #
REASON
INSPECTED BY
COMMENTS
DISCREPANCY TAG
This part deviates from drawing
No. rev . It is accept-
able to use except:
Production: Engr:

APPENDIX E

INSPECTION FORMS

INSPECTION DATA FORM

PO # **Date:**

Part Number **Inspector:**

Drawing Number

Part Description

Number of Parts Inspected

Number of Parts Accepted

Number of Parts Rejected

Reason(s) for rejection:

Supplier provided records places with purchase order:

Confirmed / Date:

Unmatched Reliability Vaughan® Chopper Triton Rotamix

MOTOR RECEIVING INSPECTION QA SHEET

TO BE COMPLETED WHEN MOTOR IS RECEIVED AT VAUGHAN COMPANY

DROP DEAD TESTING REQUIRED YES ☐ NO ☐ Note: Fill out page 2 for drop dead orders

Name of Inspector: ____________________ Date of Inspection: ____________________

Pump Model: ____________________ Pump S/N: ____________________

Motor Brand: ____________________ Motor S/N ____________________

Motor Model Number ____________________ Motor Service Factor ____________________

Motor Spec Number ____________________ Motor RPM ____________________

Motor HP ____________________ Motor Voltage ____________________

Motor Frame Size ____________________ Motor Full Load Amps ____________________

Motor Hazard Rating: (i.e. none, Class 1 Div 2, Class 1 Div 1, or other) ____________________

Motor matches purchase order description YES ☐ NO ☐ INIT ________

Motor matches packing slip YES ☐ NO ☐ INIT ________

Motor completely matches BOM description YES ☐ NO ☐ INIT ________

Motor Tests: If tests were called out, did we receive them? YES ☐ NO ☐ INIT ________

Motor deviates from BOM description. Describe discrepancies below.
Note if BOM calls out single voltage and motor nameplate calls out dual voltage, this is a discrepancy and must be noted.

____

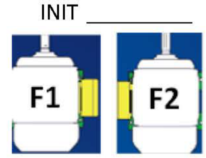
____

How many wires in main conduit box (big cord on submersibles): __________ INIT ________

How many wires are in auxiliary conduit box if equipped (small cord on subs): __________ INIT ________

Are all wires labeled? YES ☐ NO ☐ INIT ________

Is conduit box on correct side of motor? YES ☐ NO ☐ INIT ________



Moisture sensors are standard on all submersibles. If this is a submersible, what is resistance between the W1 and W2 wires? Ohms ______ INIT ________

Thermostats are standard on all submersibles, all Class 1 Div 2, and all Class 1 Div 1 motors. If equipped with thermostats, what is the resistance between P1 and P2 wires? Ohms ______ INIT ________

Space Heaters are standard on immersible motors, and optional on all TEFC motors. If equipped with space heaters, what is the resistance between heater wires H1 and H2? Ohms ______ INIT ________

Thermistors are optional on all motors. If equipped, what is the resistance between Thermistor wires?
Ohms __________ INIT ________

Resistance Temperature Devices are optional on all motors. If equipped, what is the resistance between RTD wires? __________ Ohms INIT ________

Winding Resistance Test: L1 – L2 ________ L2 – L3 ________ L1 – L3 ________ INIT ________

Megger Tests: Do all power wires pass a Megger Test to ground with 1.5 mega ohms (1,500,000 ohms) or greater? YES ☐ NO ☐ INIT ________

Page 1 of 2

Form V587 Rev. 0, 3/2/18

THIS PAGE TO BE COMPLETED FOR DROP DEAD MOTORS ONLY

Measured Amperage: ___________ Amps INIT _________

Tested Voltage: ____________ INIT _______

Amperage: L1: ___________

L2: ___________

L3: ___________ INIT _________

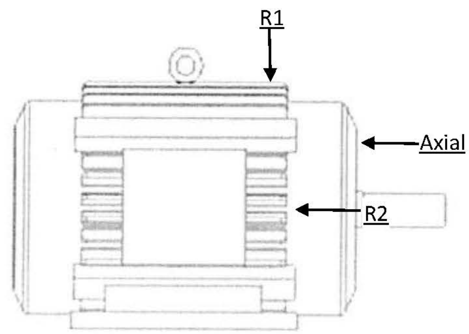
Is measured no load amperage less than half of nameplate full load amps? YES ☐ NO ☐ INIT _________

Allowable Vibration Limits: 0.08 in/s velocity for IEEE-841 and special balance motor. .015 in/s for all others.
Measured Vibration:

RADIAL1: _________________

RADIAL2: _________________

AXIAL:____________________ INIT _________



NOTES:

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Page 2 of 2

Form V587 Rev. 0, 3/2/18

SUBMERSIBLE MOTOR RECEIVING QA INSPECTION SHEET

Checked By ______________________________ Date ______________

Vaughan PO# ____________________________

Motor Manufacturer ____________________ Spec/Model No. ____________________

Serial Number ____________________________ Horsepower ______________

Nameplate RPM ___________ Frame Size __________

Gauge of Power Cable __________ Length of Power Cable __________

Nameplate Voltage: 230V ☐ 230V/460V ☐ 460V ☐ other ☐ _________ (list voltage)

Cable Cap Thread Size: 2 1/2" ☐ 3" ☐

Cable Cap leak check performed per SOP. ________ Initials

Motor cable length matches the Purchase Order Yes _____ No ______

Cable has been inspected for cuts. Yes _____ No ______

Motor assembly bolts confirmed to be tight Yes _____ No ______

Cable cap assembly removed for inspection? Yes _____ No ______

An adequate amount of oil in the motor been verified? Yes _____ No ______

Motor is properly wired for __________ (list voltage)

Resistance across P1 and P2 is approximately 6 ohms __________ (list ohms)

Resistance across W1 and W2 is 330K ohms __________ (list ohms)

Resistance from W1 to Ground should be infinite __________ (list ohms)

Resistance from W2 to Ground should be infinite __________ (list ohms)

Resistance from P1 to Ground should be infinite __________ (list ohms)

Resistance from P2 to Ground should be infinite __________ (list ohms)

Resistance of 3 main power leads to Ground should be infinite

L1 _________ ohms L2 _________ ohms L3 _________ ohms

Motor Amp Measurements: __________ Amps

Nameplate Full Load Amps __________

Reading is below half of full load? Yes _____ No ______

Shaft Endplay Per Motor Manufacturer ______________

Copy of this sheet has been attached to motor Yes _____ No

Below measurements only required if motor is repotted by Vaughan.

Cap OD 1 _____________ Cap OD 2 ____________ (measure 90° apart)

Motor ID 1 ____________ Motor ID 2 ____________ (measure 90° apart)

Form V684, Rev. 2 Page 1 of 1 12/18/23, ECN5460

APPENDIX F

PRODUCTION FLOOR TRAVELER

Vaughan

Production Floor Traveler

Date: 07/17/2017
Time - 10:43:36
Page # 1

Sales Order 137853

Facility Default
Job Order 00GN9-0000

Part Number SAMPLE ITEM
Drawing No:
COMPLETE BY 07/18/2017

Revision NS U / M EA Quantity 1.00
FINAL QTY COMPLETE__________

RETURN TO TED/NICK FOR RESCHEDULE
IF NOT DONE BY COMPLETE BY DATE.

Sales Order Coordinator TRS Ship Item Y Ship Early Y Split Ship Y

Customer VAUGHAN CO INC

Operation	Work Center	Operation Quantity	Setup Time	Pieces per Hour	Operation Time	Move Time	Elapsed Time
10	MSHOP	1.00	0.00	1.00	1.00	0.00	1.00
20	WELD	1.00	0.00	1.00	1.00	0.00	1.00
30	STOCK	1.00	0.00	10.00	0.10	0.00	0.10

Description SAMPLE ITEM FOR QA DOC

Operation 10 – MSHOP – MACHINE SHOP

Machined or welded by_____ Quantity Completed ________ Qty Inspected________
10% minumum
Operation Description Detail : Time per part ____HR_____MIN Date inspected _____________ First part inpected by ______

Operation 20 – WELD – WELSHOP

Machined or welded by_____ Quantity Completed ________ Qty Inspected________
10% minumum
Operation Description Detail : Time per part ____HR_____MIN Date inspected _____________ First part inpected by ______

Operation 30 – STOCK – TO STOCK

Machined or welded by_____ Quantity Completed ________ Qty Inspected________
10% minumum
Operation Description Detail : Time per part ____HR_____MIN Date inspected _____________ First part inpected by ______

Total Times - 0.00 2.10 0.00

JO Elapsed Time 2.10

Job Order Materials List

Part Number	Rev	Description	Qty Reqd	U/M	Source	Item #	Used In Oper

End Of Report

* Represents Sub-Contract days, these days are not included in the column total.

PRODUCTION MANAGER Sales Order 137853

APPENDIX G

ORDER ENTRY FORM

H1

Unmatched Reliability Vaughan®	PUMP ORDER FORM	DATE				
	Vaughan Co., Inc.	REF QUOTE #	Quote#			
	364 Monte Elma Road	SALES ORDER #	000000			
	Montesano, WA. 98563	BID DATE				
	Version 4/9/2019	INITIALS				

Project Name	Pump Project Name
Destination Representative	Rep name
Order-Taking Representative	Rep name
Main Engineering Representative	Rep name
Sub Engineering Representative	Rep name
Startup	Rep name
Consulting Engineer	
PO Number	
Tag Number	

Bill to		Ship to	
Name		Name	
Attention		Attention	
Address		Address	
Address		Address	
ity, State, ZIP		ity, State, Zip	
Phone / Fax		Phone / Fax	

Submittals to			
Name		Notes	
Attention			
Address			
Address			
ity, State, ZIP			
Phone / Fax			

Pricing Summary	
List Price	$ -
Sold For Price	$ -
Commission Sale Summary	
Commission	$ -
Overage/(Disc)	$ -
Commission Payable	$ -
Startup	$ -
Total Payable	$ -
Vaughan Net	$ -

	Rep name	Rep name	Rep name	Rep name	Rep name
	Destination	Order-taking	lain Engineerin	Sub Engineerin	Startup
Percentage of Total Commission	25%	25%	25%	25%	0%
Commission	$ -	$ -	$ -	$ -	
Overage/(Disc)	$ -	$ -	$ -	$ -	
Commission Payable	$ -	$ -	$ -	$ -	$ -
Startup					$ -
Total Payable	$ -	$ -	$ -	$ -	$ -

Buy / Resale Summary	
Net Due	$ -

Application			Motor HP		Motor RPM		
Industry			Belt or Direct		Pump RPM		
Pump Model			Voltage		Phase		
Fluid	% Solids		Temperature		PH		
GPM		TDH		Efficiency		BHP	#DIV/0!

Do not edit formulas

Project | Pricing | Bid Day | Engineering Time

Ready 70%

APPENDIX H

SALES ORDER FORM

Unmatched Reliability Vaughan®

Chopper Rotamix Triton Vaughan

364 Monte Elma Rd. Montesano, WA 98563
Phone: 360-249-4042 Fax: 360-249-0532

Sales Order 137853-00

Customer 000556

To :

VAUGHAN CO INC
IN-HOUSE ACCOUNT
MONTESANO WA 98563
United States

Ship to :

VAUGHAN CO INC
364 MONTE-ELMA ROAD
MONTESANO WA 98563
United States

Phone (360)249-4042 Fax ( ) -

Billing Purchase Order	Order Date	Terms	FOB	Ship Via	Salesperson
REQUIRED ON ORDER	07/17/2017	NO CHARGE	OUR PLANT	BEST WAY	JA

Item	Facility / Part / Rev / Description / Details	Quantity		
001	**SAMPLE ITEM** Rev NS U/M EA SAMPLE ITEM FOR QA DOC Approximate Ship Date: 07/18/2017	1.00		

ALL SHORTAGES MUST BE CLAIMED WITHIN FIVE (5) DAYS OF RECEIPT OF GOODS

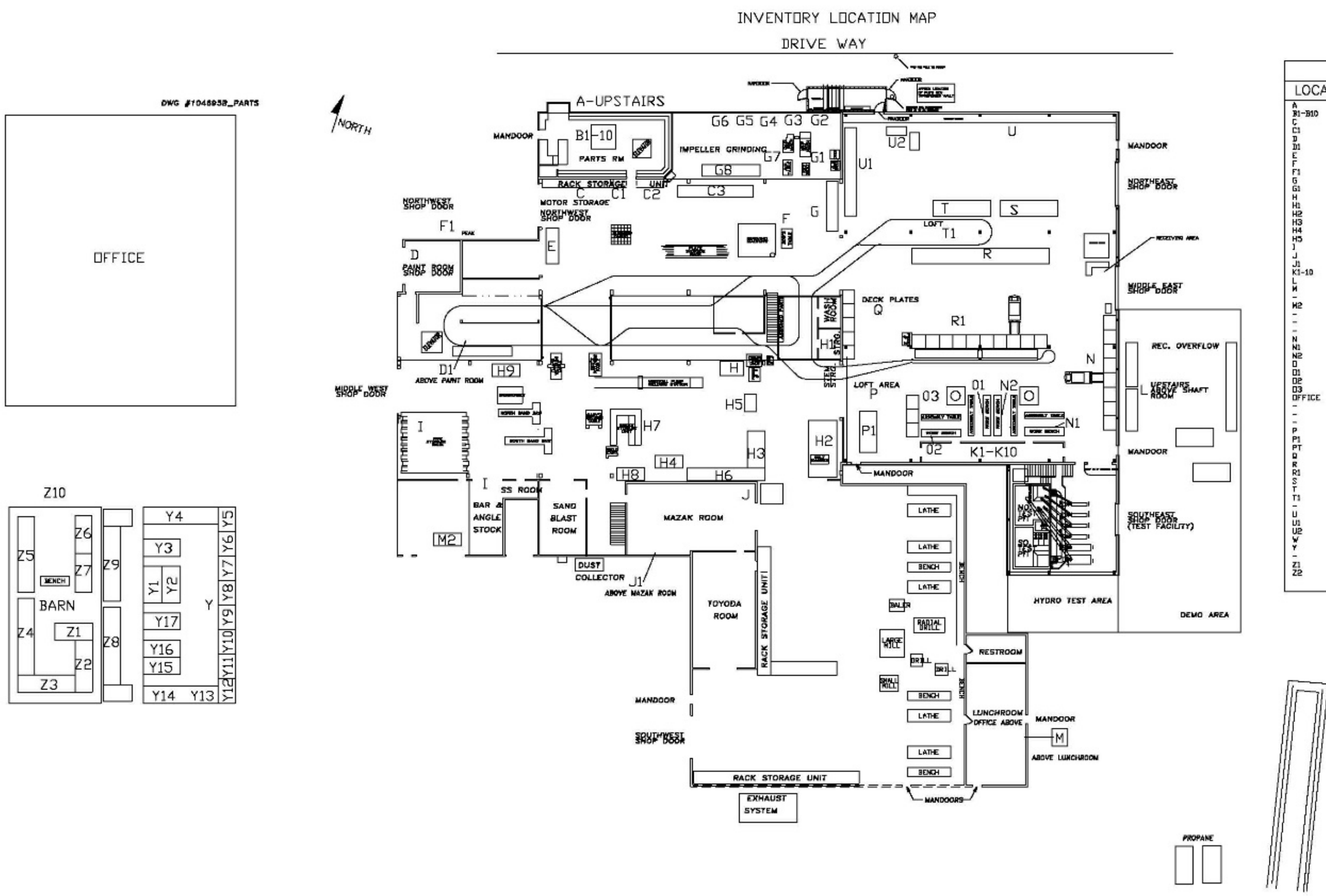
CUSTOMER COPY

Page 1

Authorized Signature **SO# 137853-00**

APPENDIX I

PLANT LAYOUT
&
PRODUCTION INSPECTION STATIONS



LOCATION	DESCRIPTION
A	UPSTAIRS ABOVE PARTS
B1-B10	PARTS ROOM UPS/FED EX
C	MOTOR STORAGE RACK
C1	MOTOR STORAGE RACK/BY SHAWN R
D	PAINT ROOM STORAGE
D1	ABOVE PAINT ROOM LOFT AREA
E	MOTOR STORAGE RACK #2
F	INDOOR PLATE STEEL
F1	OUTDOOR PLATE STEEL
G	IMPELLER/CUTTER BAR
G1	IMPELLER/CUTTER B./BLANK,BASE
H	WELDSHOP PART STORAGE
H1	RECIRC PART STORAGE
H2	MOTOR STOOL BODYS AND STEMS
H3	MOTOR STOOL FLANGES
H4	WELDSHOP STOCK BINS
H5	WELDSHOP BASE BLANKS
I	WELDSHOP TUBING RACK
J	MAZAK ROOM
J1	MISC STORAGE ABOVE MAZAK ROOM
K1-10	SHAFT/ BEARING ROOM
L	UPSTAIRS SEAL ASSEMBLY
M	MACHINE SHOP/ABOVE LUNCHROOM
-	MACHINE SHOP/RACK BY MAZAK RM
M2	MACHINE SHOP CDV/SSV
-	-
-	-
-	-
N	SELF PRIMER PARTS
N1	SELF PRIMER PARTS/DAVES BENCH
N2	SELF PRIMER PARTS/BILL BENCH
O	HORIZONTAL/VERTICAL PARTS
O1	HORIZONTAL/SCOTTS BENCH
O2	HORIZONTAL/RICH'S BENCH
O3	VERTICAL/ERIC'S BENCH
OFFICE	PARTS KEPT IN OFFICE
-	-
-	-
-	-
P	LOFT ASSEMBLIE
P1	CUTTER BARS/IMPELL FINISHED
PT	PARTS TEMPLATE ENGINEERING
Q	DECK PLATES
R	MIDDLE BAY NORTH
R1	MIDDLE BAY SOUTH
S	MISC PARTS STORAGE
T	MOTOR STOOLS
T1	MOTOR STOOLS OVERFLOW ABOVE
-	TO BE DETERMINED
U	MISC PARTS STORAGE
U1	BELT GUARDS/FINAL ASSEMBLY
U2	COUPLINGS/FINAL ASSEMBLY
W	WOOD SHED BEHIND SHOP
Y	BARN STORAGE #2
-	BARN STORAGE #1
Z1	BARN STORAGE ROTAMIX
Z2	BARN STORAGE ROTAMIX

Pole Building
February 2000
W LOC

APPENDIX J

MECHANICAL SEAL
FACE INSPECTION CRITERIA

Mechanical Seal Face Inspection Procedure

J.1 Packing slip for seal faces will be compared to drawing at receipt of parts to confirm parts are constructed of correct material. If material shown on packing slip does not match material shown on drawing, entire lot of parts will be rejected and returned to manufacturer.

J.2 During seal assembly, 100% of seal faces will be visually inspected for cracks, chips, and imperfections. Any parts found with cracks, chips, or imperfections will be rejected and returned to the manufacturer.

J.3 During seal assembly 10% of seal faces will be inspected dimensionally for conformance to drawing. If any parts inspected do not conform to the drawing, then 100% of the seal faces in the given lot will be inspected for dimensional conformance. Any non-conforming parts will be rejected and returned to the manufacturer.

J.4 During seal assembly 10% of seal faces with reflective finish will be inspected for flatness, using helium light bands, to ensure conformance to specification shown on drawing. During seal assembly, 10% of seal faces with matte finish will be inspected with profilometer to ensure conformance to specification shown on drawing. If any parts inspected do not conform to the drawing specification, then 100% of the seal faces in the given lot will be inspected for conformance to flatness specification. Any non-conforming parts will be rejected and returned to the manufacturer.

J.5 Assembled seal faces will be pressure tested with 10 psi for a duration of 30 seconds. If no leakage is observed during this time, then the seal faces will be considered acceptable. If any pressure loss is observed due to leakage at the seal faces, then the seal faces will be rejected and returned to the manufacturer.

J.6 All findings for each lot of seal faces will be documented using the Inspection Data Form (Appendix E).

APPENDIX K

CASTING INSPECTION
&
TESTING PROCEDURE

Casting Inspection Procedure and Testing Procedure

K.1 Receiving personnel will mark incoming castings with the following information, which is found on the packing slip:

- Purchase order number
- Originating foundry
- Color code for alloy identification
    - Ductile Iron—orange
    - Gray Iron—gray
    - Chrome Iron—green
    - CD4MCu—yellow
    - CF8M—blue
    - Red—cast steel

Small castings that can't be individually marked will be marked on their storage packaging.

Any gross defects discovered during receiving operations shall be reported to Engineering, who will determine whether the parts can be used with or without repair, or should be scrapped.

K.2 Machinists who uncover casting defects during their operations shall stop work and contact their Supervisor. Engineering will make a determination on the disposition of the part. Defects include, but are not limited to, shrinkage voids, sand inclusions, core/pattern shifts and hard castings.

The machinists shall mark each machined casting with its production work order number and re-mark them with their purchase order numbers, in the event the original markings are machined off. This will ensure traceability of the castings.

K.3 All machined casings delivered to the test facility will be inspected for cracks, voids, and other visible defects. Casings will be marked as passing a visual inspection. If a casing fails a visual inspection, it will be hydrotested.

Fifty percent of every production run of ductile iron machined casings will be hydro-tested in the test facility. If any failures are found in this 50% sample, then the remaining casings will be hydro-tested.

All non-tangential stainless steel casings will be hydro-tested.

Casings which pass the hydrotest will be marked with “100 psi” and the tester's initials. Casings receiving only visual tests will be marked with the tester's initials only. Casings failing the hydrotest will either be repaired or rejected and returned to the foundry. Repaired casings will be placed in inventory, only after passing a hydro retest.

K.4 All machined castings will be inspected by assembly personnel as part of the process of building a pump. Any casting defects will be reported to Engineering for disposition of the part. Also, failure of sub-assemblies during pressure/vacuum testing will be reported to Engineering, if initial attempts to correct the leak fail. Engineering will determine whether the leak is related to casting quality, and decide if the part should be repaired or scrapped.

K.5 During surface preparation on the paint line, the painters can repair superficial, non-structural cracks and small surface defects. Metal-filled epoxy compounds are suitable for this. Repairs on the paint line shall be limited to cosmetic defects only. Engineering should be contacted to determine if the casting defect impairs the physical strength of the casting or its pressure tight barrier.

K.6 All defective castings will be clearly marked with fluorescent red paint and will be accompanied by either a Discrepancy, Scrap or Repairable tag (Appendix D), per section 9.4. They will be stored in a separate "SCRAP" or "REWORK" area, out of the normal production flow, for disposition by the production and engineering managers.

K.7 Engineering will make a record of all casting defects, report them to the appropriate foundries, and will ensure the defects are corrected.

APPENDIX L

ROTAMIX NOZZLE INSPECTION PROCEDURE

Nozzle Component Inspection and Testing Procedure

L.1 Receiving personnel will inspect incoming nozzle component castings prior to painting with an epoxy coating. Any gross defects discovered during receiving operations shall be reported to Engineering, who will determine whether the parts can be used with or without repair, or should be scrapped. Parts will be inspected for continuous glass lining without pits or holes. Internal and external surfaces will be inspected for good surface finish without voids, cracks, or obvious casting defects.

L.2 All epoxy coated nozzle assembly castings will be inspected by assembly personnel as part of the process of building a nozzle assembly. Any casting defects as discussed above in M.1 will be reported to Engineering for disposition of the part.

L.3 All defective castings will be clearly marked with fluorescent red paint and will be accompanied by either a Discrepancy, Scrap or Repairable tag (Appendix D), per section 9.4. They will be stored in a separate "SCRAP" or "REWORK" area, out of the normal production flow, for disposition by the production and engineering managers.

L.4 Engineering will make a record of all casting defects, report them to the appropriate foundries, and will ensure the defects are corrected.

L.5 Epoxy coating shall be tested with an ETG (electronic thickness gage) for minimum powder coat thickness established by the coating product manufacturer specification. Each component will be measured at a minimum of 3 different random locations and the thickness at each location will be noted by the inspector on the inspection form, Appendix E.

L.6 Glass lining shall be tested with an ETG (electronic thickness gage) for minimum glass lining thickness established by the coating product manufacturer specification. Each component will be measured at 3 different random locations and the glass thickness at each location will be noted by the inspector on the inspection form, Appendix E.

APPENDIX M

PUMP TEST
&
CERTIFICATION CHECKLIST

PUMP TEST & CERTIFICATION CHECKLIST

Job Order #	**I_Test**	Spec #	**Spec #**
Model #	**Model #**	PO #	**PO #**
Quantity	**Quantity**		
Customer	**Customer**	Application	**Application**
Rep	**Rep**	Motor HP	**Motor HP**
Send Tests To	**attn:@ Rep**	Motor Volts	**Motor Volts**
@ address	**address:, city/st/zip**	Motor RPM	**Motor RPM**
@ fax #	**fax #, phone # phone #**	***WANT***	***Saturday, January 00, 1900***

TESTING REQUIREMENTS ((#VALUE! L/s @ M3/HR @ #VALUE! M TDH) APPROVAL DATES

Yes ____ No **X** ***Performance Testing*** **#VALUE!** GPM @ **#VALUE!** ' TDH @ **Motor RPM** RPM w/
na % Efficiency and **na** Feet of NPSHA

yes ____ no **X** Witnessing Required? ________ Test w/ Test Rig?
yes ____ no **X** NPSH Test Required? ________ TEST w/ CUSTOMER MOTOR !
yes ____ no **X** Vaughan Approval Req'd? ________ Test Parts or Assembled?
yes ____ no **X** Customer Approval Req'd? ________ : ___/___/___
yes ____ no **X** Send Cert w/ Pump? ________ : ___/___/___
yes ____ no **X** Raw Data / Test Logs Required? ________ : ___/___/___
yes ____ no **X** Calibration Certificates Required? ________ : ___/___/___
yes ____ no **X** P.E. Stamp Required? ________ : ___/___/___

Yes ____ No **X** ***Hydrostatic Testing Required?*** **none** PSI for **10** minutes.
(#VALUE! M tdh) ________ Casing Only? . . .or
Shutoff Head = #VALUE! **FT tdh** ________ Entire Pump? **test nothing**
yes ____ no **X** Vaughan Approval Req'd? No EIM Motor?
yes ____ no **X** Customer Approval Req'd? ________ : ___/___/___
yes ____ no **X** Send Cert w/ Pump? ________ : ___/___/___

Yes ____ No **X** ***DRY RUN VIBRATION TEST Required? Max*** 0.14 inches/second overall peak vibration velocity allowed
yes ____ no **X** Vaughan Approval Req'd? ________ : ___/___/___ Wet Run Req'd? NO
yes ____ no **X** Customer Approval Req'd? ________ : ___/___/___ Dry Run Req'd? YES
yes ____ no **X** Send Cert w/ Pump? ________ : ___/___/___

Yes ____ No **X** ***Impeller Balance Required?*** **1** MIL IMPELLER IMBALANCE ALLOWED.
yes ____ no **X** Vaughan Approval Req'd? ________ : ___/___/___
yes ____ no **X** Customer Approval Req'd? ________ : ___/___/___
yes ____ no **X** Send Cert w/ Pump? ________ : ___/___/___

Yes ____ No **X** ***Paint Testing Required?*** ________ mils of primer required
min ________ mils of topcoat required
min ________ mils total required
yes ____ no **X** Vaughan Approval Req'd? ________ : ___/___/___
yes ____ no **X** Customer Approval Req'd? ________ : ___/___/___
yes ____ no **X** Send Cert w/ Pump? ________ : ___/___/___

Yes ____ No **X** ***Motor Testing Required?*** ________ no X complete test required?
yes ________ no X routine test req'd?
yes ________ no X amp readings required?
yes ________ no X megger tests required?
yes ________ no X run pump submerged in test pit?
yes ____ no **X** Vaughan Approval Req'd? ________ : ___/___/___
yes ____ no **X** Customer Approval Req'd? ________ : ___/___/___
yes ____ no **X** Send Cert w/ Pump? ________ : ___/___/___

Yes ____ No **X** ***Noise Testing Required?*** ________ decibels allowed
______ A-Weighted, Overall, Type 1 Meter? . . . Or
______ A-Weighted , Octave Bance, Type 1 Meter?
yes ____ no **X** Vaughan Approval Req'd? ________ : ___/___/___
yes ____ no **X** Customer Approval Req'd? ________ : ___/___/___
yes ____ no **X** Send Cert w/ Pump? ________ : ___/___/___

Note, be sure to include: bill of materials, list of equip, list of exceptions, testing page from spec & performance page from spec

s:\engr\perfdata\testcert\excel_test_cert_template.xls Rev 11

APPENDIX N

QUALITY ASSURANCE VERIFICATION FORMS

VERTICAL WETWELL CHOPPER PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER#__________________________ SALES ORDER #________________________

MODEL#________________________________DRAWING#____________________________

SECTION 1: PUMP ASSEMBLY STATION:

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#__________Rev#_____ _________
2. Impeller has _____ blades and diameter measures _________ inches. _________
3. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ _________
4. Impeller has been balanced? _________
5. Pump length is: ________feet. This matches Job Order ☐ BOM ☐ Drawing ☐ _________
6. Measurement from top of deckplate to end of shaft ___________ _________
   All 3-6”, 8K & 8L: 5.25” +/– 0.25” 8N, 8P, All 10-16”: 7.0” +/– 0.50”
7. If pump is built with Tapered Roller Bearings, shaft endplay is ________inch _________
   3-6”: 0.001- 0.004” 8-16”: 0.001- 0.006”
8. All nuts and bolts tightened with lock washers? Yes ☐ No ☐ _________

SECTION 2: All 3-6” & 8L PUMPS

Initial Below

1. The cutter nut/disintegrator tool/external tool (if req’d) is properly torqued to 125 ft/lb _________
2. Impeller/Cutter bar gap to close vane measures _________ inch **0.015”- 0.025”** _________
3. Cutterbar bolts are torqued to 45 ft/lb. _________
4. Clearance between tip of cutter bar finger and hub tooth. **0.010”- 0.030”** _________
   Measurements for 2 locations: a)____________ inch b)___________ inch
5. Axial clearance between external tool wings and cutter bar fingers ______ **0.015-0.025”** ________
6. Clearance between tips of external tool and anvil __________ **0.010–0.030”** _________
7. Pump Out Vane/Back Cutter gap to close vane:__________inch **0.005”- 0.050”** _________
8. Cutterbar inlet diameter is correct per below _________
   3F/3G: 4.25” ☐ 3L/3M: 4.63” ☐ 3P: 4.3” ☐ 3V: 4.0” ☐ 3W: 4.63” ☐
   4K/4L: 5.13 ☐ 4M/4S: 6.0” ☐ 4T/6U/6X: 7.25” ☐ 4V: 8.50” ☐ 5300” 6” ☐

SECTION 3: 8N, 8P, All 10-16” PUMPS

Initial Below

1. Impeller is torqued to 1000 ft/lb _________
2. The external tool/disintegrator tool (if req’d) is properly torqued to 250 ft/lb _________
3. Impeller/Cutter bar gap to close vane measures _________ inch **0.015”- 0.025”** _________
4. Cutterbar bolts are torqued to 95 ft/lb. _________
5. Clearance between tip of cutterbar finger and external tool tooth _________
   Measurements for 2 locations: ____________ inch and ___________ inch
   8N & 8P: 0.025”- 0.040” All 10”: 0.035”- 0.050” 12” & 16”: 0.040”- 0.060”
6. Pump Out Vane/Back Cutter gap to close vane: __________inch **0.030”- 0.050”** _________
7. Verify clearance between tips of external tool and anvil: ___________ inch _________
   8N & 8P: 0.025”- 0.040” All 10”: 0.035”- 0.050” 12” & 16”: 0.040”- 0.060”
8. Axial clearance between external tool wings and cutterbar fingers: _________ inch _________
   0.015” – 0.025”

Job Order #________________________

Cutterbar inlet diameter is correct per below____________ _________
8M/8N: 9.0" ☐ **8P: 9.75"** ☐ **10R/10W: 11.0"** ☐ **12W: 13.5"** ☐ **16W: 16.0"** ☐

SECTION 4: 8K PUMPS

Initial Below

1. The Impeller is properly torqued to 500 ft/lb. _________
2. The Cutter Nose is properly torqued to 40 ft/lb. _________
3. The External Tool (if required) is properly torqued to 125 ft/lb. _________
4. Impeller/Cutter Bar gap to close vane measures _________ inch **0.015"- 0.025"** _________
5. Cutterbar bolts are torqued to 45 ft/lb. _________
6. Pumpout Vane/Back Cutter gap to close vane: ________inch. **0.005"- 0.050"** _________
7. Clearance between tip of cutter bar finger and cutter nose or external tool hub. **0.025"- 0.040"**
   Measurements for 2 locations: a)____________ inch b)___________ inch _________
8. Axial clearance between tips of external tool wings & cutter bar fingers______**0.015"-0.025"**______
9. Clearance between tips of external tool and cutter bar: __________ **0.025"- 0.040"** _________
10. Cutter Bar inlet diameter is correct **8K – 8.75"** _________

SECTION 5: WELDING STATION

Initial Below

1. Does pump match Job Order ☐ BOM ☐ Drawing ☐ _________
2. Below deck discharge (if req'd) is in the correct position as shown on the drawing _________
3. If required is Companion Flange (export pumps) ☐ or Blind Flange ☐ installed? _________
4. If a custom deckplate is required does it match the Drawing? Yes ☐ No ☐ _________
5. Gauge port is located opposite side from oil monitor. _________
6. Set collar is tack welded to the nozzle extension handle below the deckplate. _________
7. Oil ports in top stem are clocked toward the oil monitor. _________

SECTION 6: PAINTING STATION

Initial Below

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐? _________
2. Paint type:________________________ Required paint thickness:_________mils _________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of deckplate between stem and discharge _____MWFT
b) top of deckplate adjacent to oil monitor _____MWFT
c) midway on bearing column _____MWFT
d) top of casing at the discharge throat _____MWFT
e) bottom of cutter bar flange _____MWFT
Initials _____ Date _____

DRY
a) underside of deckplate between stem and discharge _____MDFT

Job Order # ___________________________

b) top of deckplate adjacent to oil monitor _____MDFT
c) midway on bearing column _____MDFT
d) top of casing at the discharge throat _____MDFT
e) bottom of cutter bar flange _____MDFT

Initials _____ Date _____

SECTION 7: FINAL ASSEMBLY STATION

Initial Below

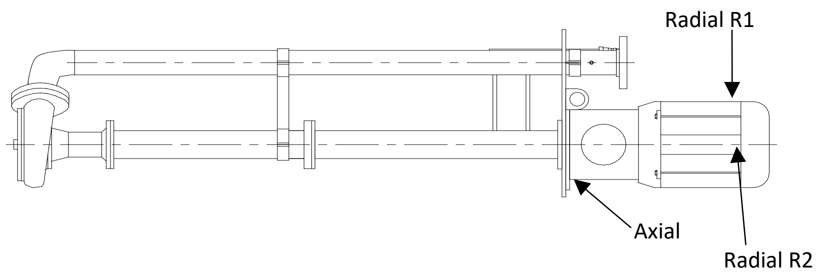
1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ _________
2. Pump matches: Job Order ☐ Drawing ☐ Drawing#_________Rev#____ _________
3. Pump and drive assembly match drawing referenced above _________
4. Is the Disintegrator Tool (if required) installed and tight? _________
5. Check coupling alignment after motor mounting (direct drive). _________
6. Verify that the pump shaft turns by hand _________
7. Oil is filled to the proper level _________
8. Hole in vent cap is drilled through and intersects roof? (Use 0.156" drill bit to check) _________
9. Customer control voltage for oil level relay is ______________ Volts _________
10. Automatic Oil Monitor checked for proper operation _________
11. Is Relay ☐ Base ☐ Female OLM Connector ☐ supplied with pump? _________
12. Intrinsically safe relay supplied if motor is explosion proof? Yes ☐ No ☐ _________
13. Check sheaves for proper alignment (belt drive). _________
14. If customer supplied motor, are Motor Plates ☐ Bushings ☐ Sheaves ☐ _________
    Belts ☐ Brackets ☐ Tensioners ☐ Guards ☐ Pump Driveshaft Key ☐
    Mounting Hardware ☐ supplied?
15. Motor greased with ____________________________ Not Greasable ☐ _________
    Check motor manual for grease type when not listed on nameplate.
16. Motor Manufacturer _______________________________
    Motor Serial Number ______________________________
    Spec/Model/Catalog#______________________________
    Motor HP__________ Speed__________RPM Frame Size__________
17. All guards are in place. _________
18. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ _________
19. Decals are installed on the pump per decal placement drawing _________

Vibration Readings

Axial________________

Radial R1____________

Radial R2____________



Job Order #____________________ **Initial Below**

Motor Amp Measurement__________ amps _________

Nameplate full load amps___________ _________

Reading is below 1/2 of full load. Yes ☐ No ☐ _________

Verify pump speed with tachometer: __________RPM (belt drive only) _________

FOR WET-WELL RECIRCULATORS

20. Handle length____________________
    ____

21. Valve actuator system operated and functions properly.

22. Decals installed per decal placement drawing.

23. Operating voltage of Auto Valve Actuator matches Job Order ☐ BOM ☐ Drawing ☐
    ____

SECTION 8: FINAL SHIPPING STATION

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ _________
2. Pump matches: Job Order ☐ Drawing ☐ Drawing#_________Rev#____ _________
3. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ _________
4. Decals are installed on the pump per decal placement drawing _________
5. When installed on floating platform, weight of completed unit is ____________ lbs _________
   Advise Engineering of weight prior to shipment.
6. The IO&M Manual for wet-well pumps is attached to pump at time of shipping. _________

FOR WET-WELL RECIRCULATORS

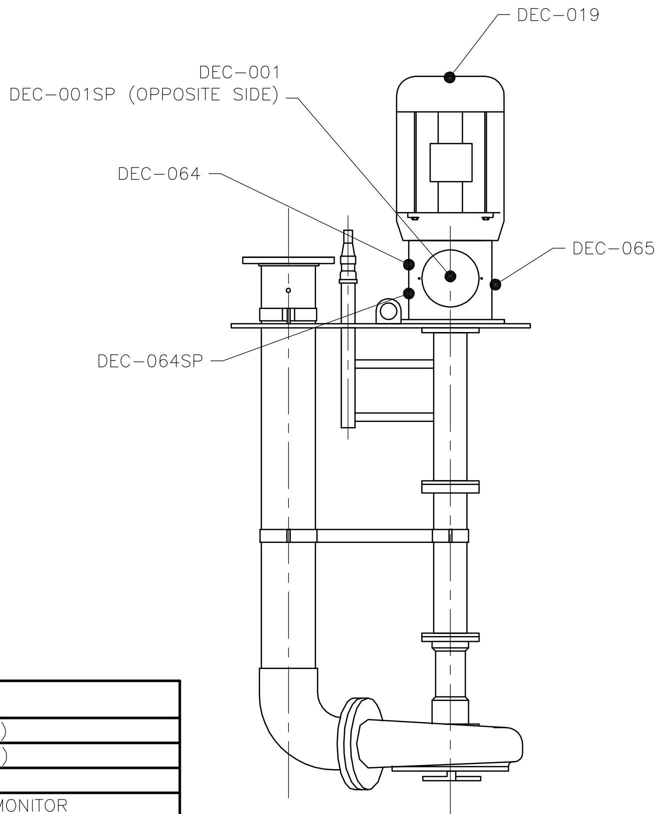
7. Decals installed per decal placement drawing. _________

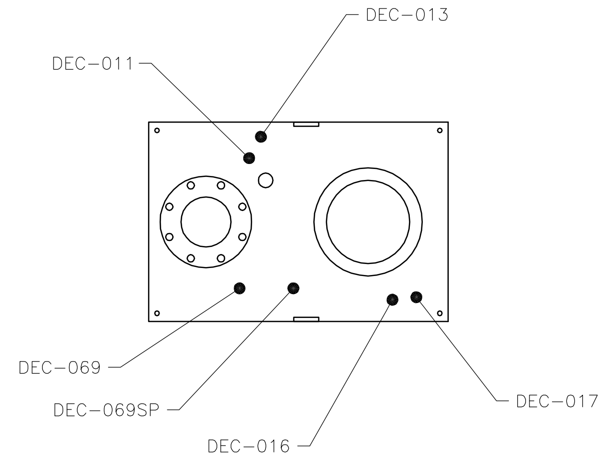
8. QA Sheet is complete:

   ____
   (Shipping Clerk) (Date)

REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/14/16	2	UPDAATED FOREIGN PUMP TABLE	KCW





FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC–065 WITH DEC–037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC–065CAN	CANADA
DEC–037CZ	CZECHOSLOVAKIA
DEC–037DK	DENMARK
DEC–037NL	HOLLAND
DEC–037FI	FINLAND
DEC–037FR	FRANCE
DEC–037DE	GERMANY
DEC–037GR	GREECE
DEC–037IT	ITALY
DEC–037NOR	NORWAY
DEC–037PT	PORTUGAL
DEC–037ES	SPAIN
DEC–037SE	SWEDEN
DEC–037UAE	UNITED ARAB EMIRATES

DOMESTIC PUMPS

PART #	DESCRIPTION
DEC–001	5.375" X 6" ATTENTION (ENGLISH)
DEC–001SP	5.375" X 6" ATTENTION (SPANISH)
DEC–011	2" X 4" OIL REQUIREMENTS
DEC–013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC–016	1.25" X 3.5" QUALITY ASSURED
DEC–017	2" X 4" QUALITY CRAFTED IN USA
DEC–019	.5 X 5.5" CLOCKWISE ROTATION
DEC–064	3.5" X 5" WARNING (ENGLISH)
DEC–064SP	3.5" X 5" WARNING (SPANISH)
DEC–065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC–069	2" X 4" WARNING SUFFOCATION HAZARD (ENGLISH)
DEC–069SP	2" X 4" WARNING SUFFOCATION HAZARD (SPANISH)

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

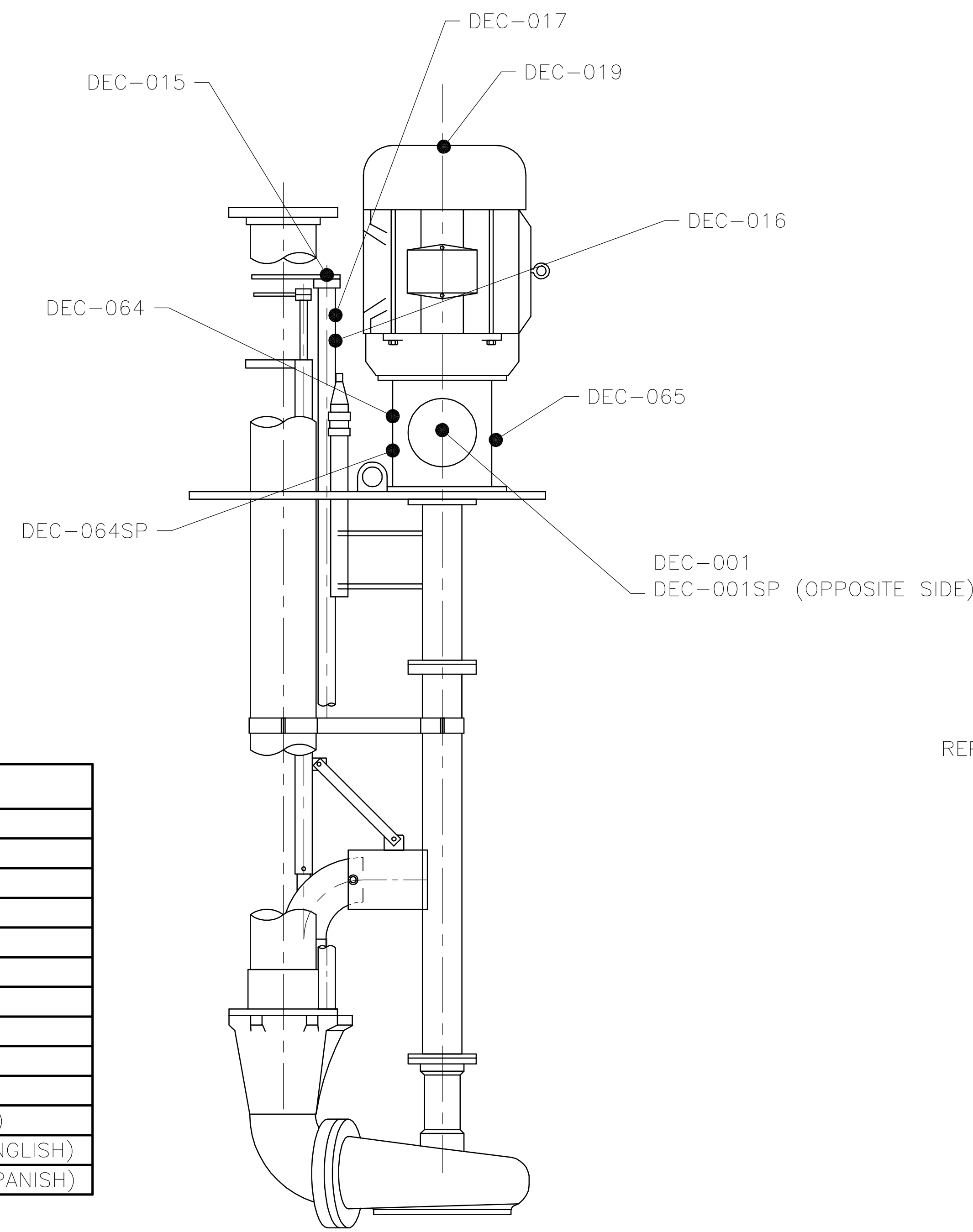
TITLE: VERTICAL WET–WELL DECAL PLACEMENT CHOPPER & SCREW PUMPS

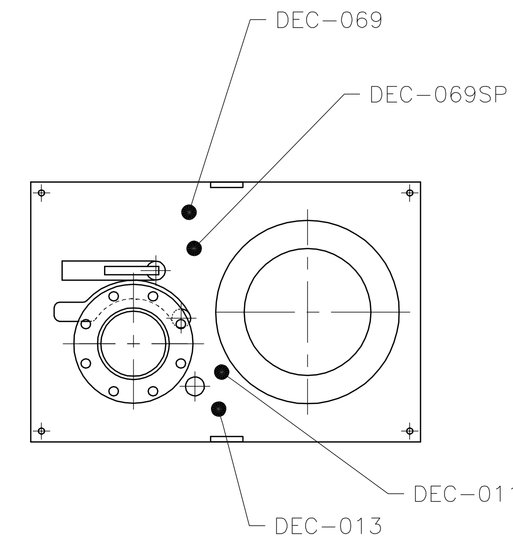
CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	3/17/16	118414	2

REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	2	UPDAATED FOREIGN PUMP TABLE	KCW





DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-011	2" X 4" OIL REQUIREMENTS
DEC-013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC-015	1.75" DIA., PUMP OUT/AGITATE
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING SUFFOCATION HAZARD (ENGLISH)
DEC-069SP	2" X 4" WARNING SUFFOCATION HAZARD (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE:
VERTICAL WET-WELL RECIRC.
DECAL PLACEMENT
CHOPPER & SCREW PUMPS

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
- SURFACE FINISH: 125 MICROINCHES
- DIMENSIONS, WELDING: X.XX = 1/16"
- DIMENSIONS, MACHINING: X.XX = 0.020"
- DIMENSIONS, MACHINING: X.XXX = 0.010"
- GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
- GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	3/21/16	118417	2

HORIZONTAL CHOPPER PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER#_________________________ **SALES ORDER #** _______________________

MODEL#_____________________________ **DRAWING#**____________________________

SECTION 1: PUMP ASSEMBLY STATION:

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
2. Impeller has ____ blades and measures ______ inches ________
3. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ ________
4. Impeller has been balanced ________
5. If pump is built with Tapered Roller Bearings, shaft endplay is ________inch ________
   3-6": 0.001-0.004", 8–16": 0.001–0.006"
6. Shaft run out at impeller: ____________ **(0.004" max.)** ________
7. Inboard and Outboard Bearing Isolators have been installed with drain-back at 6:00. ________
8. Inboard oil containment cup is installed ________
9. All nuts and bolts tightened with lock washers. ________
10. Discharge orientation matches Job Order ☐ BOM ☐ Drawing ☐ ________
11. Vent port is present and plugged in casing Yes ☐ No ☐ ________
12. Drain port is present and plugged in casing Yes ☐ No ☐ ________

SECTION 2: All 3-6" & 8L PUMPS

Initial Below

1. The Cutter Nut ☐ or External Tool ☐ is properly torqued to 125 ft/lb. ________
2. Impeller/Cutter Bar gap to close vane measures _________ inch **0.015–0.025"** ________
3. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.005–0.050"** ________
4. Clearance between tip of cutter bar finger and hub tooth. **0.010-0.030"** ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
5. Clearance between tips of external tool and anvil ____________ **0.010–0.030"** ________
6. Axial clearance between external tool wings and cutter bar fingers _________inch ________
   0.015 – 0.025"
7. Cutter Bar inlet diameter is correct per below ________
   3F/3G/3P: 4.25" ☐ **3L/3M/3W: 4.63"** ☐ **3V: 4.0"** ☐ **4K/4L: 5.25"** ☐
   4P/4S: 6.50" ☐ **4R/4T/6U/6W/6X: 7.40"** ☐ **4V: 8.50"** ☐ **8L – 9.25"** ☐

SECTION 3: 8N, 8P, All 10-16" PUMPS

Initial Below

1. The impeller is properly torqued to 1000 ft/lb. ________
2. The External Tool is properly torqued to 250 ft/lb. ________
3. Impeller/Cutter Bar gap to close vane measures _________ inch ________
   8N, All 10–16": 0.015–0.025" 8P: 0.010–0.020"
4. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.030–0.050"** ________
5. Clearance between tip of cutter bar finger and external tool tooth ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
   8N & 8P: 0.025–0.040", All 10": 0.035–0.050" All 12:&16": 0.040–0.060"
6. Clearance between tips of external tool and anvil: ______________ inch ________
   8N, 8P: 0.025–0.040" 10": 0.035–0.050" 12" &16": 0.040–0.060"

JOB ORDER# ______________________________ **Initial Below**

7. Axial clearance between external tool wings and cutter bar fingers __________inch **0.015 – 0.025”** ________
8. Cutter Bar inlet diameter is correct per below ________
   8N – 9.0” ☐ **8P – 9.75”** ☐ **10R/10W – 11.0”** ☐ **12W – 13.5”** ☐ **16W – 16.0”** ☐

SECTION 4: 8K PUMPS

Initial Below

1. The Impeller is properly torqued to 500 ft/lb. ________
2. The Cutter Nose is properly torqued to 40 ft/lb. ________
3. The External Tool (if required) is properly torqued to 125 ft/lb. ________
4. Impeller/Cutter Bar gap to close vane measures __________ inch **0.015-0.025”** ________
5. Pump Out Vane/Back Cutter gap to close vane: _________inch. **0.005-0.050”** ________
6. Clearance between tip of cutter bar finger and cutter nose or external tool hub. **0.030-0.040”**
   Measurements for 2 locations: a)_____________ inch b)____________ inch ________
7. Axial clearance between external tool wings and cutter bar fingers __________inch ________
   0.015-0.025”
8. Clearance between tips of external tool and anvil: _______________ inch **0.020–0.045”** ________
9. Cutter Bar inlet diameter is correct **8K – 8.75”** ________

SECTION 5: PAINTING STATION

Initial Below

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Paint type:__________________________ Required paint thickness:__________mils _________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MWFT
b) top of baseplate between casing and bearing housing _____MWFT
c) side of suction manifold _____MWFT
d) side of bearing housing adjacent to oil site glass _____MWFT
e) casing discharge throat on side nearest bearing housing _____MWFT
Initials _____ Date _____

DRY
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MDFT
b) top of baseplate between casing and bearing housing _____MDFT
c) side of suction manifold _____MDFT
d) side of bearing housing adjacent to oil site glass _____MDFT
e) casing discharge throat on side nearest bearing housing _____MDFT
Initials _____ Date _____

SECTION 6: FINAL ASSEMBLY STATION

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________

JOB ORDER#__________________________ **Initial Below**

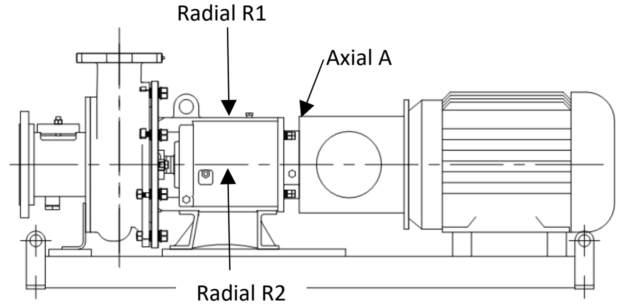
2. Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ ________
3. Pump matches: Job Order ☐ Drawing ☐ Drawing#__________Rev#____ ________
4. Pump and drive assembly match drawing referenced above ________
5. If a special base is required does it match the drawing? Yes☐ No☐ ________
6. Verify pump shaft turns over by hand. ________
7. Check coupling alignment after motor mounting (direct drive). ________
8. Sheaves have been checked for proper alignment (belt drive). ________
9. Inboard and outboard oil containment cups are installed. ________
10. Bearing housing oil level is half way up sight glass. ________
11. Sight glass is installed on left side when looking at pump suction, or opposite side of motor on belt drive pumps. ________
12. Bearing housing vent plug and cap are installed ________
13. All guards are in place. ________
14. If customer supplied motor, are (4) proper size motor adjuster feet (with hardware) ☐ and pump drive shaft key ☐ supplied? ________
15. If customer supplied motor on belt-drive pump, are the Motor Adjustable Bases ☐
    Belts ☐ Tensioners ☐ Bushings ☐ Sheaves ☐ Belt Guard ☐
    Pump Drive Shaft Key ☐ supplied with the pump? ________
16. Companion flanges and gaskets (if required) supplied with export pumps. ________
17. Verify pump speed of belt drive pumps with tachometer ___________ RPM ________
18. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
19. If required the Rotamix trademark has been installed on base of pump. ________
20. Tags and nameplates are installed on the pump per the decal placement drawing. ________
21. Motor Manufacturer ______________________________ ________
    Motor: Serial No.____________________________ Frame Size ____________
    Spec/Model/Catalog Number ________________________ Speed___________RPM
22. Motor greased with ______________________________ Not Greasable ☐ ________
    Note: Check motor manual for grease type when not listed on nameplate.

Vibration Measurements

Axial A ___________________

Radial R1 ________________

Radial R2 ________________



JOB ORDER#__________________________

Nameplate Full Load Amps ______________ _________ (Init)

Motor Amp Measurements: ______________ _________ (Init)

Reading is below half of full load. _________ (Init)

SECTION 7: FINAL SHIPPING STATION

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ ________
3. Pump matches: Job Order ☐ Drawing ☐ Drawing#__________Rev#____ ________
4. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
5. Tags and nameplates are installed on the pump per the decal placement drawing. ________
6. Is the IO&M Manual attached to the pump at the time of shipping? ________
7. QA Sheet is complete:

____

(Shipping Clerk) (Date)

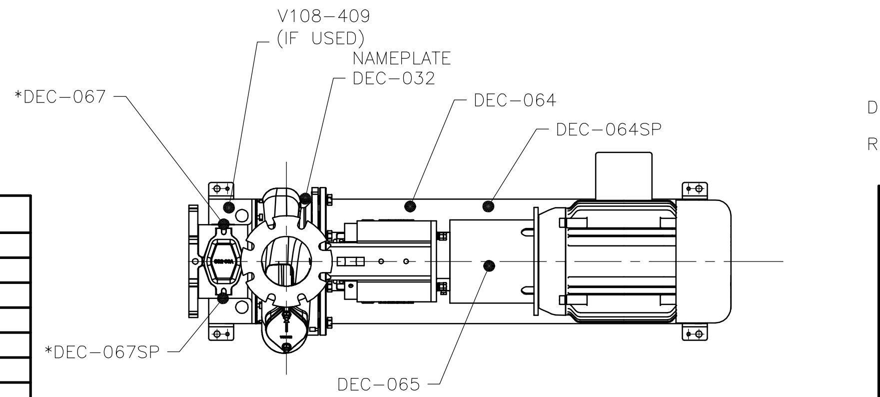
REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
6/20/16	1	ADDED NAME PLATE DEC-32	KCW
7/13/16	2	RENAMED DRAWING & ADDED NOTE	KCW
7/20/16	3	ADDED FOREIGN PUMP TABLE	KCW
10/14/16	4	UPDAATED FOREIGN PUMP TABLE	KCW

DOMESTIC PUMPS

* DEC-067 NOT USED IF PUMP HAS NO CLEANOUT

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-032	NAME PLATE
DEC-061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC-067	2" X 4" DANGER (ENGLISH)
*DEC-067SP	2" X 4" DANGER (SPANISH)
V108-409	2.75" X 6.5" SS ROTAMIX LOGO PLATE

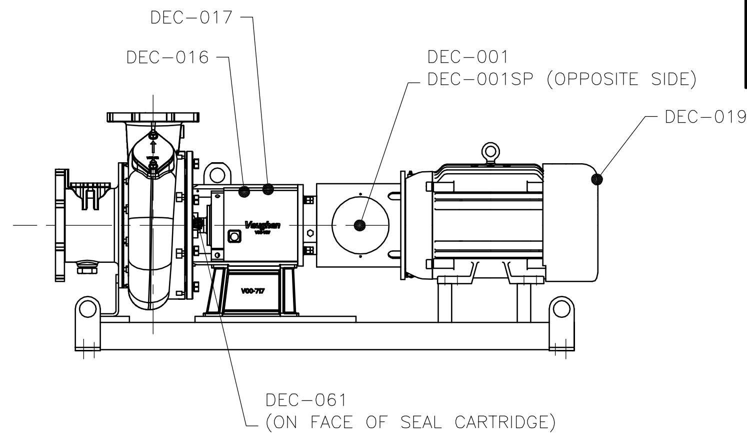


FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES



Vaughan

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: HE/HSC DIRECT DRIVE DECAL PLACEMENT

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	2/8/16	118370	4

SE SERIES SUBMERSIBLE PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER #______________________ SALES ORDER #____________________________

MODEL #___________________________ DRAWING #_________________________________

SECTION 1: PUMP ASSEMBLY STATION, ALL PUMPS

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#_____ _ ___
2. Impeller has _____ blades and diameter measures _________ inches _____
3. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ _____
4. Impeller has been balanced. _____
5. List number and thickness of seal length shims used: _________ @ _____________ _____
6. List number and thickness of stub shaft shims used _________ @ _____________ _____
7. Motor Shaft Endplay __________ inch is within allowable range listed below. _____
   180TY: 0.011" max 210TY: 0.013" max 250TY: 0.015" max 320TY/360TY: 0.001-0.002"
8. Form V684 motor receiving sheet is attached to the production packet _____
9. Cap pulled to verify voltage matches Production Traveler. _____

SECTION 2: PUMP ASSEMBLY STATION, All 3"- 6" PUMPS

Initial Below

1. Is the cutter nut/external tool (if reqd) properly torqued? _____
   Cutternut 125 ft/lb ☐ 180TY bolt 40 ft/lb ☐ 210TY & 250TY bolt 90 ft/lb ☐
   320TY bolt 125 ft/lb ☐ 360TY w/ 320TY shaft bolt 125 ft/lb ☐
   440TY with 320TY shaft bolt 125 ft/lb ☐
2. Impeller/Cutter bar gap to close vane measures _______ inch **0.015"- 0.025"** _____
3. Verify clearance between tip of cutterbar finger and hub tooth **0.010"- 0.030"** _____
   Measurements for 2 locations: ______________ inch and _____________ inch
4. Pump Out Vane/Back Cutter Clearance: ___________ inch **Max. allowable is 0.050"** _____
   Minimum clearance to be Shaft Endplay __________ + 0.005".= ____________
5. Axial clearance between external tool wings and cutter bar fingers ________ **0.015"-0.025"** _____
6. Clearance between tips of external tool and anvil __________ **0.010"–0.030"** _____
7. Cutter bar inlet diameter is correct per below _____
   3F/3G/3P: 4.25" ☐ 3L/3M/3W: 4.63" ☐ 3V: 4.0" ☐ 4K/4L: 5.13" ☐
   4S: 6.5" ☐ 4T/6U/6X: 7.25" ☐ 6W: 7.40" ☐

SECTION 3: PUMP ASSEMBLY STATION, 8N, 8P All 10 – 16"

Initial Below

1. The External Tool is properly torqued to _____
   180TY-320TY 100 ft/lb. ☐ 360TY 150 ft/lb. ☐
   440TY with 320TY shaft 100 ft/lb. ☐ 440TY with 360TY shaft 150 ft/lb. ☐
2. Impeller/Cutter bar gap to close vane measures __________ inch **0.015"-0.025"** _____
3. Verify clearance between tip of cutterbar finger and external tool tooth _____
   Measurements for 2 locations: __________inch and _________ inch
   8": 0.025"- 0.040" 10": 0.035"- 0.050" 12/16: 0.040"- 0.060"
4. Axial clearance between tips of external tool wings & cutter bar fingers ________ inch _____
   0.015"- 0.025"
5. Clearance between tips of external tool and anvil: ____________ inch _____
   8": 0.025"- 0.040" 10": 0.035"- 0.050" 12-16": 0.040"- 0.060"
6. Pump Out Vane/Back Cutter Clearance: _________ inch **Max. allowable is 0.050"** _____
   Minimum clearance to be Shaft Endplay __________+ 0.030" = __________.

SE SERIES SUBMERSIBLE PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER#___________________

7. Cutterbar inlet diameter is correct per below ______
   8N: 9.0" ☐ **8P: 9.75"** ☐ **10P/10R: 11.0"** ☐ **12W: 13.5"** ☐ **16W: 16.0"** ☐

SECTION 4: PUMP ASSEMBLY STATION: All PUMPS

Initial Below

1. All nuts and bolts tightened with lock washers? ______
2. Motor Cable Quick Disconnects: If applicable, cables fully inserted, gland nuts torqued per spec. located on cable tags, and set screws tightened? ______
3. Adapter bracket (if reqd) is attached to the pump discharge. ______
   Note: If pump requires testing do not glue bracket.
4. Pump was run at assembly station? Yes ☐ No ☐ ______
5. Any rubbing? Yes ☐ No ☐ ______

SECTION 5: PAINTING STATION

Initial Below

1. Paint applied matches: Job Order ☐ BOM ☐ Drawing ☐? ______
2. Paint type:_________________________ Required paint thickness:_________mils ________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) backplate ______MWFT
b) top of casing, opposite of discharge flange ______MWFT
c) bottom of casing, opposite of discharge flange ______MWFT
d) top of casing at the discharge throat ______MWFT
e) bottom of cutter bar flange ______MWFT
f) elbow, under flange ______MWFT
Initials ______ Date ______

DRY
a) backplate ______MDFT
b) top of casing, opposite of discharge flange ______MDFT
c) bottom of casing, opposite of discharge flange ______MDFT
d) top of casing at the discharge throat ______MDFT
e) bottom of cutter bar flange ______MDFT
f) elbow, under flange ______MDFT
Initials ______ Date ______

SECTION 6: FINAL ASSEMBLY STATION

Initial Below

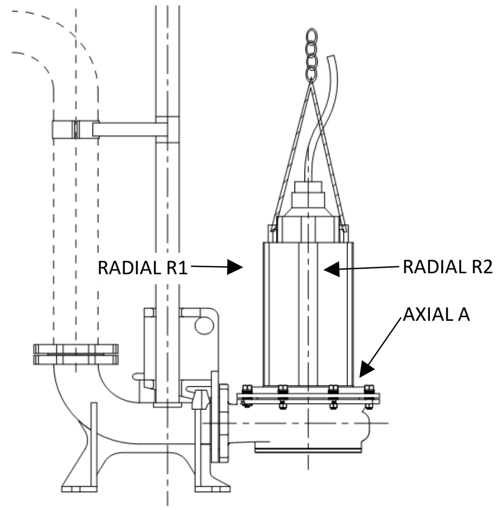
1. Motor matches: Job Order ☐ BOM ☐ Drawing ☐ ______
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#_____ ______
3. Verify the pump turns over by hand. ______

SE SERIES SUBMERSIBLE PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER#____________________ **Initial Below**

4. Verify correct rotation of pump. ______
5. If required is VPMR supplied? Yes ☐ No ☐ ______
6. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ______
7. Decals are installed per the decal placement drawing 118384. ______

Vibration Measurements:
Axial A ________________________________
Radial R1 ________________________________
Radial R2 ________________________________



Motor Amp Measurements: ____________ Amps

Nameplate Full Load Amps ___________

Reading is below half of full load? Yes ☐ No ☐

SUBMERSIBLE RECIRCULATORS

Initial Below

8. Form V838 Sub Recirculator Supplement QA is included in the production packet. ______
9. Goes with Sales Order # _______________________ ______

SECTION 7: FINAL SHIPPING STATION

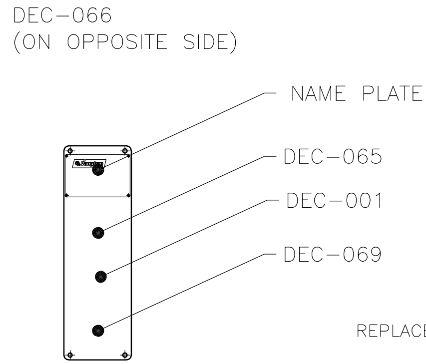
Initial Below

1. Motor matches: Job Order ☐ BOM ☐ Drawing ☐ ______
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ______
3. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ______
4. Decals are installed per the decal placement drawing 118384. ______
5. Form V684 motor receiving sheet is included in the production packet. ______
6. The IO&M Manual is attached to the pump at the time of shipping. ______
7. QA Sheet is complete:

____
(Shipping Clerk) (Date)

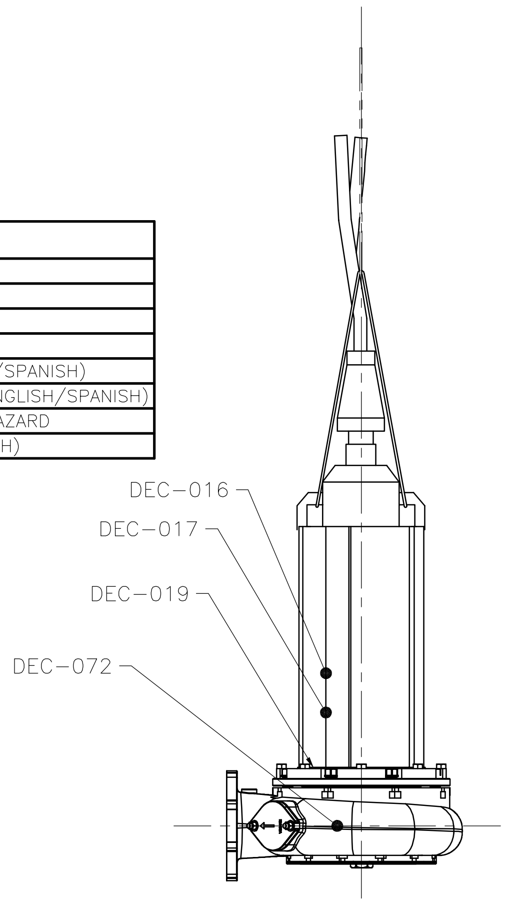
REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	2	UPDAATED FOREIGN PUMP TABLE	KCW



DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-066	4" X 10" ATTENTION INSTALLER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING, SUFFOCATION HAZARD
DEC-072	2" X 6" DANGER, (ENGLISH/SPANISH)



FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

Vaughan

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE: SE SERIES SUBMERSBLE PUMP DECAL PLACEMENT

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XX ± 0.020"
DIMENSIONS, MACHINING: X.XXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	
JOB ORDER NUMBER:		DATE: 2/23/16	DRAWING NUMBER: 118384	REVISION NUMBER: 2

SUBMERSIBLE RECIRCULATOR SUPPLEMENT QA VERIFICATION

To be used in addition to V321/V531

SALES ORDER # ______________

JOB ORDER#____________ MODEL#____________ DRAWING#______________

SECTION 1: FINAL ASSEMBLY STATION

Initial Below

1. Recirculator matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_______Rev#____ _____
2. Handle Length ______________________________ _____
3. Valve actuator system operated and functions properly. _____
4. Decals are installed per the decal placement drawing 118501 / 118502. _____
5. Operating voltage of Auto Valve Actuator matches: Job Order ☐ BOM ☐ Drawing ☐ _____
6. Confirm left hand or right hand system matches the BOM. _____

SECTION 2: PAINTING STATION

Initial Below

1. Paint applied matches: Job Order ☐ BOM ☐ Drawing ☐ _____
2. Paint type:_______________________ Required paint thickness:_________mils _____

Dry Film Thickness				
Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET

a) Deckplate, bottom center _____MWFT

b) Valve handle, middle _____MWFT

c) Valve body, underside _____MWFT

Initials _____ Date _____

DRY

a) Deckplate, bottom center _____MDFT

b) Valve handle, middle _____MDFT

c) Valve body, underside _____MDFT

Initials _____ Date _____

SECTION 3: FINAL SHIPPING STATION

Initial Below

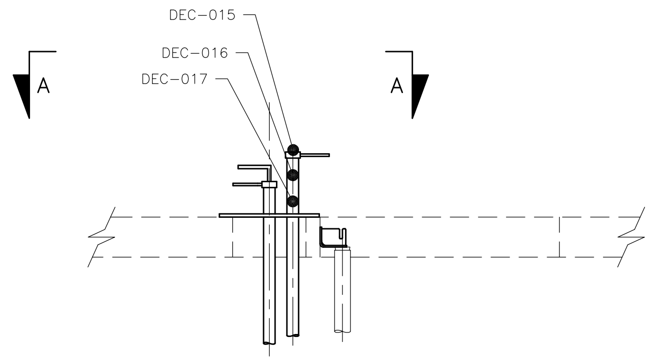
1. Form V321 / V531 is included in the production packet. _____
2. Outline dimension drawing # __________ matches pump assembly. Engineer only: _____
   (Mark N/A if no outline dimension drawing; if N/A, no Eng initials required)
3. Decals are installed per the decal placement drawing 118501 / 118502. _____
4. Supplement QA Sheet is complete:

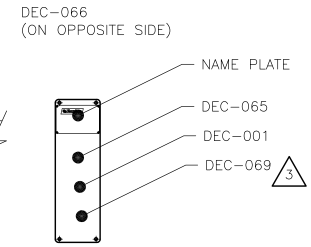
____

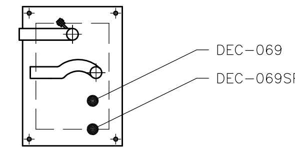
(Shipping Clerk) (Date)

REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	2	UPDAATED FOREIGN PUMP TABLE	KCW
1/3/24	3	ADDED SECOND DEC-069 DECAL & PLACEMENT; ECN: 5471	EAB



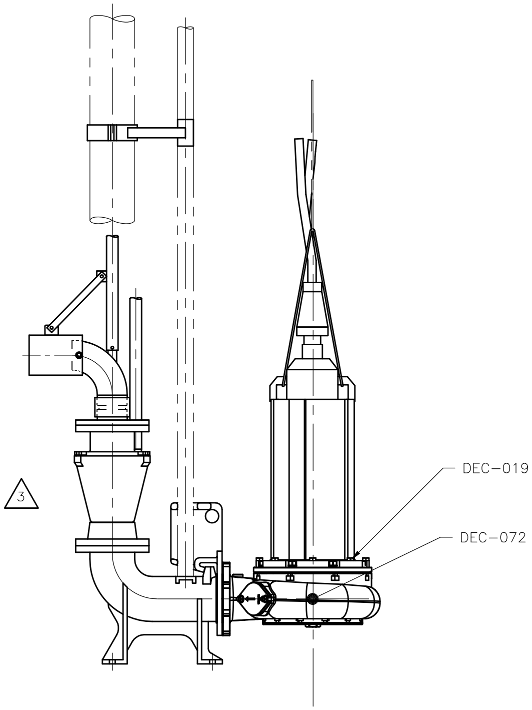




VIEW A-A

DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-015	1.75" DIAMETER, PUMPOUT/AGITATE
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-066	4" X 10" ATTENTION INSTALLER (ENGLISH/SPANISH)
(2X) DEC-069	2" X 4" WARNING, SUFFOCATION HAZARD (ENGLISH) 3
DEC-069SP	2" X 4" WARNING, SUFFOCATION HAZARD (SPANISH)
DEC-072	2" X 6" DANGER, (ENGLISH/SPANISH)



FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE: SE SERIES SUBMERSBLE RECIRC. DECAL PLACEMENT

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	5/5/16	118502	3

VERTICAL MIXER QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ **SALES ORDER # _________________**

MODEL #_____________________________ **DRAWING # _____________________**

SECTION 1: PUMP ASSEMBLY STATION

1. Mixer length is ________feet ________ (Init)
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#__ ________ (Init)
3. Use Loctite 243 (blue) and lock washers on bearing housing to casing bolts. ________ (Init)
4. Torque bearing housing to casing bolts to 80 ftlb. ________ (Init)
5. All nuts and bolts are tightened with lock washers. ________ (Init)
6. Propeller has been balanced. ________ (Init)
7. Measurement from top of deckplate to end of shaft ___________. ________ (Init)

Turbo and 100-S Mixer 6.0” +/– 0.50”

SECTION 2: WELDING STATION

1. Mixer matches Job Order ☐ BOM ☐ Drawing ☐? ________ (Init)
2. Elbow (if equipped) has internal cutter, correct height. ________ (Init)
3. Lifting eyes and rotation arrow on deckplate. ________ (Init)
4. If a custom deckplate is required does it match the Drawing? Yes ☐ No ☐ ________ (Init)
5. Rotating Turbo: Locking mechanism installed? Yes ☐ No ☐ ________ (Init)
6. Rotating Turbo: Locking mechanism in place prior to lifting? Yes ☐ No ☐ ________ (Init)

SECTION 3: PAINTING STATION

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)
2. Paint type:_________________________ Required paint thickness:_________mils ________ (Init)

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of deckplate between stem and discharge _____MWFT
b) top of deckplate adjacent to oil monitor _____MWFT
c) midway on bearing column _____MWFT
d) top of casing at the discharge throat _____MWFT
e) bottom of cutter bar flange _____MWFT
Initials _____ Date _____

DRY
a) underside of deckplate between stem and discharge _____MDFT
b) top of deckplate adjacent to oil monitor _____MDFT
c) midway on bearing column _____MDFT
d) top of casing at the discharge throat _____MDFT
e) bottom of cutter bar flange _____MDFT
Initials _____ Date _____

JOB ORDER# ___________________________________

SECTION 4: FINAL ASSEMBLY STATION

1. Motor/drive matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#__________Rev#__________ (Init)
3. Rotating Turbo: Locking mechanism in place prior to lifting? Yes ☐ No ☐ ________ (Init)
4. Verify that the shaft turns by hand ________ (Init)
5. Propeller is tack welded to shaft ________ (Init)
6. Vent hole in propeller hub is not plugged ________ (Init)
7. Propeller to cutter clearance __________ **0.010" – 0.030"** ________ (Init)
8. Hole in OLM vent cap is drilled through and intersects roof? ________ (Init)
   (Use 0.156" drill bit to check)
9. Customer control voltage for oil level relay is ___________ Volts ________ (Init)
10. Is Relay ☐ Base ☐ Female Connector ☐ supplied with pump? ________ (Init)
11. Oil reservoir is full ________ (Init)
12. Automatic Oil Monitor checked for proper operation ________ (Init)
13. Check sheaves for proper alignment (belt drive) ________ (Init)
14. If customer supplied motor, are Motor Plates ☐ Bushings ☐ Sheaves ☐ ________ (Init)
    Belts ☐ Brackets ☐ Tensioners ☐ Guards ☐ Pump Driveshaft Key ☐
    Mounting Hardware ☐ supplied?
15. Mixer speed __________ RPM. (belt drive) ________ (Init)
16. Motor Manufacturer: _____________________________ ________ (Init)
    Spec/Model/Catalog# ____________________ Speed __________ RPM
    Serial Number __________________________________ Frame Size __________
17. Motor greased with ___________________________ Not Greasable ☐ ________ (Init)
    Check motor manual for grease type when not listed on nameplate.
18. Gearbox: Serial Number _________________________ ________ (Init)
    Model Number _________________________________ Speed__________RPM
19. All guards are in place ________ (Init)
20. Rotation viewed from motor end matches direction arrow on deckplate & chart below. ________ (Init)

Turbo	**100S**	**Reverse Turbo**
5, 7.5, 10, 15, 25 HP TEFC w/ Gearbox **CW** ☐	7.5, 10, 15, 20 HP TEFC w/ Gearbox **CCW** ☐	5, 7.5, 10, 15, 25 HP TEFC w/ Gearbox **CCW** ☐
20 HP TEFC w/ Gearbox **CCW** ☐		20 HP TEFC w/ Gearbox **CW** ☐
30 HP Gearmotor **CCW** ☐		30 HP Gearmotor **CW** ☐
Belt Drive (All) **CW** ☐	Belt Drive (All) **CCW** ☐	Belt Drive (All) **CCW** ☐

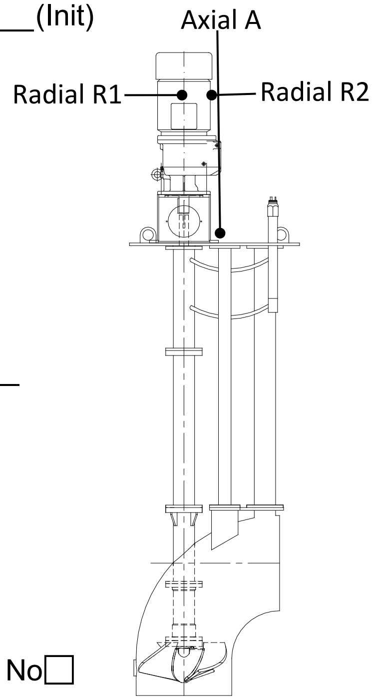
JOB ORDER# ___________________________________

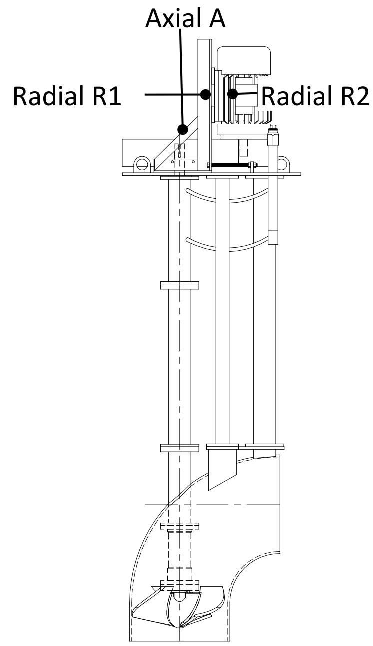
Vibration Measurements _______(Init)

Axial A _______________

Radial R1_____________

Radial R2_____________





Nameplate full load amps______________

Motor Amps

L1 __________

L2 __________

L3 __________

All readings below 1/2 of full load Yes☐ No☐

21. Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)
22. When 20 HP motor with gearbox or 30 HP gearmotor is used, the special nameplate showing motor and pump rotate in opposite directions is installed. **(Turbo only)** ________ (Init)
23. Lifting eyes and correct rotation arrow are installed on deckplate. ________ (Init)
24. Rotating Turbo: Locking mechanism in place prior to shipping? Yes ☐ No ☐ ________ (Init)
25. Decals installed per decal placement drawings ________ (Init)
26. When 100S is installed on floating platform, advise Engineering of weight prior to shipment.

    Weight of completed unit: _________________________ pounds. ________ (Init)

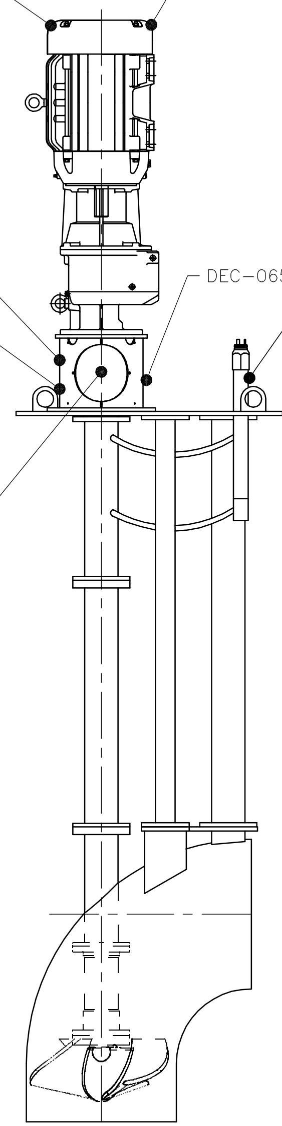
SECTION 5: FINAL SHIPPING STATION

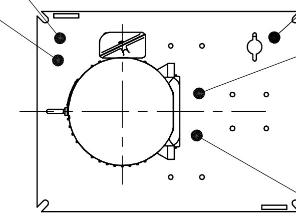
1. Motor/drive matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#__________ (Init)
3. Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)
4. When 20 HP motor with gearbox or 30 HP gearmotor is used, the special nameplate showing motor and pump rotate in opposite directions is installed. **(Turbo only)** ________ (Init)
5. Lifting eyes and correct rotation arrow are installed on deckplate. ________ (Init)
6. Rotating Turbo: Locking mechanism in place prior to shipping? Yes ☐ No ☐ ________ (Init)
7. Decals installed per decal placement drawings ________ (Init)
8. When 100S is installed on floating platform, advise Engineering of weight prior to shipment.

   Weight of completed unit: _________________________ pounds. ________ (Init)
9. The IO&M Manual is attached to the pump at time of shipping. ________ (Init)
10. QA Sheet is complete:

    ____

    (Shipping Clerk) (Date)





DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-011	2" X 4" OIL REQUIREMENTS
DEC-013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-020	.5 X 5.5" COUNTER CLOCKWISE ROTATION
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING SUFFOCATION HAZARD (ENGLISH)
DEC-069SP	2" X 4" WARNING SUFFOCATION HAZARD (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

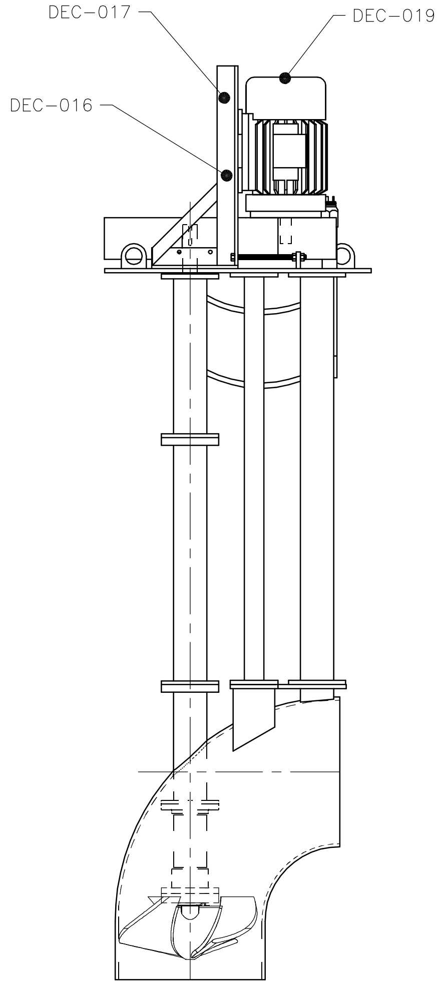
VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

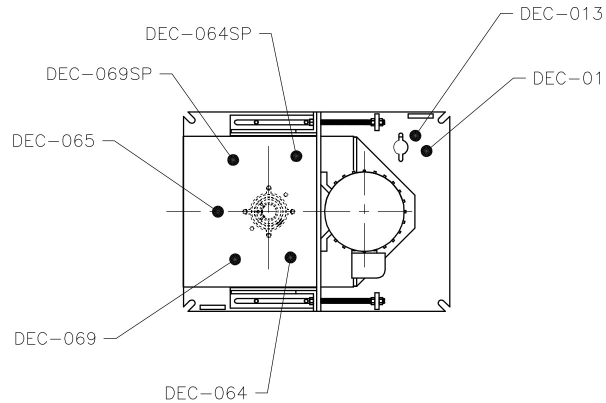
UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: TURBO DIRECT DRIVE DECAL PLACEMENT

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	
JOB ORDER NUMBER:		DATE: 11/16/16	DRAWING NUMBER: 118849	REVISION NUMBER: 0





FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-011	2" X 4" OIL REQUIREMENTS
DEC-013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING SUFFOCATION HAZARD (ENGLISH)
DEC-069SP	2" X 4" WARNING SUFFOCATION HAZARD (SPANISH)

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE:
TURBO-S
DECAL PLACEMENT
BELT DRIVE

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XXX ± 0.020"
DIMENSIONS, MACHINING: X.XXXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	6/12/17	119216	0

PEDESTAL CHOPPER PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ SALES ORDER #____________________

MODEL #____________________________ DRAWING #________________________

SECTION 1: PUMP ASSEMBLY STATION

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
2. Impeller has _____ blades and measures ________ inches ________
3. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ ________
4. Impeller has been balanced. ________
5. If pump is built with Tapered Roller Bearings, shaft endplay is ________inch ________
   3-6": 0.001-0.004", 8-16": 0.001-0.006"
6. Shaft run out at impeller: ____________ **(0.004" max)** ________
7. All nuts and bolts tightened with lock washers. ________
8. Discharge orientation matches Job Order ☐ BOM ☐ Drawing ☐ ________

SECTION 2: All 3-6" & 8L PUMPS

Initial Below

1. The Cutter Nut ☐ or External Tool ☐ is properly torqued to 125 ft/lb. ________
2. Impeller/Cutter bar gap to close vane measures _________ inch. **0.015-0.025"** ________
3. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.005-0.050"** ________
4. Clearance between tip of cutter bar finger and hub tooth. **0.010-0.030"** ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
5. Clearance between tips of external tool and anvil ____________ **0.010–0.030"** ________
6. Axial clearance between external tool wings and cutter bar fingers ________ ________
   0.015–0.025"
7. Cutter Bar inlet diameter is correct per below ________
   3F/3G/3P – 4.25" ☐ **3L/3M/3W – 4.63"** ☐ **3V – 4.0"** ☐ **4K/4L – 5.25"** ☐
   4P/4S – 6.5" ☐ **4R/4T/6U/6W/6X – 7.4"** ☐ **4V – 8.5"** ☐ **8L – 9.25"** ☐

SECTION 3: 8N, 8P, All 10 – 16" PUMPS

Initial Below

1. The impeller is properly torqued to 1000 ft/lb ________
2. The External Tool is properly torqued to 250 ft/lb. ________
3. Impeller/Cutter bar gap to close vane measures _________ inch. ________
   8N, All 10-16": 0.015-0.025" 8P: 0.010-0.020"
4. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.030-0.050"** ________
5. Clearance between tip of cutter bar finger and external tool tooth ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
   8N & 8P: 0.025-0.040", All 10": 0.035-0.050" All 12:&16": 0.040-0.060"
6. Clearance between tips of **external tool** and **anvil**: ______________ inch ________
   8N, 8P: 0.025-0.040" All 10": 0.035-0.050" All 12" &16": 0.040-0.060"
7. Axial clearance between tips of external tool wings and cutter bar fingers ________
8. Cutter Bar inlet diameter is correct per below ________
   8N – 9.0" ☐ **8P – 9.75"** ☐ **10R/10W – 11.0"** ☐ **12W – 13.5"** ☐ **16W – 16.0"** ☐

JOB ORDER#___________________________

SECTION 4: 8K PUMPS

Initial Below

1. The Impeller is properly torqued to 500 ft/lb. ________
2. The Cutter Nose is properly torqued to 40 ft/lb. ________
3. The External Tool (if required) is properly torqued to 125 ft/lb. ________
4. Impeller/Cutter bar gap to close vane measures _________ inch. **0.015-0.025"** ________
5. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.005-0.050"** ________
6. Clearance between tip of cutter bar finger and cutter nose **0.030-0.040"** ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
7. Axial clearance between external tool wings and cutter bar fingers __________inch **0.015–0.025"** ________
8. Clearance between tips of external tool and anvil __________ inch **0.020–0.045** ________
9. Cutter Bar inlet diameter is correct **8K – 8.75"** ☐ ________

SECTION 5: PAINTING STATION

Initial Below

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Paint type:________________________ Required paint thickness:_________mils ________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET

a) underside of pedestal beneath the Vaughan nameplate _____MWFT
b) adjacent to nameplate _____MWFT
c) underside of elbow _____MWFT
d) side of bearing housing adjacent to oil hose _____MWFT
e) casing discharge throat on side nearest bearing housing _____MWFT

Initials _____ Date _____

DRY

a) underside of pedestal beneath the Vaughan nameplate _____MDFT
b) adjacent to nameplate _____MDFT
c) underside of elbow _____MDFT
d) side of bearing housing adjacent to oil hose _____MDFT
e) casing discharge throat on side nearest bearing housing _____MDFT

Initials _____ Date _____

SECTION 6: FINAL ASSEMBLY STATION

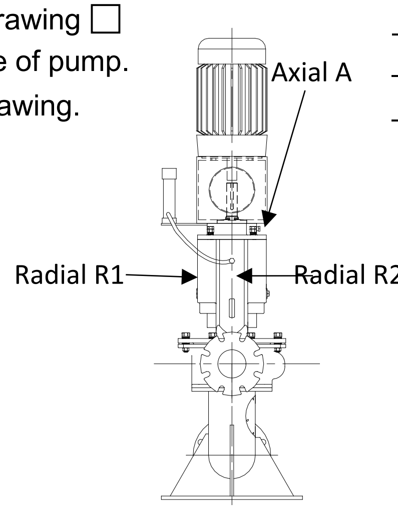
Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ ________
3. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
4. Pump and drive assembly match drawing referenced above ________
5. If a special pedestal base is required does it match the drawing? Yes ☐ No ☐ ________
6. Verify pump shaft turns over by hand. ________
7. Check coupling alignment after motor mounting. ________
8. Oil level in reservoir is 40 – 60% full. ________
9. Hole in vent cap is drilled through and intersects roof. *(use 0.156 drill bit to check)* ________

JOB ORDER#___________________________

Initial Below

10. Motor: Manufacturer___________________________ ________
    Spec/Model/Catalog Number ________________________ Speed__________ RPM
    Serial No. ______________________________ Frame Size ____________
11. Motor greased with ___________________________ Not Greasable ☐ ________
    Note: Check motor manual for grease type when not listed on nameplate.
12. Verify all guards are in place. ________
13. Companion flanges (with gasket) supplied with export pumps? ________
14. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
15. If required the Rotamix trademark has been installed on base of pump. ________
16. Decals are installed on the pump per the decal placement drawing. ________



Vibration Measurements

Motor amps measurement___________

Nameplate full load amps ___________

All readings are below half of full load Yes ☐ No ☐

SECTION 7: FINAL SHIPPING STATION

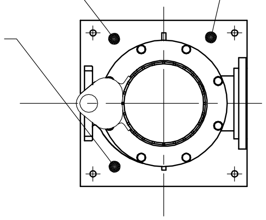
Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ ________
3. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
4. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
5. Decals are installed on the pump per the decal placement drawing. ________
6. IO&M Manual is attached to the pump at the time of shipping ________
7. QA Sheet is complete:

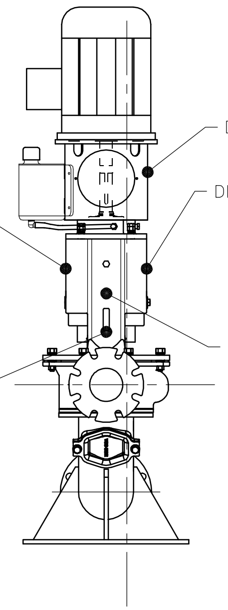
   ____
   (Shipping Clerk) (Date)

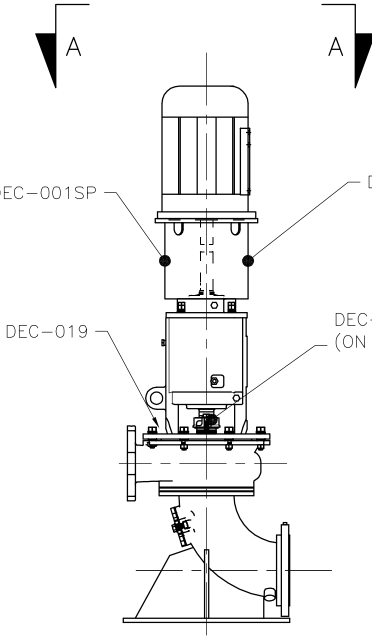
REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
7/13/16	1	RENAMED DRAWING & ADDED NOTE & DEC-032 FOR NAMEPLATE	KCW
9/14/16	2	ADDED FOREIGN PUMP TABLE	KCW
10/17/16	3	UPDAATED FOREIGN PUMP TABLE	KCW



VIEW A-A





DOMESTIC PUMPS

* DEC-070 NOT USED IF PUMP HAS NO CLEANOUT

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-032	NAME PLATE
DEC-061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC-070	4" X 6" WARNING (ENGLISH)
*DEC-070SP	4" X 6" WARNING (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE: PE/PSC DECAL PLACEMENT

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XX ± 0.020"
DIMENSIONS, MACHINING: X.XXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	3/3/16	118396	3

DMA QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ **SALES ORDER # ___________________**

MODEL #_____________________________ **DRAWING #________________________**

SECTION 1: WELD/ASSEMBLY STATION

1. Mixer length is __________ _________(Init)
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#______Rev#__ _________(Init)
3. Lift eyes are installed on deckplate _________(Init)
4. Zinc anode is installed _________(Init)
5. All nuts and bolts are tightened with lock washers. _________(Init)

SECTION 2: PAINT STATION

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ _________(Init)
2. Zinc anode is masked and unpainted. _________(Init)
3. Paint type:________________ Required paint thickness:_________mils _________(Init)

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of deckplate _____MWFT
b) top of deckplate _____MWFT
c) midway on pipe _____MWFT
d) top of elbow _____MWFT
e) bottom of elbow _____MWFT
Initials_____ Date _____

DRY
a) underside of deckplate _____MDFT
b) top of deckplate _____MDFT
c) midway on pipe _____MDFT
d) top of elbow _____MDFT
e) bottom of elbow _____MDFT
Initials_____ Date _____

JOB ORDER# ___________________________

SECTION 3: FINAL ASSEMBLY

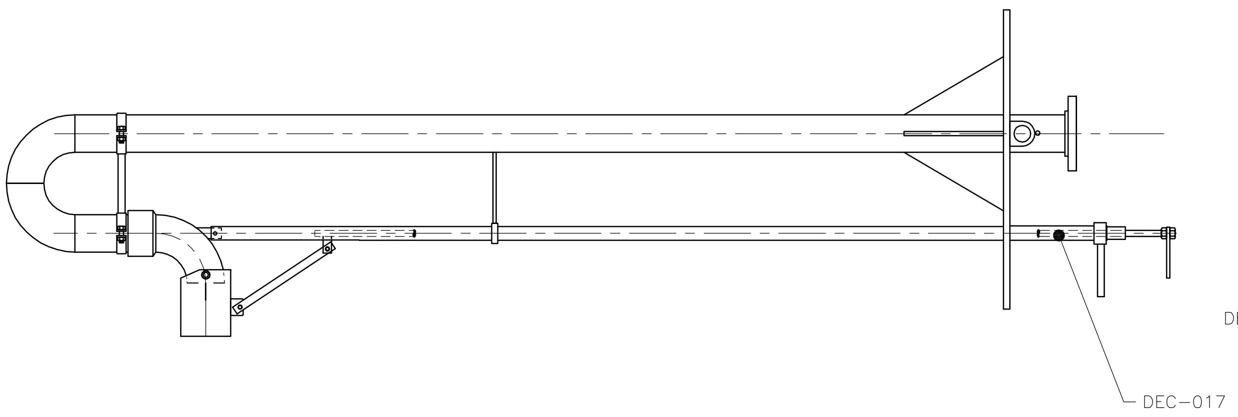
1. Unit matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#______Rev#__ _______(Init)
2. Nameplate matches Job Order ☐ BOM ☐ Drawing ☐ __________(Init)
3. Zinc anode is installed and unpainted. __________(Init)
4. Decals are placed per the decal placement drawing. __________(Init)

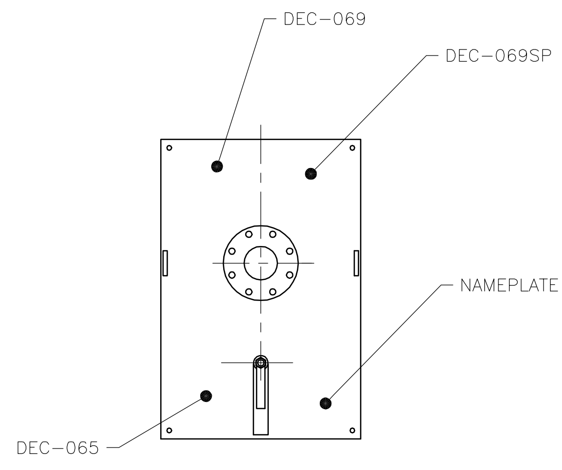
SECTION 4: FINAL SHIPPING STATION

1. Unit matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#______Rev#__ _______(Init)
2. Nameplate matches Job Order ☐ BOM ☐ Drawing ☐ __________(Init)
3. Decals are placed per the decal placement drawing. __________(Init)
4. QA Sheet is complete:

____

(Shipping Clerk) (Date)





DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING SUFFOCATION HAZARD (ENGLISH)
DEC-069SP	2" X 4" WARNING SUFFOCATION HAZARD (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XX ± 0.020"
DIMENSIONS, MACHINING: X.XXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE:	DMA DECAL PLACEMENT
CHECKED:	JK
ENGINEER:	
JOB ORDER NUMBER:	
DRAWN BY:	KCW
DATE:	6/1/17
SCALE:	1"=1'
DRAWING NUMBER:	119188
PART NUMBER:	
REVISION NUMBER:	0

ROTAMIX QUALITY ASSURANCE VERIFICATION

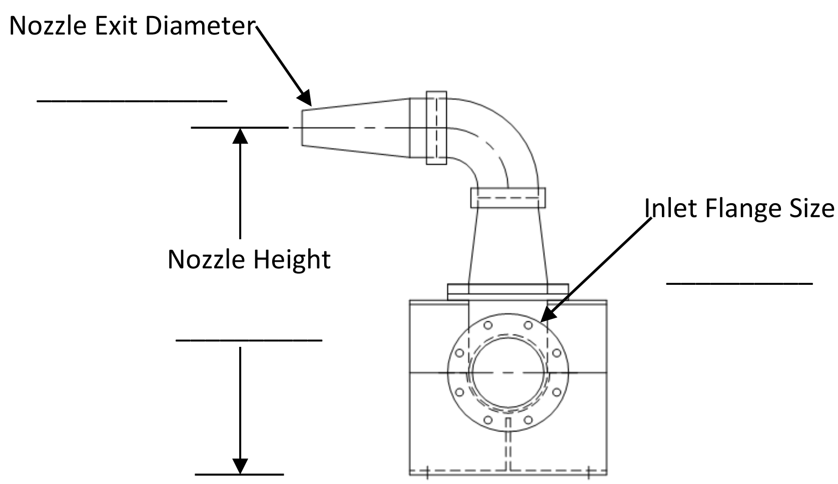
JOB ORDER# ________________________

MODEL# ____________________________ **DRAWING#** ________________________

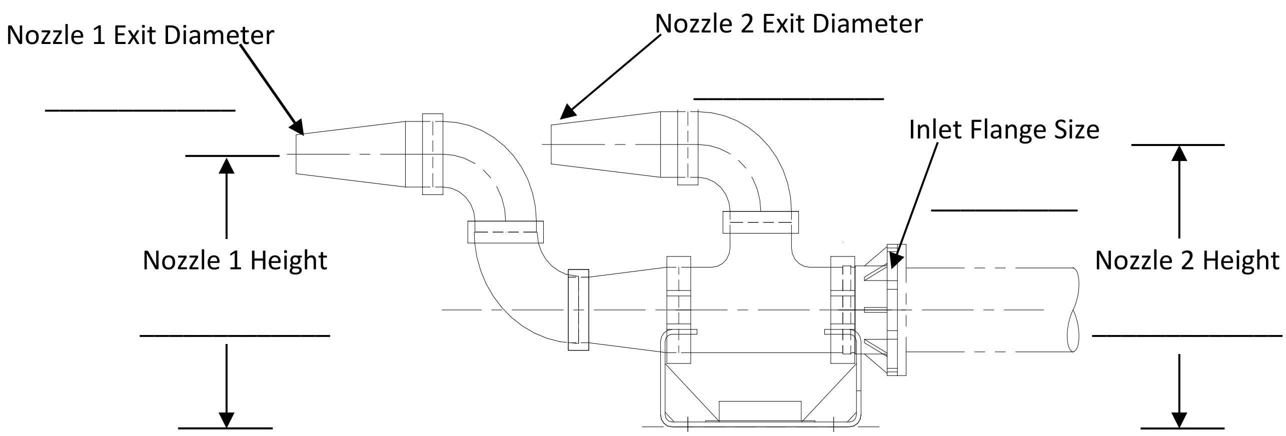
SECTION 1: ASSEMBLY STATION:

1. Nozzle material: Polyurethane ☐ Ductile Iron ☐ ________(Init)
2. Number of nozzles ________ ________(Init)
3. Assembly is Floor Mount ☐ Pipe Mount ☐ Wall Mount ☐ ________(Init)
4. Nozzle one (1) discharge angle 0° ☐ 22.5° ☐ ________(Init)
5. Nozzle two (2) discharge angle 0° ☐ 22.5° ☐ ________(Init)

Single Nozzle Assembly

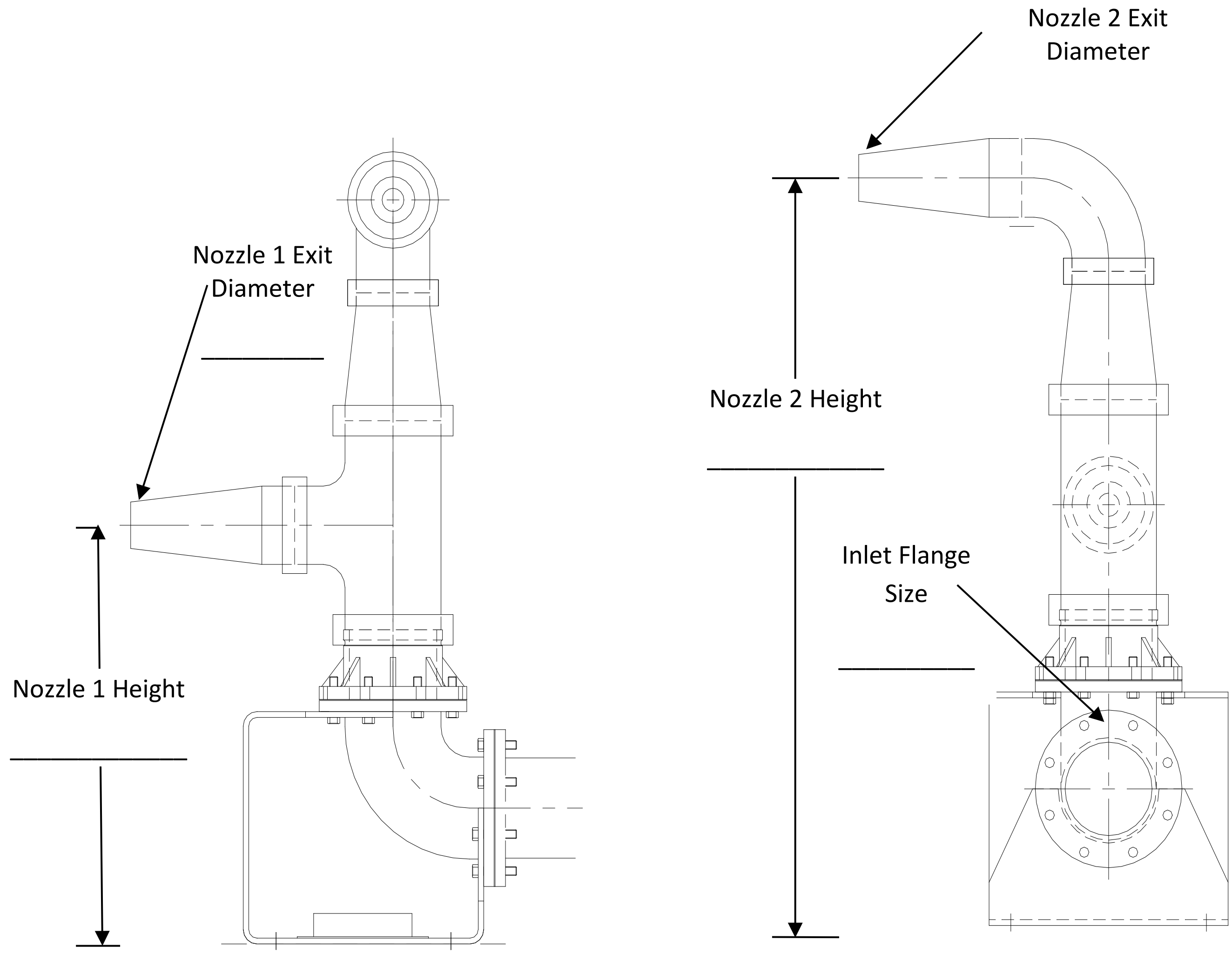


XLP Low Profile Dual Nozzle Assembly



JOB ORDER# ____________________________

DUAL NOZZLE ASSEMBLY



SECTION 2: FINAL ASSEMBLY

1. Assembly matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_____Rev#_________ (Init)
2. All white circles have been removed from glass lining in all nozzles. _________ (Init)
3. Zinc anode is installed. _________ (Init)
4. Rotamix stickers have been applied. _________ (Init)

SECTION 3: FINAL SHIPPING STATION

1. Assembly matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_____Rev#_________ (Init)
2. Rotamix stickers have been applied. _________ (Init)
3. Nozzle matches drawing. Engineer only: _________ (Init)
4. The Rotamix IO&M manual is attached at time of shipping. _________ (Init)
5. QA Sheet is complete:

____

(Shipping Clerk) (Date)

SUBMERSIBLE SCREW PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ SALES ORDER # ____________________

MODEL #______________________________ DRAWING #________________________

SECTION 1: PUMP ASSEMBLY STATION

Initial Below

1) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ _________
2) Impeller model matches Job Order ☐ BOM ☐ Drawing ☐ _________
3) Impeller has been balanced _________
4) Shaft endplay is measured at ____________ _________
   (3DS, 4DH, 6EM, 6ES: 0.001 – 0.004”) (8F, 8x8B, 10D, 12D: 0.001 – 0.006”)
5) Shaft run out at impeller __________ _________
6) Impeller bolt is properly torqued with Loctite 242. _________
   3DS, 4DH: 30# ☐ **6EM, 6ES: 40#** ☐ **8F, 10D: 80#** ☐ **8x8B, 12D: 100#** ☐
7) Hub Cutter/Back Cutter Clearance: ________ inch **(0.010 - 0.020" acceptable)** _________
8) Impeller/Suction Cone gap measures _________ inch **(0.015 - 0.025” acceptable)** _________
9) Bearing housing has been filled 85% full with oil. _________
10) All nuts and bolts tightened with lock washers _________
11) Adapter bracket (if required) is attached to the pump discharge. _________
    Note: If pump requires testing do not glue bracket.
12) Pump was run at assembly station? Yes ☐ No ☐ _________
13) Any rubbing? Yes ☐ No ☐ _________
14) Form V684 motor receiving sheet is attached to the production packet. _________
15) Cap pulled to verify voltage matches Production Traveler. _________

SECTION 2: PAINTING STATION

Initial Below

1) Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ _________
2) Paint type:________________________ Required paint thickness:_________mils _________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

JOB ORDER# ________________________

WET

a) bearing housing, above upper plug _____MWFT
b) top of casing, opposite of discharge flange _____MWFT
c) bottom of casing, opposite of discharge flange _____MWFT
d) top of casing at the discharge throat _____MWFT
e) bottom of cone _____MWFT
f) elbow, under flange _____MWFT

Initials _____ Date _____

DRY

a) bearing housing, above upper plug _____MDFT
b) top of casing, opposite of discharge flange _____MDFT
c) bottom of casing, opposite of discharge flange _____MDFT
d) top of casing at the discharge throat _____MDFT
e) bottom of cone _____MDFT
f) elbow, under flange _____MDFT

Initials _____ Date _____

SECTION 3: FINAL ASSEMBLY STATION

Initial Below

1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
3) Verify pump shaft turns over by hand. ________
4) If required is VPMR supplied? ________
5) Oil Level Monitor ☐ Female Plug ☐ Relay ☐ Hose ☐ Reservoir ☐ are supplied. ________
6) If spark proof bracket is OLM relay intrinsically safe? ________
7) Hole in OLM vent cap is drilled through and intersects roof? (Use 0.156” drill bit to check) ________
8) Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
9) If required is baffle ring installed? Yes ☐ No ☐ ________

   Diameter of baffle ring ______________
10) Decals are installed per the decal placement drawing 118384 ________
11) Blind flanges have been installed on both suction & discharge before vibration testing. ________

Vibration Measurements

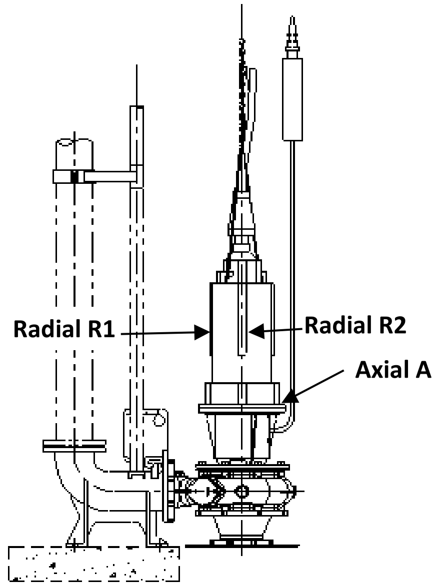
Axial A ________________

Radial R1 ______________

Radial R2 ______________

NOTE: Maximum speed for dry run vibration test of 12” pumps is 30 Hz.

JOB ORDER# ___________________________



Nameplate full load amps __________

Motor Amp Measurement: __________

Initial Below

Reading is below half of full load Yes ☐ No ☐ ________

SECTION 4: FINAL SHIPPING STATION

Initial Below

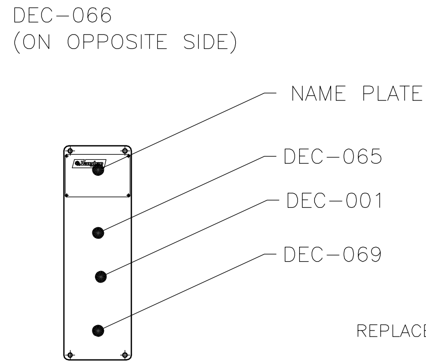
1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
3) Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
4) Decals are installed per the decal placement drawing 118384 ________
5) The IO&M Manual is attached to the pump at the time of shipping. ________
6) Form V684 motor receiving sheet is included in the production packet ________
7) QA Sheet is complete:

____

(Shipping Clerk) (Date)

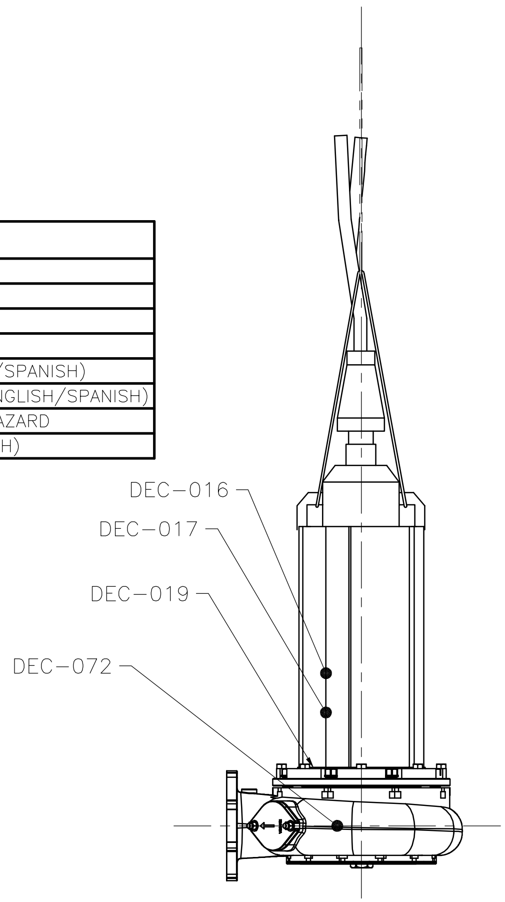
REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	2	UPDAATED FOREIGN PUMP TABLE	KCW



DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-066	4" X 10" ATTENTION INSTALLER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING, SUFFOCATION HAZARD
DEC-072	2" X 6" DANGER, (ENGLISH/SPANISH)



FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

Vaughan

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE: SE SERIES SUBMERSBLE PUMP DECAL PLACEMENT

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: XXX ± 1/16"
DIMENSIONS, MACHINING: XXX ± 0.020"
DIMENSIONS, MACHINING: XXXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	2/23/16	118384	2

S SERIES SUBMERSIBLE PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER #________________________ SALES ORDER # ____________________

MODEL #____________________________ DRAWING #__________________________

SECTION 1: PUMP ASSEMBLY STATION: ALL PUMPS

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
3. Impeller has _____ blades and diameter measures _________ inches ________
4. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ ________
5. Impeller has been balanced. ________
6. When assembled with tapered roller bearings shaft endplay is measured at________
   3-6”: 0.001- 0.004” 8-16”: 0.001-0.006” ________
7. Form V684 motor receiving sheet is attached to the production packet. ________
8. Cap pulled to verify voltage matches Production Traveler. ________

SECTION 2: PUMP ASSEMBLY STATION: All 3”- 6” & 8L

Initial Below

1. The Cutter Nut ☐ or External Tool ☐ is properly torqued to **125 ft/lb** ________
2. Impeller/Cutter bar gap to close vane measures _________ **0.015”- 0.025”** ________
3. Clearance between tip of cutterbar finger and hub tooth **0.010”– 0.030”** ________
   Measurements for 2 locations: __________ and _________
4. Axial clearance between external tool wings and cutter bar fingers ________ **0.015-0.025”** _____
5. Clearance between tips of external tool and anvil __________ **0.010–0.030”** ________
6. If pump uses cutterbar shims, bolts are torqued to **45 ft/lb**. ________
7. Cutter bar inlet diameter is correct per below ________
   3F/3G/3P: 4.25” ☐ **3L/3M: 4.63”** ☐ **3V/3W: 4.0”** ☐ **4K/4L: 5.13”** ☐
   4S: 6.0” ☐ **4T/6U/6X: 7.25”** ☐ **4V: 8.50”** ☐ **8L: 9.25”** ☐

SECTION 3: PUMP ASSEMBLY STATION: 8N, 8P, 8S, All 10 – 16”

Initial Below

1. The External Tool is properly torqued to ________
   180TY-320TY 125 ft/lb. ☐ **360TY tool 150 ft/lb.** ☐ **440TY tool 250 ft/lb.** ☐
2. Impeller/Cutter bar gap to close vane measures _________ inch **0.015 - 0.025”** ________
3. Clearance between tip of Cutter Bar finger and External Tool tooth.
   Measurements for 2 locations: __________ and _________ ________
   8”: 0.025”- 0.040” 10”: 0.035”- 0.050” 12-16”: 0.040”- 0.060”
4. Axial clearance between tip of external tool wings & cutter bar finger __________ ________
   0.015”- 0.025”
5. Clearance between tips of external tool and anvil: ____________ ________
   8”: 0.025”- 0.040” 10”: 0.035”- 0.050” 12-16”: 0.040”- 0.060”
6. If pump uses cutterbar shims, bolts are torqued to **95 ft/lb.** ________
7. Cutter bar inlet diameter is correct per below ________

8N: 9.0” ☐ **8P: 9.75”** ☐ **8S: 11.5”** ☐ **10R/10W: 11.0”** ☐ **12W: 13.50”** ☐ **16W: 16.0”** ☐

SECTION 4: PUMP ASSEMBLY STATION: 8K

Initial Below

1. The External Tool is properly torqued to **125 ft/lb** ________
2. Impeller/Cutter bar gap to close vane measures _________ inch. **0.015”- 0.025”** ________
3. Clearance between tip of cutterbar finger and external tool hub**.** **0.025”- 0.040”**
   Measurements for 2 locations: ___________ and ___________ ________
4. Axial clearance between tips of external tool wings & cutter bar fingers______**0.015”-0.025”** _____
5. Clearance between tips of external tool and cutter bar: __________ **0.020”- 0.045”** ________

S SERIES SUBMERSIBLE PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER #____________________

Initial Below

6. Cutter bar inlet diameter is correct. **8K: 8.75”** ☐ ________

SECTION 5: PUMP ASSEMBLY STATION: All PUMPS

Initial Below

1. Bearing housing has been filled 85% full with oil. ________
2. All nuts and bolts tightened with lock washers? ________
3. Motor Cable Quick Disconnects: If applicable, cables fully inserted, gland nuts torqued per spec. located on cable tags, and set screws tightened? ________
4. Adapter bracket (if reqd) is attached to the pump discharge. ________
   Note: If pump requires testing do not glue bracket.
5. Pump was run at assembly station? Yes ☐ No ☐ ________
6. Any rubbing? Yes ☐ No ☐ ________

SECTION 6: PAINTING STATION

Initial Below

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Paint type:________________________ Required paint thickness:_________mils ________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) bearing housing, above upper plug _____MWFT
b) top of casing, opposite of discharge flange _____MWFT
c) bottom of casing, opposite of discharge flange _____MWFT
d) top of casing at the discharge throat _____MWFT
e) bottom of cutter bar flange _____MWFT
f) elbow, under flange _____MWFT
Initials _____ Date _____

DRY
a) bearing housing, above upper plug _____MDFT
b) top of casing, opposite of discharge flange _____MDFT
c) bottom of casing, opposite of discharge flange _____MDFT
d) top of casing at the discharge throat _____MDFT
e) bottom of cutter bar flange _____MDFT
f) elbow, under flange _____MDFT
Initials _____ Date _____

SECTION 7: FINAL ASSEMBLY STATION

Initial Below

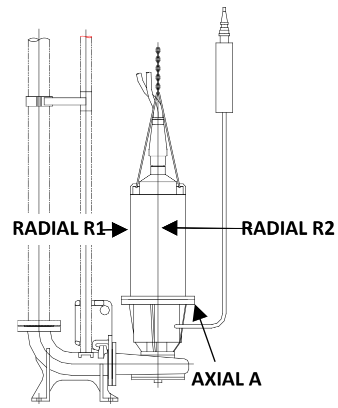
1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
3. Verify the pump turns over by hand. ________
4. Verify correct rotation of pump. ________
5. If required is VPMR supplied? ________
6. Automatic Oil Monitor checked for proper operation. ________
7. Oil Level Monitor ☐ Female Plug ☐ Relay ☐ Hose ☐ Reservoir ☐ are supplied. ________

S SERIES SUBMERSIBLE PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER #___________________ **Initial Below**

8. If spark proof bracket is OLM relay intrinsically safe? ________
9. Hole in vent cap is drilled through and intersects roof? (Use 0.156" drill bit to check) ________
10. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
11. Decals are installed per the decal placement drawing 118383. ________

Vibration Measurements:
Axial A ____________________
Radial R1 __________________
Radial R2 __________________



Motor Amp Measurements: ________________Amps

Nameplate Full Load Amps ______________

Reading is below half of full load? Yes ☐ No ☐

FOR SUBMERSIBLE RECIRCULATORS

Initial Below

12. Form V838 Sub Recirculator Supplement QA is included in the production packet. ________
13. Goes with Sales Order # _______________________ ________

SECTION 8: FINAL SHIPPING STATION

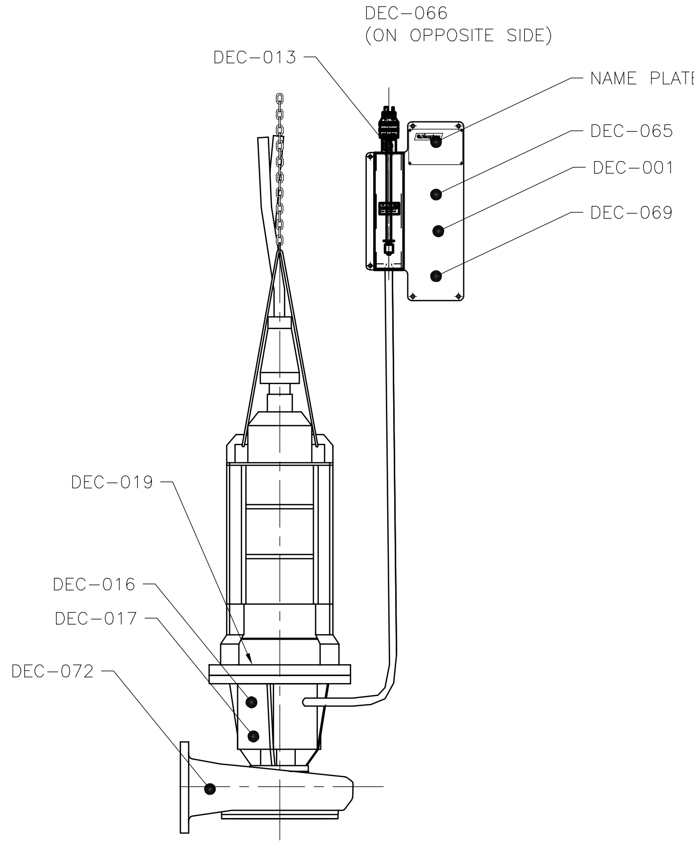
Initial Below

1. Motor or drive matches: Job Order ☐ BOM ☐ Drawing ☐ ________
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
3. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
4. Decals are installed per the decal placement drawing 118383. ________
5. Form V684 motor receiving sheet is included in the production packet. ________
6. The IO&M Manual is attached to the pump at the time of shipping. ________
7. QA Sheet is complete:

____
(Shipping Clerk) (Date)

REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	2	UPDAATED FOREIGN PUMP TABLE	KCW



DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-066	4" X 10" ATTENTION INSTALLER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING, SUFFOCATION HAZARD
DEC-072	2" X 6" DANGER, (ENGLISH/SPANISH)

MOTOR REPAIR TAGS		
TAG#	PUMP SIZE	FASTEN TAG TO PUMP WITH CUTTER BAR OR SUCTION CONE BOLT
V115-105	ALL 3"-6"	
V118-392	ALL 8" & 10"	
V118-387	ALL 12" & 16"	

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XX ± 0.020"
DIMENSIONS, MACHINING: X.XXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: S SERIES SUBMERSBLE PUMP DECAL PLACEMENT CHOPPER & SCREW PUMPS

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	2/22/16	118383	2

SUBMERSIBLE RECIRCULATOR SUPPLEMENT QA VERIFICATION

To be used in addition to V321/V531

SALES ORDER # ______________

JOB ORDER#____________ MODEL#____________ DRAWING#______________

SECTION 1: FINAL ASSEMBLY STATION

Initial Below

1. Recirculator matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_______Rev#____ _____
2. Handle Length ______________________________ _____
3. Valve actuator system operated and functions properly. _____
4. Decals are installed per the decal placement drawing 118501 / 118502. _____
5. Operating voltage of Auto Valve Actuator matches: Job Order ☐ BOM ☐ Drawing ☐ _____
6. Confirm left hand or right hand system matches the BOM. _____

SECTION 2: PAINTING STATION

Initial Below

1. Paint applied matches: Job Order ☐ BOM ☐ Drawing ☐ _____
2. Paint type:________________________ Required paint thickness:_________mils _____

Dry Film Thickness				
Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) Deckplate, bottom center _____MWFT
b) Valve handle, middle _____MWFT
c) Valve body, underside _____MWFT
Initials _____ Date _____

DRY
a) Deckplate, bottom center _____MDFT
b) Valve handle, middle _____MDFT
c) Valve body, underside _____MDFT
Initials _____ Date _____

SECTION 3: FINAL SHIPPING STATION

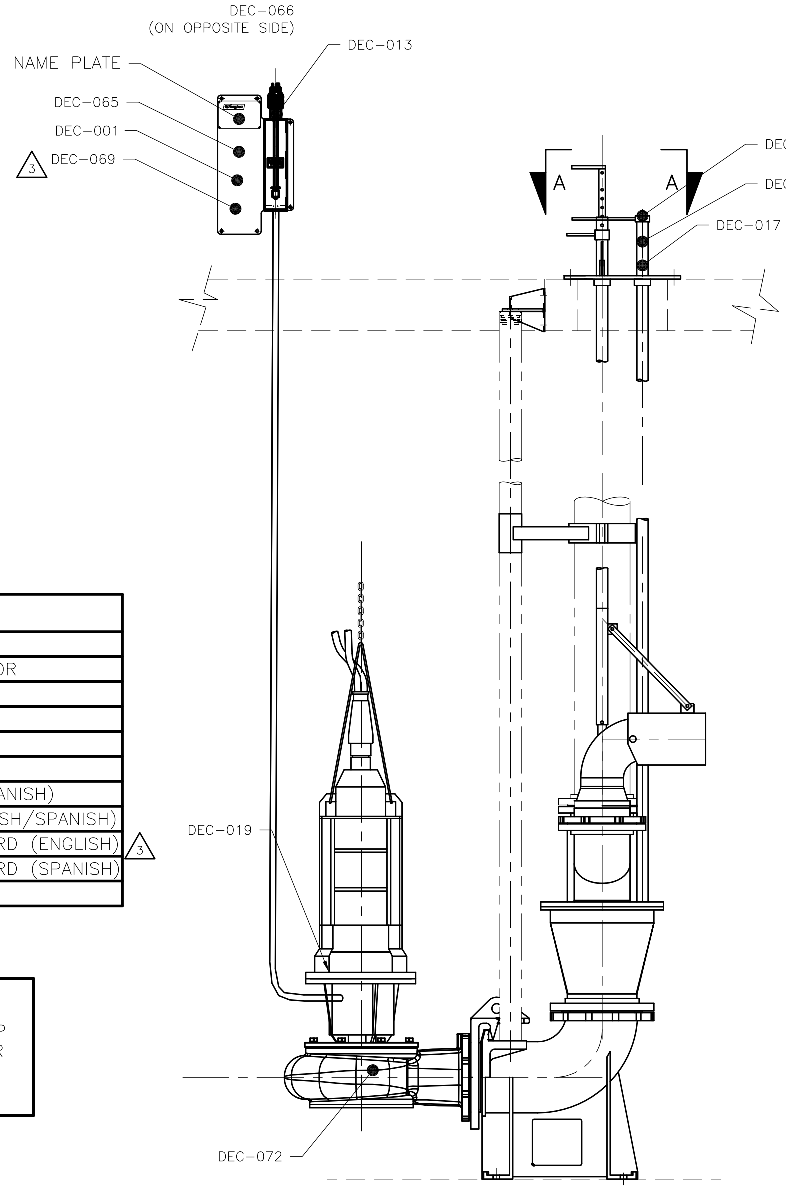
Initial Below

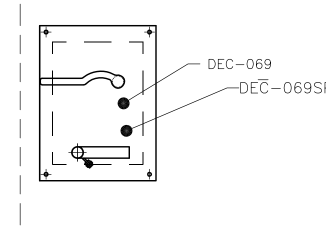
1. Form V321 / V531 is included in the production packet. _____
2. Outline dimension drawing # __________ matches pump assembly. Engineer only: _____
   (Mark N/A if no outline dimension drawing; if N/A, no Eng initials required)
3. Decals are installed per the decal placement drawing 118501 / 118502. _____
4. Supplement QA Sheet is complete:

____
(Shipping Clerk) (Date)

REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	2	UPDAATED FOREIGN PUMP TABLE	KCW
1/3/24	3	ADDED SECOND DEC-069 DECAL & PLACEMENT; ECN: 5471	EAB





VIEW A-A

DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC-015	1.75" DIAMETER, PUMPOUT/AGITATE
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-066	4" X 10" ATTENTION INSTALLER (ENGLISH/SPANISH)
(2X) DEC-069	2" X 4" WARNING, SUFFOCATION HAZARD (ENGLISH) 3
DEC-069SP	2" X 4" WARNING, SUFFOCATION HAZARD (SPANISH)
DEC-072	2" X 6" DANGER, (ENGLISH/SPANISH)

MOTOR REPAIR TAGS		
TAG#	PUMP SIZE	FASTEN TAG TO PUMP WITH CUTTER BAR OR SUCTION CONE BOLT
V115-105	ALL 3"-6"	
V118-392	ALL 8" & 10"	
V118-387	ALL 12" & 16"	

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

Vaughan

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE:
S SERIES SUBMERSBLE RECIRC.
DECAL PLACEMENT
CHOPPER & SCREW PUMPS

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	
JOB ORDER NUMBER:		DATE:	DRAWING NUMBER:	REVISION NUMBER:
		5/5/16	118501	3

TRITON VERTICAL WETWELL QUALITY ASSURANCE VERIFICATION

JOB ORDER #________________________ **SALES ORDER #** __________________

MODEL #____________________________ **DRAWING #** ______________________

SECTION 1: PUMP ASSEMBLY STATION:

Initial Below

1) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ _________
2) Impeller model matches Job Order ☐ BOM ☐ Drawing ☐ _________
3) Impeller has been balanced? _________
4) Pump length is_______feet. Pump length matches Job Order☐ BOM☐ Drawing☐ _________
5) Is a custom deck plate required? Yes☐ No☐ _________
6) Deckplate matches the drawing? Yes☐ No☐ _________
7) Shaft endplay is measured at ________ _________
   (3DS, 4DH, 6EM, 6ES: 0.001 – 0.004”) (8F, 8x8B, 10D, 12D: 0.001 – 0.006”)
8) Extension shaft runout measures__________________ _________
9) Measurement from top stem flange to end of shaft ___________ _________
   (3DS, 4DH, 6EM, 6ES: 5.75” +/- 0.25”) (8F, 8x8B, 10D, 12D: 7.75” +/- 0.25”)
10) Impeller bolt is properly torqued with Loctite 243. _________
    3DS, 4DH: 30# ☐ 6EM, 6ES: 40# ☐ 8F, 10D: 80# ☐ 8x8B, 12D: 100# ☐
11) Hub Cutter/Back Cutter Clearance: __________inch **(0.010 - 0.020" acceptable)** _________
12) Impeller/Suction Cone gap measures __________ inch **(0.015 - 0.025” acceptable)** _________
13) Number of suction cone shims used. Metal __________ Paper __________ _________
14) All nuts and bolts tightened with lock washers? _________

SECTION 2: WELDING STATION

Initial Below

1) Does pump match drawing? _________
2) Below deck discharge (if reqd) is in the correct position as shown on the drawing? _________
3) The companion flange for export pumps is supplied. Yes ☐ No ☐ _________
4) The blind flange if required is installed. _________
5) Gauge port is located opposite side from oil monitor. _________
6) Set collar is tack welded to the nozzle extension handle below the deckplate. _________
7) Oil ports in top stem are clocked toward the oil monitor. _________

SECTION 3: PAINTING STATION

Initial Below

1) Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ _________
2) Paint type:________________________ Required paint thickness:_________mils _________

JOB ORDER# ___________________________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of deckplate between stem and discharge _____MWFT
b) top of deckplate adjacent to oil monitor _____MWFT
c) midway on bearing column _____MWFT
d) top of casing at the discharge throat _____MWFT
e) bottom of cutter bar flange _____MWFT
Initials _____ Date _____

DRY
a) underside of deckplate between stem and discharge _____MDFT
b) top of deckplate adjacent to oil monitor _____MDFT
c) midway on bearing column _____MDFT
d) top of casing at the discharge throat _____MDFT
e) bottom of cutter bar flange _____MDFT
Initials _____ Date _____

SECTION 4: FINAL ASSEMBLY STATION

Initial Below

1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ _________
2) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ _________
3) Pump and drive assembly match drawing referenced above? _________
4) Verify that the pump shaft turns by hand. _________
5) Oil is filled to the proper level. _________
6) Check coupling alignment after motor mounting. (direct drive) _________
7) Check sheaves for proper alignment. (belt drive) _________
8) If customer supplied motor, are Motor Plates ☐ Bushings (both) ☐ Sheaves ☐ Belts ☐ Brackets ☐ Tensioners ☐ Guards ☐ Pump Driveshaft Key ☐ Mounting Hardware ☐supplied? _________
9) Customer control voltage for oil level relay is ___________ Volts _________
10) Is Relay ☐ Base ☐ and Female OLM connector ☐ supplied with pump? _________
11) Hole in vent cap is drilled through and intersects roof? *(Use 0.156" drill bit to check)* _________
12) Are all guards in place? _________
13) Lift eyes and rotation arrow are installed on deck plate. _________
14) Motor Manufacturer _________

    Spec/Model/Catalog#__________________________

    Serial Number _______________________________ Frame Size __________

    What is the motor HP? ___________ Speed? ___________RPM _________

JOB ORDER# ______________________________

Initial Below

15) Motor greased with: __________________________________ Not Greasable ☐ _________

Note: Check motor manual for grease type when not listed on nameplate.

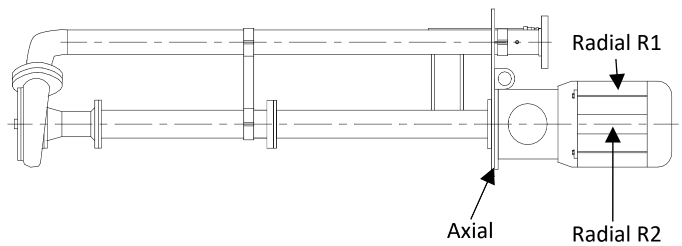
16) Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ _________

17) Are the tags and nameplates installed on the pump per decal placement drawing _________

18) If required is baffle ring installed? Yes ☐ No ☐ _________

Diameter of baffle ring _______________

Vibration Readings _____________(Init)



Axial_________________

Radial R1_____________

Radial R2_____________

Nameplate full load amps____________

Motor Amp Measurement ___________

Reading is below 1/2 of full load Yes ☐ No ☐

Verify pump speed with tachometer:___________RPM (Belt drive pumps) _________

SECTION 5: FINAL SHIPPING STATION

Initial Below

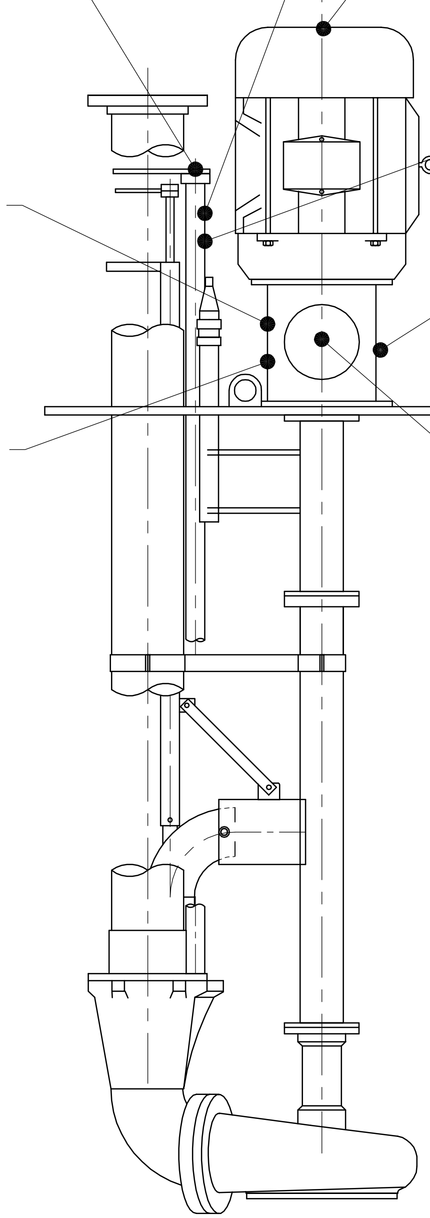
1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ _________
2) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#__________Rev#_____ _________
3) Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ _________
4) Are the tags and nameplates installed on the pump per decal placement drawing _________
5) Is the IO&M Manual attached to the pump at the time of shipping? _________
6) QA Sheet is complete:

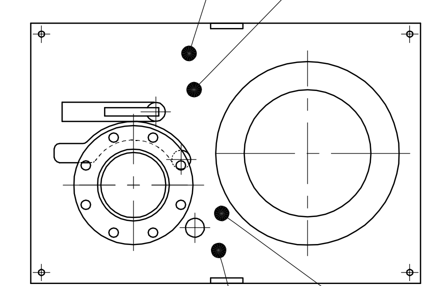
____

(Shipping Clerk) (Date)

REVISION LIST

DATE:	REV #:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	2	UPDAATED FOREIGN PUMP TABLE	KCW





DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-011	2" X 4" OIL REQUIREMENTS
DEC-013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC-015	1.75" DIA., PUMP OUT/AGITATE
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING SUFFOCATION HAZARD (ENGLISH)
DEC-069SP	2" X 4" WARNING SUFFOCATION HAZARD (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

Vaughan

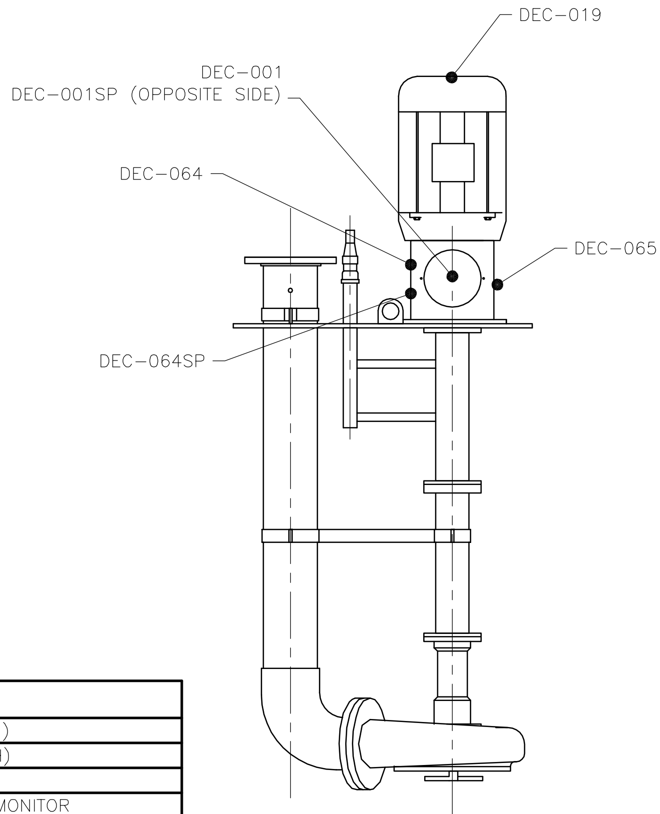
VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

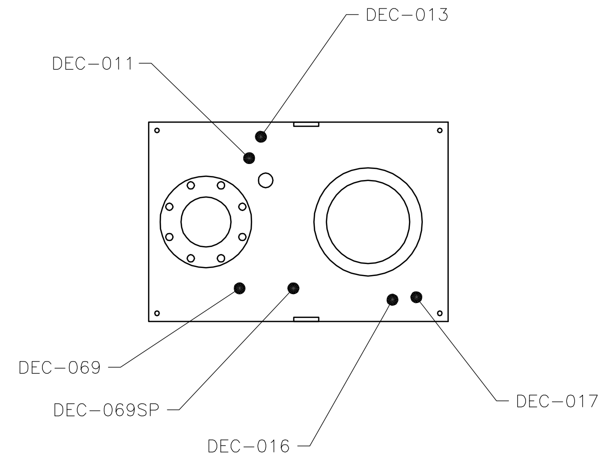
TITLE:
VERTICAL WET-WELL RECIRC. DECAL PLACEMENT CHOPPER & SCREW PUMPS

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	
JOB ORDER NUMBER:		DATE: 3/21/16	DRAWING NUMBER: 118417	REVISION NUMBER: 2

REVISION LIST			
DATE:	REV#:	DESCRIPTION:	BY:
9/6/16	1	ADDED FOREIGN PUMP TABLE	BSS
10/14/16	2	UPDAATED FOREIGN PUMP TABLE	KCW





FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-011	2" X 4" OIL REQUIREMENTS
DEC-013	1.5" X 2.375" AUTO OIL LEVEL MONITOR
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-069	2" X 4" WARNING SUFFOCATION HAZARD (ENGLISH)
DEC-069SP	2" X 4" WARNING SUFFOCATION HAZARD (SPANISH)

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XX ± 0.020"
DIMENSIONS, MACHINING: X.XXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE:
VERTICAL WET-WELL DECAL PLACEMENT CHOPPER & SCREW PUMPS

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	3/17/16	118414	2

HORIZONTAL SCREW PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ SALES ORDER # ____________________

MODEL #_____________________________ DRAWING #________________________

SECTION 1: PUMP ASSEMBLY STATION

1) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#___ ________ (Init)

2) Impeller model matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

3) Impeller has been balanced ________ (Init)

4) Inboard and outboard bearing isolators have been installed with drain back at 6:00. ________ (Init)

5) Inboard oil containment cup is on shaft. ________ (Init)

6) Shaft endplay is measured at ________ ________ (Init)
   (3DS, 4DH, 6EM, 6ES: 0.001 – 0.004") **(8F, 8x8B, 10D, 12D: 0.001 – 0.006")**

7) Shaft run out at impeller __________ ________ (Init)

8) Impeller bolt is properly torqued with Loctite 242. ________ (Init)
   3DS, 4DH: 30# ☐ **6EM, 6ES: 40#** ☐ **8F, 10D: 80#** ☐ **8x8B, 12D: 100#** ☐

9) Hub Cutter/Back Cutter Clearance: ________ inch **(0.010 - 0.020" acceptable)** ________ (Init)

10) Impeller/Suction Cone gap measures _________ inch **(0.015 - 0.025" acceptable)** ________ (Init)

11) All nuts and bolts tightened with lock washers ________ (Init)

12) Discharge orientation matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

13) Vent port is present and plugged in casing Yes ☐ No ☐ ________ (Init)

14) Drain port is present and plugged in casing Yes ☐ No ☐ ________ (Init)

SECTION 2: PAINTING STATION

1) Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

2) Paint type:________________________ Required paint thickness:_________mils ________ (Init)

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MWFT
b) top of baseplate between casing and bearing housing _____MWFT
c) side of suction manifold _____MWFT
d) side of bearing housing adjacent to oil site glass _____MWFT
e) casing discharge throat on side nearest bearing housing _____MWFT

Initials _____ Date _____

JOB ORDER# ___________________________

DRY
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MDFT
b) top of baseplate between casing and bearing housing _____MDFT
c) side of suction manifold _____MDFT
d) side of bearing housing adjacent to oil site glass _____MDFT
e) casing discharge throat on side nearest bearing housing _____MDFT

Initials _____ Date _____

SECTION 3: FINAL ASSEMBLY STATION

1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

2) Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

3) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#___ ________ (Init)

4) Pump and drive assembly match drawing referenced above ☐ ________ (Init)

5) If a special base is required does it match the drawing? Yes ☐ No ☐ ________ (Init)

6) Verify pump shaft turns over by hand. ________ Init)

7) Check coupling alignment after motor mounting (direct drive). ________ (Init)

8) Sheaves have been checked for proper alignment (belt drive). ________ (Init)

9) Inboard and outboard oil containment cups are installed. ________ (Init)

10) Bearing housing oil level is half way up sight glass. ________ (Init)

11) Sight glass is installed on left side when looking at pump suction, or opposite side of motor on belt drive pumps. ________ (Init)

12) Bearing housing vent plug and cap are installed. ________ (Init)

13) All guards are in place. ________ (Init)

14) Companion flanges and gaskets (if required) supplied with export pumps. ________ (Init)

15) If customer supplied motor, are (4) of the proper size Motor Adjuster Feet (with set screws & mounting hardware) ☐ Pump Drive Shaft Key ☐ supplied? ________ (Init)

16) If customer supplied motor on belt-drive pump, are Adjustable Motor Base ☐ Belts ☐ Sheaves ☐ Bushings (both) ☐ Belt Guard ☐ Pump Drive Shaft Key ☐ supplied? ________ (Init)

17) Motor Manufacturer: ________________________________ ________ (Init)

Spec/Catalog/Model Number _________________________ Speed__________ RPM

Serial number ____________________________ Frame Size _________

18) Motor greased with: ________________________________ Not Greasable ☐ ________ (Init)
Note: Check motor manual for grease type when not listed on nameplate

19) Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

20) Decals and nameplate are installed per decal placement drawing. ________ (Init)

JOB ORDER# ___________________________

Nameplate full load amps __________

Motor Amp Measurement: __________

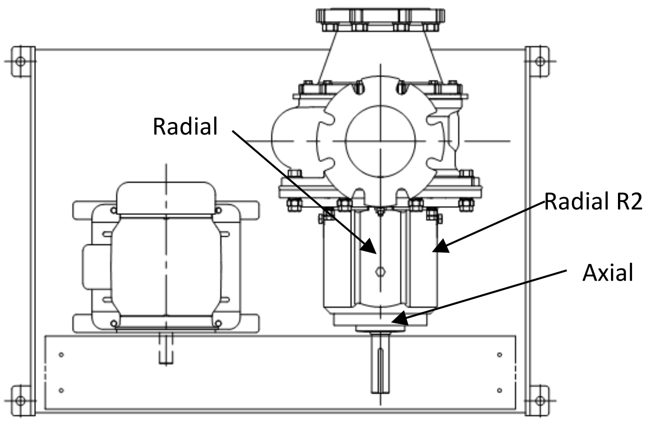
Reading is below half of full load Yes ☐ No ☐ _______ (Init)

Vibration Measurements

Axial A _________________

Radial R1 _______________

Radial R2 _______________



SECTION 4: FINAL SHIPPING STATION

1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)
2) Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)
3) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#___ _______ (Init)
4) Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)
5) Decals and nameplate are installed per decal placement drawing. _______ (Init)
6) The IO&M Manual is attached to the pump at time of shipping. _______ (Init)
7) QA Sheet is complete:

   ____
   (Shipping Clerk) (Date)

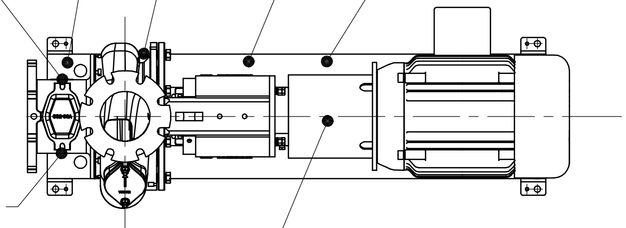
REVISION LIST

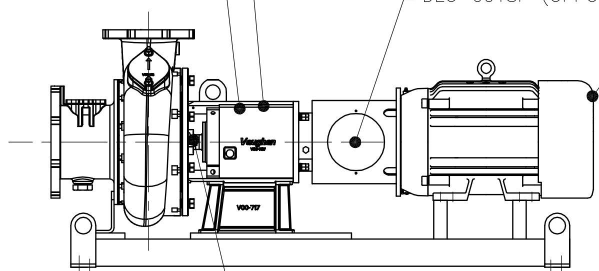
DATE:	REV#:	DESCRIPTION:	BY:
6/20/16	1	ADDED NAME PLATE DEC-32	KCW
7/13/16	2	RENAMED DRAWING & ADDED NOTE	KCW
7/20/16	3	ADDED FOREIGN PUMP TABLE	KCW
10/14/16	4	UPDAATED FOREIGN PUMP TABLE	KCW

DOMESTIC PUMPS

* DEC-067 NOT USED IF PUMP HAS NO CLEANOUT

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-032	NAME PLATE
DEC-061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC-067	2" X 4" DANGER (ENGLISH)
*DEC-067SP	2" X 4" DANGER (SPANISH)
V108-409	2.75" X 6.5" SS ROTAMIX LOGO PLATE





FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

Vaughan
VAUGHAN CO., INC.
364 MONTE-ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249-4042
FAX: (360) 249-6155

TITLE: HE/HSC DIRECT DRIVE DECAL PLACEMENT

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE: SURFACE FINISH 125 MICROINCHES; DIMENSIONS, WELDING ± 1/16"; DIMENSIONS, MACHINING ± 0.020"; DIMENSIONS, MACHINING ± 0.010"; GEOMETRIC TOLERANCE, RUNOUT OR CIRCULARITY: 0.010"; GEOMETRIC TOLERANCE, PARALLELISM 0.005"; BREAK ALL MACHINE EDGES 0.005" IF UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	
JOB ORDER NUMBER:		DATE: 2/8/16	DRAWING NUMBER: 118370	REVISION NUMBER: 4

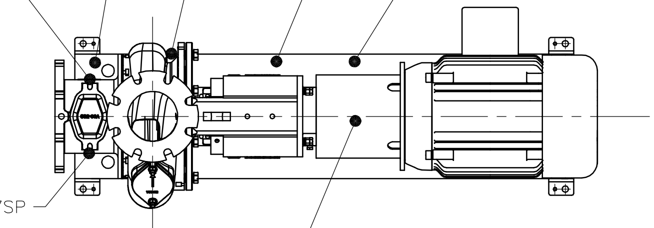
REVISION LIST

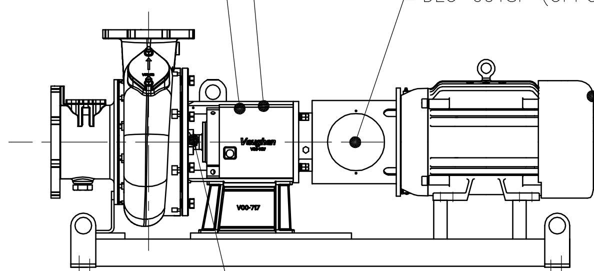
DATE:	REV#:	DESCRIPTION:	BY:
6/20/16	1	ADDED NAME PLATE DEC-32	KCW
7/13/16	2	RENAMED DRAWING & ADDED NOTE	KCW
7/20/16	3	ADDED FOREIGN PUMP TABLE	KCW
10/14/16	4	UPDAATED FOREIGN PUMP TABLE	KCW

DOMESTIC PUMPS

* DEC-067 NOT USED IF PUMP HAS NO CLEANOUT

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-032	NAME PLATE
DEC-061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC-067	2" X 4" DANGER (ENGLISH)
*DEC-067SP	2" X 4" DANGER (SPANISH)
V108-409	2.75" X 6.5" SS ROTAMIX LOGO PLATE





FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XXX ± 0.020"
DIMENSIONS, MACHINING: X.XXXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: HE/HSC DIRECT DRIVE DECAL PLACEMENT				
CHECKED: JK	ENGINEER:	DRAWN BY: KCW	SCALE: 1"=1'	PART NUMBER:
JOB ORDER NUMBER:		DATE: 2/8/16	DRAWING NUMBER: 118370	REVISION NUMBER: 4

PEDESTAL SCREW PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ____________________ SALES ORDER # ____________________

MODEL # ________________________ DRAWING # ________________________

SECTION 1: PUMP ASSEMBLY STATION:

1) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#___ _______(Init)

2) Impeller model matches Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)

3) Impeller has been balanced. _______ (Init)

4) Shaft endplay is measured at ________ _______ (Init)
(3DS, 4DH, 6EM, 6ES: 0.001 – 0.004”) (8F, 8x8B, 10D, 12D: 0.001 – 0.006”)

5) Shaft run out at impeller __________ _______ (Init)

6) Impeller bolt is properly torqued with Loctite 242. _______ (Init)
3DS, 4DH: 30# ☐ 6EM, 6ES: 40# ☐ 8F, 10D: 80# ☐ 8x8B, 12D: 100# ☐

7) Hub Cutter/Back Cutter Clearance: ________ inch *(0.010 - 0.020" acceptable)* _______ (Init)

8) Impeller/Suction Cone gap measures _________ inch (*0.015 - 0.025” acceptable)* _______ (Init)

9) All nuts and bolts tightened with lock washers _______ (Init)

10) Discharge orientation matches Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)

SECTION 2: PAINTING STATION

1) Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)

2) Paint type:________________________ Required paint thickness:_________mils _______ (Init)

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of pedestal beneath the Vaughan nameplate _____MWFT
b) adjacent to nameplate _____MWFT
c) underside of elbow _____MWFT
d) side of bearing housing adjacent to oil hose _____MWFT
e) casing discharge throat on side nearest bearing housing _____MWFT
Initials _____ Date _____

DRY
a) underside of pedestal beneath the Vaughan nameplate _____MDFT
b) adjacent to nameplate _____MDFT
c) underside of elbow _____MDFT
d) side of bearing housing adjacent to oil hose _____MDFT
e) casing discharge throat on side nearest bearing housing _____MDFT
Initials _____ Date _____

JOB ORDER#___________________________________

SECTION 3: FINAL ASSEMBLY STATION

1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

2) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_______Rev#__ ________(Init)

3) Motor stool matches the motor shown on Drawing ☐ Job Order ☐ ________ (Init)

4) Pump and drive assembly match Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

5) If a special base is required does it match the drawing? Yes ☐ No ☐ ________ (Init)

6) Verify pump shaft turns by hand. ________ (Init)

7) Check coupling alignment after motor mounting. ________ (Init)

8) Oil level in oil reservoir 40-60% full ________ (Init)

9) Hole in vent cap is drilled through and intersects roof? (Use 0.156" drill bit to check) ________ (Init)

10) "Motor Manufacturer ______________________________________ ________ (Init)

Spec/Model/Catalog Number _________________________

Serial number ___________________________

Frame Size __________ Speed__________RPM

11) Motor greased with: ________________________________ Not Greasable ☐ ________ (Init)
Note: Check motor manual for grease type when not listed on nameplate.

12) Verify all guards are in place ________ (Init)

13) Companion flanges (with gasket) supplied with export pumps? ________ (Init)

14) Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

15) Decals are installed per the Decal Placement Drawing. ________ (Init)

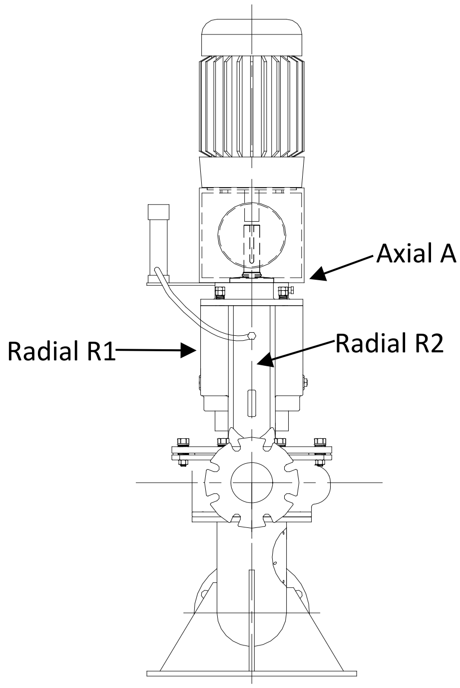
16) Vibration Test Measurements ________ (Init)

Axial A ____________________

Radial R1 __________________

Radial R2 __________________

JOB ORDER#________________________________



Nameplate Full Load Amps ____________

Motor amps measurement______________

Reading is below half of full load Yes ☐ No ☐

SECTION 4: FINAL SHIPPING STATION

1) Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)

2) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#___ ________(Init)

3) Motor stool matches the motor shown on Drawing ☐ Job Order ☐ _______ (Init)

4) Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ _______ (Init)

5) Decals are installed per the Decal Placement Drawing. _______ (Init)

6) IO&M Manual is attached to the pump at the time of shipping. _______ (Init)

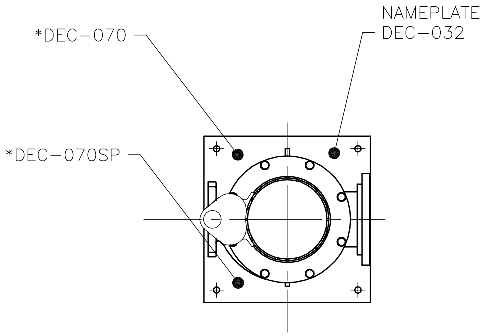
7) QA Sheet is complete:

____

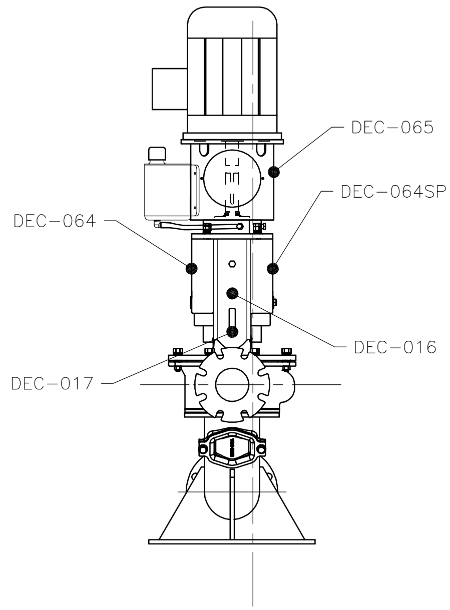
(Shipping Clerk) (Date)

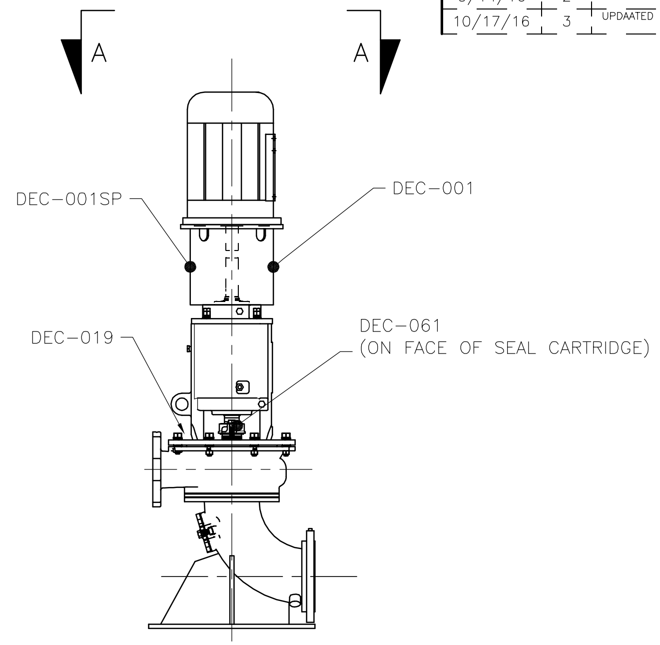
REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
7/13/16	1	RENAMED DRAWING & ADDED NOTE & DEC-032 FOR NAMEPLATE	KCW
9/14/16	2	ADDED FOREIGN PUMP TABLE	KCW
10/17/16	3	UPDAATED FOREIGN PUMP TABLE	KCW



VIEW A-A





DOMESTIC PUMPS

* DEC-070 NOT USED IF PUMP HAS NO CLEANOUT

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-032	NAME PLATE
DEC-061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC-070	4" X 6" WARNING (ENGLISH)
*DEC-070SP	4" X 6" WARNING (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: XXX ± 1/16"
DIMENSIONS, MACHINING: XXX ± 0.020"
DIMENSIONS, MACHINING: XXXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: PE/PSC DECAL PLACEMENT

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	3/3/16	118396	3

SELF-PRIMING PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER #____________________ SALES ORDER #_______________

MODEL #______________________________ DRAWING #____________________

SECTION 1: PUMP ASSEMBLY STATION

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ______
2. Impeller has ______ blades and measures _______ inches ______
3. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ ______
4. Impeller has been balanced ______
5. Shaft run out at impeller: ____________ ______
6. Pressure test: Bearing Housing, 1 min @ 10 PSI ☐ Seal Chamber 1 min @ 7 PSI ☐ ______
7. Pressure test Pump Housing, 1 min @ 10 PSI ☐ ______
8. All nuts and bolts tightened with lock washers. ______

SECTION 2: SP4B, SP4C, SP6K

Initial Below

1. The Impeller is properly torqued to 200 ft/lb. ______
2. The Cutter Nose is properly torqued to 40 ft/lb. ______
3. The External Tool (if required) is properly torqued to 125 ft/lb. ______
4. Impeller/Cutter Bar gap to close vane measures _________ inch **0.015-0.025"** ______
5. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.005-0.050"** ______
6. Clearance between tip of cutter bar finger and cutter nose or external tool hub. **0.010"-0.020"**
   Measurements for 2 locations: a)____________ inch b)___________ inch ______
7. Clearance between tips of external tool and anvil __________ **0.010"–0.020"** ______
8. Axial clearance between external tool wings and cutter bar fingers _________inch ______
   0.015 - 0.025"
7. Cutter Bar inlet diameter is correct per below ______
   SP4B/4C – 4.25" ☐ **SP6K – 6.50"** ☐

SECTION 3: SP8N, SP10R

Initial Below

1. The Impeller is properly torqued to 1000 ft/lb. ______
2. The External Tool is properly torqued to 250 ft/lb. ______
3. Impeller/Cutter Bar gap to close vane measures _________ inch **0.015-0.025"** ______
4. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.030-0.050"** ______
5. Clearance between tip of cutter bar finger and external tool tooth. **0.025-0.040"** ______
   Measurements for 2 locations: a)____________ inch b)___________ inch ______
6. Axial clearance between external tool wings and cutter bar fingers _________inch ______
   0.015 - 0.025"
7. Cutter bar inlet diameter is correct per below ______
   SP8N – 9.0" ☐ **SP10R – 11.0"** ☐

SECTION 4: PAINTING STATION

Initial Below

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ______
2. Paint type:________________________ Required paint thickness:_________mils ______

JOB ORDER#______________________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MWFT
b) top of baseplate below oil site glasses _____MWFT
c) adjacent to suction cleanout _____MWFT
d) side of bearing housing adjacent to oil site glasses _____MWFT
e) side of pump housing _____MWFT
Initials _____ Date _____

DRY
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MDFT
b) top of baseplate below oil site glasses _____MDFT
c) adjacent to suction cleanout _____MDFT
d) side of bearing housing adjacent to oil site glasses _____MDFT
e) side of pump housing _____MDFT
Initials _____ Date _____

SECTION 5: FINAL ASSEMBLY

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ _____
2. Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ _____
3. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ _____
4. Pump and drive assembly match Job Order ☐ BOM ☐ Drawing ☐ _____
5. If a special base is required does it match the drawing? Yes☐ No☐ _____
6. Verify pump shaft turns over by hand. _____
7. Check coupling alignment after motor mounting (direct drive). _____
8. Sheaves have been checked for proper alignment (belt drive). _____
9. If customer supplied motor, are (4) proper size motor adjuster feet (with hardware) ☐ and pump drive shaft key ☐ supplied? _____
10. If customer supplied motor on belt-drive pump, are the Motor Adjustable Bases ☐ Belts ☐Tensioners ☐ Bushings ☐ Sheaves ☐ Belt Guard ☐ Pump Drive Shaft Key ☐ supplied with the pump? _____
11. Companion flanges and gaskets (if required) supplied with export pumps. _____
12. Bearing housing oil level is half way up sight glass. _____
13. Seal oil reservoir is full. _____
14. All guards are in place. _____
15. Verify pump speed of belt drive pumps with tachometer _________RPM _____
16. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ _____
17. Tags and nameplates are installed on the pump per the decal placement drawing. _____
18. Motor Manufacturer __________________________ ________________ _____

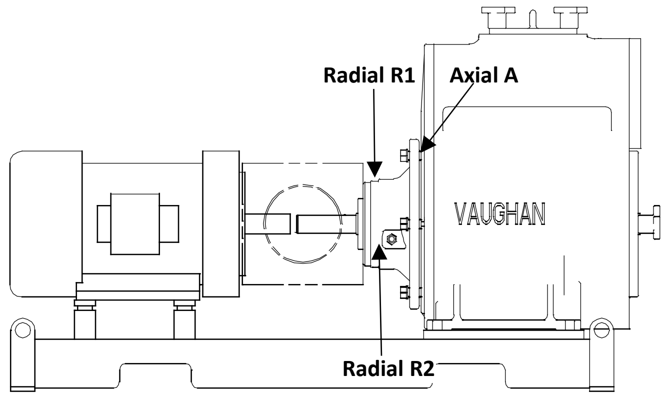
    Serial No. _____________________________ Frame Size ____________

JOB ORDER#________________________

Speed __________RPM Spec/Model/Catalog Number ________________________ **Initial Below**

19. Motor greased with ____________________________ Not Greasable ☐ ______
    Note: Check motor manual for grease type when not listed on nameplate.

Vibration Measurements



Axial A _________________

Radial R1 _______________

Radial R2 _______________

Nameplate Full Load Amps _____________

Motor Amp Measurements: _____________

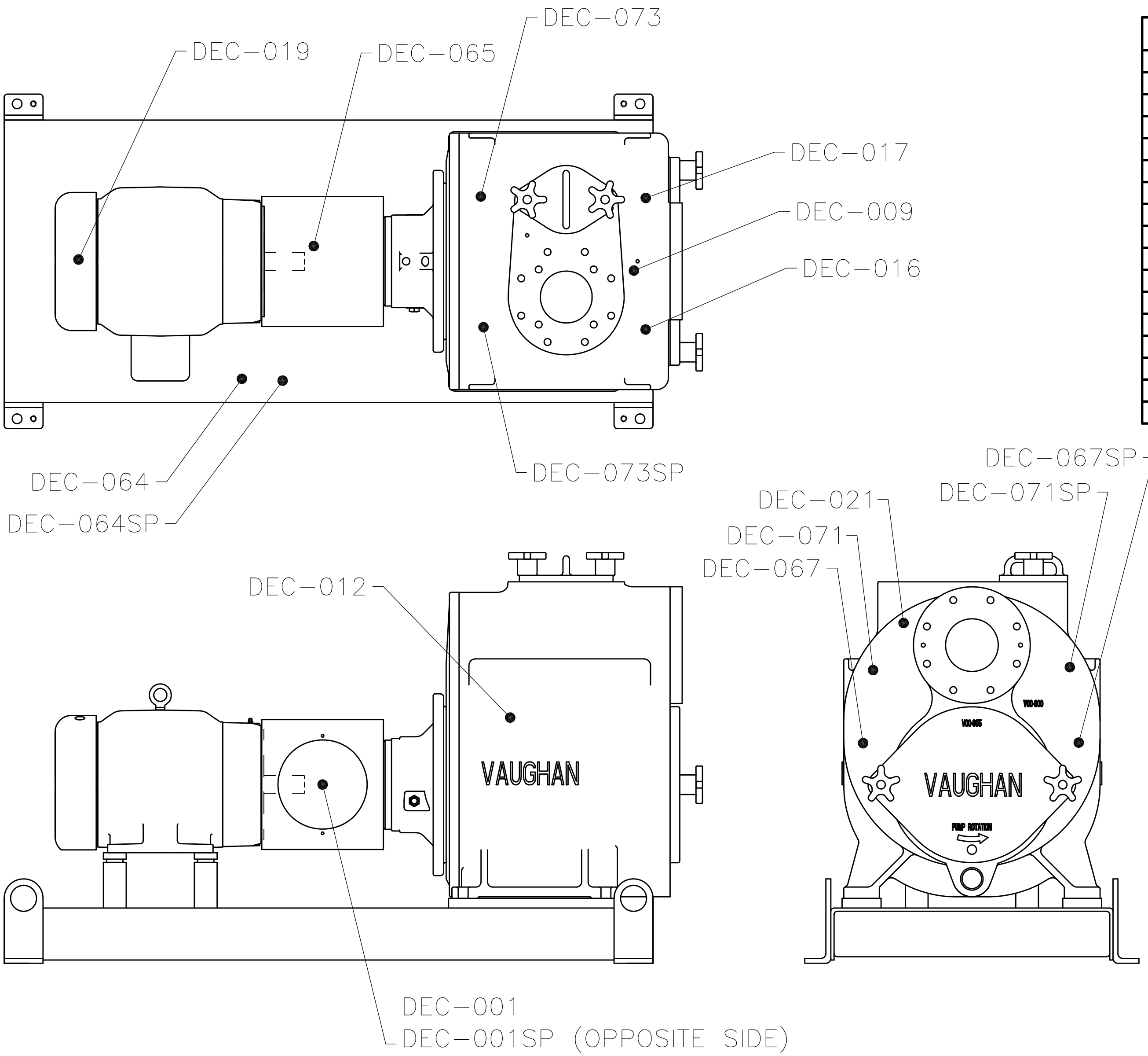
Reading is below half of full load. ______ (Init)

SECTION 6: FINAL SHIPPING STATION

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ______
2. Motor stool matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ ______
3. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ______
4. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ______
5. Tags and nameplates are installed on the pump per the decal placement drawing. ______
6. Is the IO&M Manual attached to the pump at the time of shipping? ______
7. QA Sheet is complete:

____
(Shipping Clerk) (Date)



DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-009	.5" X 1.5" DISCHARGE
DEC-012	4" X 4" LUBRICATION
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-021	.5" X 5.5", SUCTION
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (ENGLISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-067	2" X 4" DANGER (ENGLISH)
DEC-067SP	2" X 4" DANGER (SPANISH)
DEC-071	3.5" X 5" WARNING, HIGH TEMP (ENGLISH)
DEC-071SP	3.5" X 5" WARNING, HIGH TEMP (SPANISH)
DEC-073	3.5" X 5" WARNING, (ENGLISH)
DEC-073SP	3.5" X 5" WARNING, (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

Vaughan

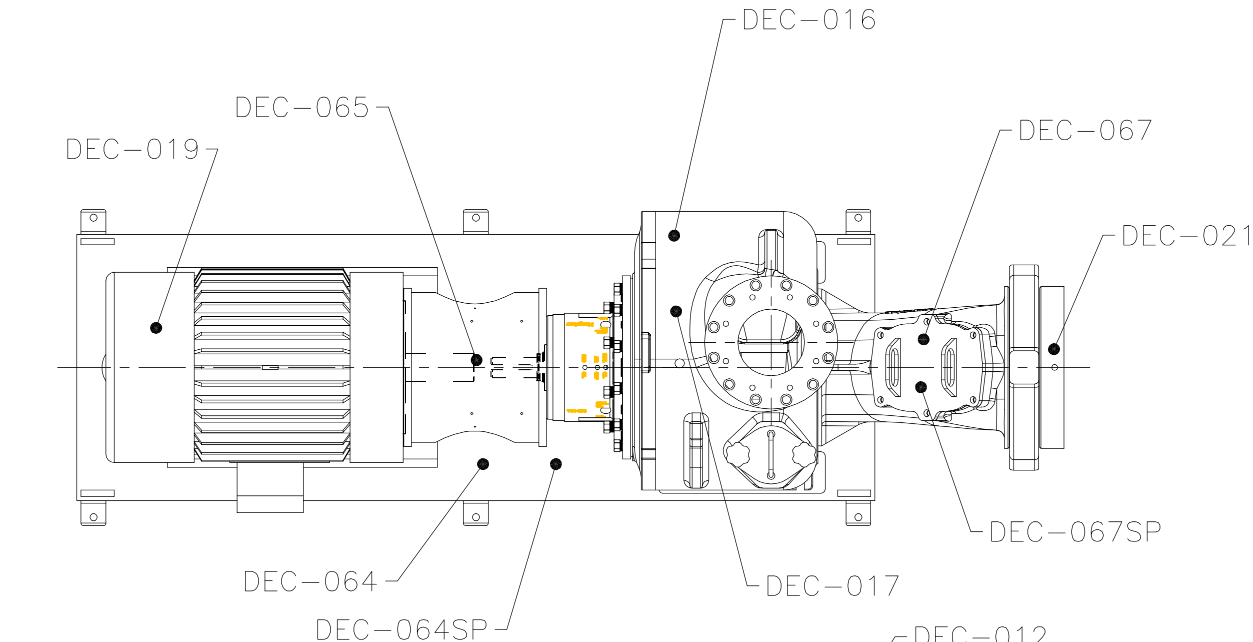
VAUGHAN CO., INC.
364 MONTE-ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249-4042
FAX: (360) 249-6155

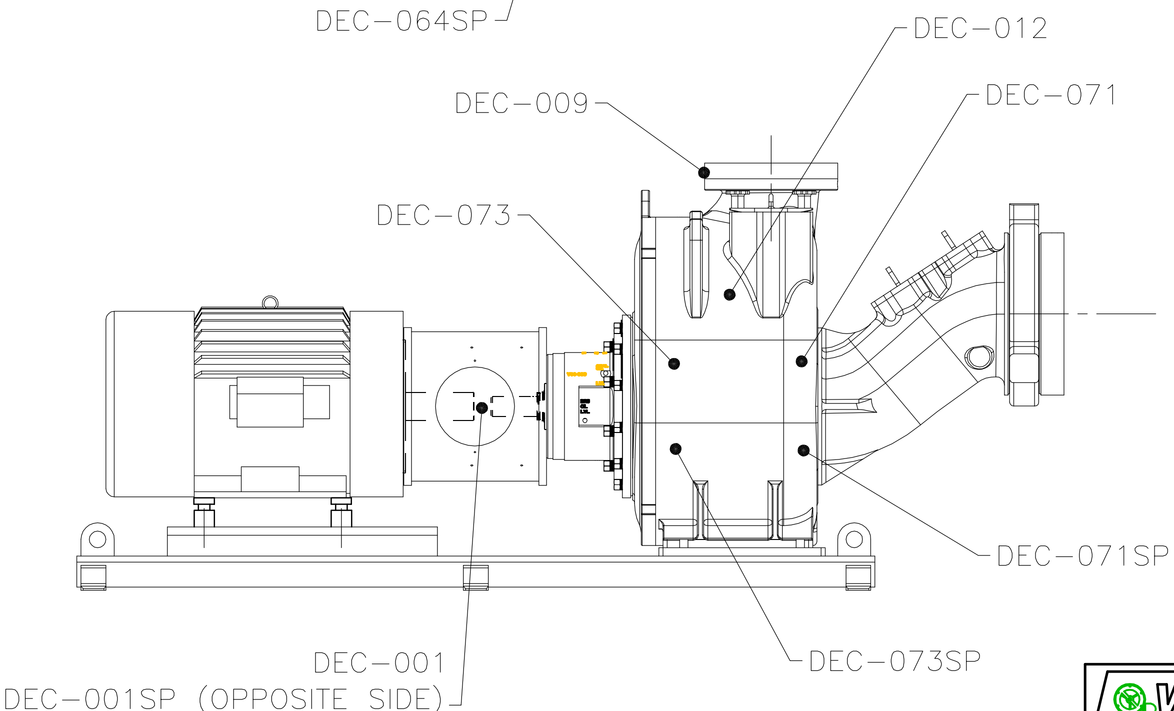
TITLE:
4-6 SELF PRIMER
DIRECT DRIVE
DECAL PLACEMENT

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		TGW		
JOB ORDER NUMBER:		DATE: 3/9/17	DRAWING NUMBER: 119036	REVISION NUMBER: 0





DOMESTIC PUMPS

PART #	DESCRIPTION
DEC-001	5.375" X 6" ATTENTION (ENGLISH)
DEC-001SP	5.375" X 6" ATTENTION (SPANISH)
DEC-009	.5" X 1.5" DISCHARGE
DEC-012	4" X 4" OIL REQUIREMENTS
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-021	.5" X 5.5", SUCTION
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (ENGLISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
DEC-067	2" X 4" DANGER (ENGLISH)
DEC-067SP	2" X 4" DANGER (SPANISH)
DEC-071	3.5" X 5" WARNING, HIGH TEMP (ENGLISH)
DEC-071SP	3.5" X 5" WARNING, HIGH TEMP (SPANISH)
DEC-073	3.5" X 5" WARNING, (ENGLISH)
DEC-073SP	3.5" X 5" WARNING, (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
7/12/17	1	ADDED DEC-012	KCW

Vaughan

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249-6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XXX ± 0.020"
DIMENSIONS, MACHINING: X.XXXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: 8-10 SELF PRIMER DIRECT DRIVE DECAL PLACEMENT

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	1/18/17	118945	1

FOAMBUSTER QUALITY ASSURANCE VERIFICATION

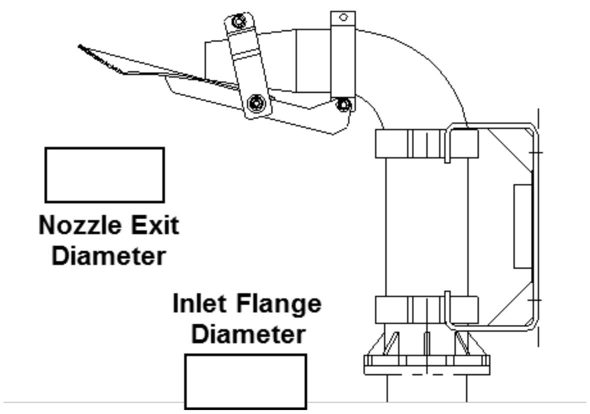
JOB ORDER# ________________________

MODEL# _________________________ **DRAWING#** ____________________

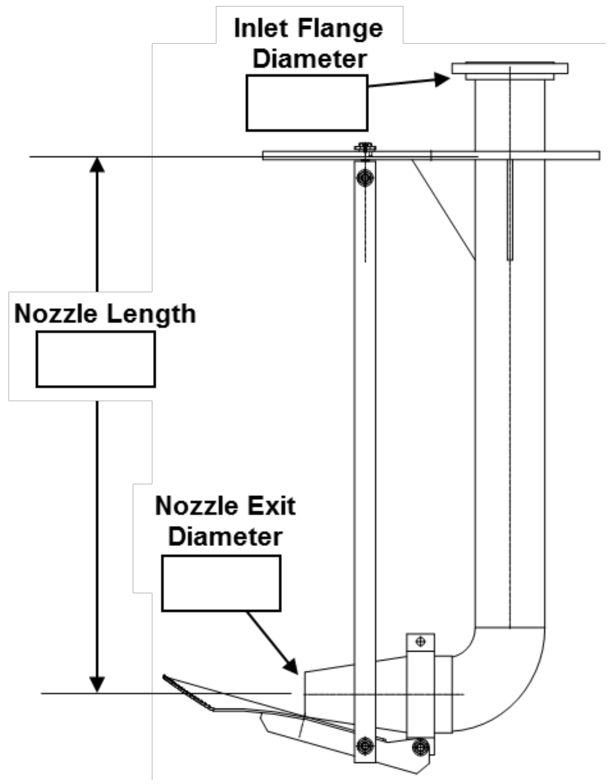
SECTION 1: ASSEMBLY STATION:

1. Nozzle material: Polyurethane ☐ Ductile iron ☐ _________(Init)
2. If roof mounted model does deckplate match the drawing? _________(Init)

Wall Mounted



Roof Mounted



SECTION 2: FINAL ASSEMBLY

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_______Rev#___ _________(Init)
2. All white circles have been removed from glass lining in all nozzles _________(Init)
3. Deckplate has lifting eyes. _________(Init)
4. Zinc anode is installed and unpainted. _________(Init)
5. Rotamix stickers have been applied? _________(Init)

SECTION 3: FINAL SHIPPING STATION

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_______Rev#___ _________(Init)
2. Rotamix stickers have been applied? _________(Init)

JOB ORDER# ___________________________

3. Is the Rotamix IO&M manual attached at time of shipping? __________(Init)
4. If roof mounted are the installation instructions included? __________(Init)
5. QA Sheet is complete:

____

(Shipping Clerk) (Date)

AUGER HOPPER QUALITY ASSURANCE VERIFICATION

JOB ORDER #______________________ **SALES ORDER #** _______________

MODEL #__________________________ **DRAWING #** ____________________

SECTION 1: PUMP ASSEMBLY STATION:

1) Motor matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

2) Auger model matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#______ Rev#___ _____ (Init)

3) Drain in bearing housing installed Yes ☐ No ☐ ________ (Init)

4) Grease zerk on bearing is extended Yes ☐ No ☐ ________ (Init)

5) Guard on gear motor is in place Yes ☐ No ☐ ________ (Init)

6) The Nord pressure vent is supplied loose with the motor Yes ☐ No ☐ ________ (Init)

7) All nuts and bolts tightened with lock washers? ________ (Init)

SECTION 2: PAINTING STATION

1) Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

2) Paint type:________________________ Required paint thickness:_________mils ________ (Init)

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MWFT
b) side of bearing housing adjacent to oil site glass _____MWFT
Initials _____ Date _____

DRY
a) underside of baseplate in the corner nearest the Vaughan nameplate _____MDFT
b) side of bearing housing adjacent to oil site glass _____MDFT
Initials _____ Date _____

SECTION 3: FINAL ASSEMBLY STATION

1) Motor matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

2) Hopper matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________ Rev#____ _______ (Init)

3) Verify that the auger turns by hand. ________ (Init)

4) Oil in gear motor is filled to the proper level. ________ (Init)

JOB ORDER#________________________________

5) The Nord pressure vent is shipping loose with the hopper Yes ☐ No ☐ ________ (Init)

6) All guards are in place ________ (Init)

7) Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

8) Are the tags and nameplates installed on the hopper per decal placement drawing ________ (Init)

9) Motor manufacturer is ________________________________ ________ (Init)

Motor HP ___________ Speed ___________RPM ________ (Init)

Motor Serial Number _______________________

Spec/Model/Catalog#_______________________

Nameplate Full Load Amps _____________ ________ (Init)

Motor Amp Measurements: ______________ amps ________ (Init)

SECTION 4: FINAL SHIPPING STATION

1) Motor matches Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

2) Hopper matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________ Rev#____ _______ (Init)

3) Nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________ (Init)

4) Are the tags and nameplates installed on the hopper per decal placement drawing ________ (Init)

5) Is the IO&M Manual attached to the hopper at the time of shipping? ________ (Init)

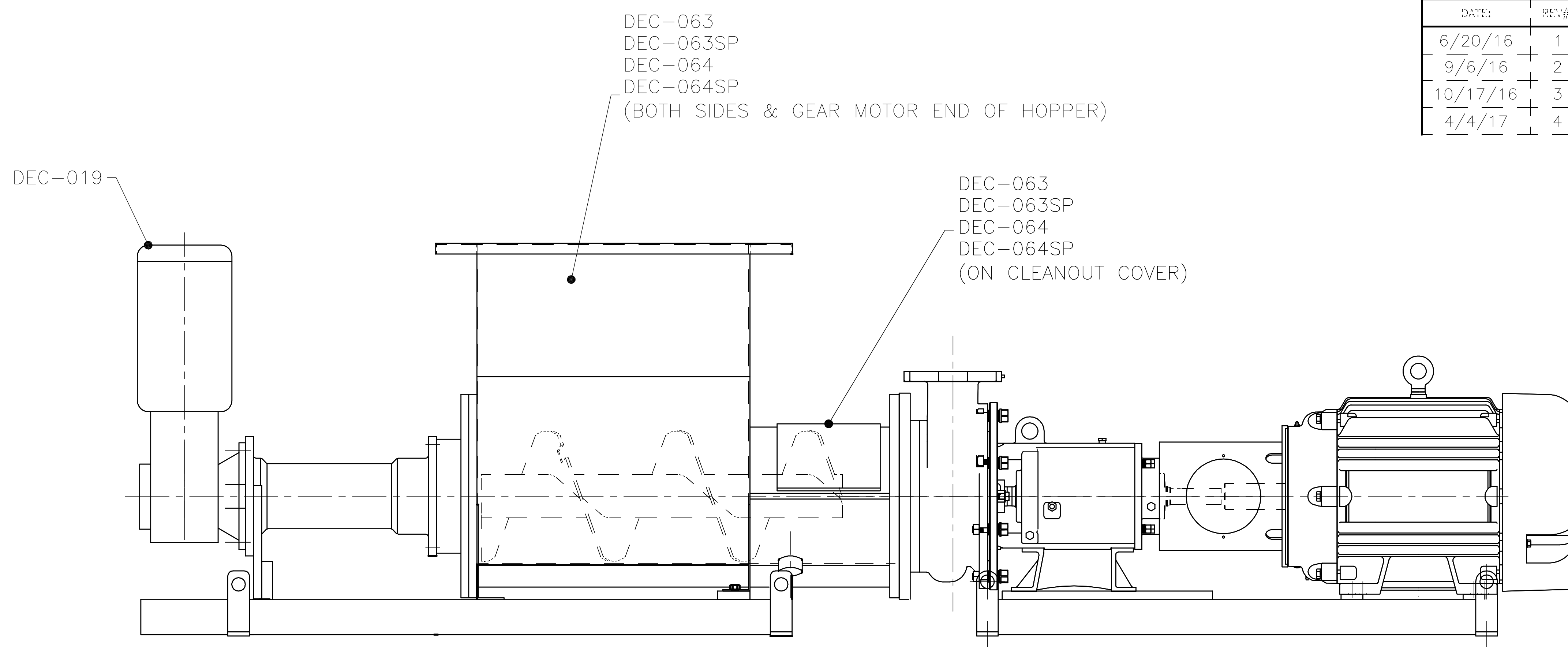
6) QA Sheet is complete:

____

(Shipping Clerk) (Date)

REVISION LIST

DATE:	REV#:	DESCRIPTION:	BY:
6/20/16	1	REMOVED PUMP DECALS AND ADDED NOTE	KCW
9/6/16	2	ADDED FOREIGN PUMP TABLE	BSS
10/17/16	3	UPDAATED FOREIGN PUMP TABLE	KCW
4/4/17	4	DEC-019 WAS DEC-020	KCW



DOMESTIC PUMPS

NOTE:

1. USE KIT 118370 FOR PUMP DECALS

PART #	DESCRIPTION
DEC-019	.5" X 5.5" CLOCKWISE ROTATION
DEC-063	3.5" X 5" DANGER (ENGLISH)
DEC-063SP	3.5" X 5" DANGER (SPANISH)
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

Vaughan

VAUGHAN CO., INC.
364 MONTE-ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249-4042
FAX: (360) 249-6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XX = 0.020"
DIMENSIONS, MACHINING: X.XXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: HOPPER CHOPPER DECAL PLACEMENT

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
KHK	KHK	KCW	1:8	
JOB ORDER NUMBER:		DATE: 1/15/16	DRAWING NUMBER: 118332	REVISION NUMBER: 4

CLOSE COUPLED PEDESTAL CHOPPER PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ___________________________ SALES ORDER # ______________________

MODEL #________________________________ DRAWING #__________________________

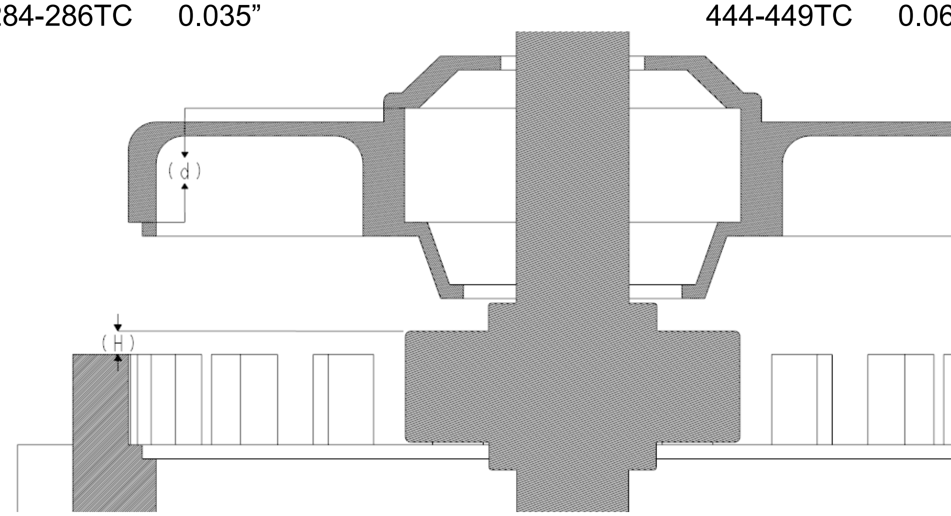
SECTION 1: PUMP ASSEMBLY STATION

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#__________Rev#____ ________
2. Prior to motor disassembly record motor no load amps_____________ ________

Shaft Thermal Growth Allowances

182-184TC	0.030”	324-326TC	0.039”
213-215TC	0.030”	364-365TC	0.041”
254-256TC	0.032”	404-405TC	0.046”
284-286TC	0.035”	444-449TC	0.068”



Original Bore Depth (**d**) from outer shoulder: ______ **.** ______ in.

− Shaft Thermal Expansion Allowance: 0.______ in.

= Useable Bore Depth (**D**): ______ **.** ______ in.

Bearing Height (**H**): ______ **.** ______ in.

If D exceeds H, OK to proceed and skip step 4. If not, machine bore depth D deeper as needed and fill out step 4.

____

3. Bore Depth (**d**) from outer shoulder after machining: ______ **.** ______ in.

− Shaft Thermal Expansion Allowance: 0.______ in.

= Useable Bore Depth (**D**): ______ **.** ______ in.

Bearing Height (**H**): ______ **.** ______ in.

If D exceeds H, OK to proceed. ________

4. Wave spring has been removed and discarded? Yes ☐ No ☐ ________

JOB ORDER# ___________________________

Initial Below

5. Radial bearing has fan end shield/seal removed. ________
6. Thrust bearing has drive end shield/seal removed. ________
7. Grease port paths are free of obstructions and prefilled with grease. (Pump grease in zerk until it appears at shaft.) ________
8. Bearings are prepacked with grease from manufacturer Yes ☐ No ☐ ________

   If no, bearings have been hand packed with grease Yes ☐ ________
9. Thrust bearing motor grease zerk has been removed and plugged. ________
10. Impeller has _____ blades and measures _________ inches ________
11. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ ________
12. Impeller has been balanced. ________
13. Shaft run out at impeller: _____________ ________
14. All nuts and bolts tightened with lock washers. ________
15. Discharge orientation matches Job Order ☐ BOM ☐ Drawing ☐ ________

SECTION 2: All 3-6" & 8L PUMPS

Initial Below

1. The Cutter Nut ☐ or External Tool ☐ is properly torqued to 125 ft/lb. ________
2. Impeller/Cutter bar gap to close vane measures __________ inch. **0.015-0.025"** ________
3. Pump Out Vane/Back Cutter gap to close vane: _________inch. **0.005-0.050"** ________
4. Clearance between tip of cutter bar finger and hub tooth. **0.010-0.030"** ________
   Measurements for 2 locations: a)_____________ inch b)____________ inch
5. Clearance between tips of external tool and anvil **0.010–0.030"** ________
6. Axial clearance between external tool wings and cutter bar fingers __________inch ________
   0.015 – 0.025"
7. Cutter Bar inlet diameter is correct per below ________

   3F/3G/3P – 4.25" ☐ **3L/3M/3W – 4.63"** ☐ **3V – 4.0"** ☐ **4K/4L – 5.25"** ☐
   4P/4S – 6.5" ☐ **4R/4T/6U/6W/6X – 7.4"** ☐ **4V – 8.5"** ☐ **8L – 9.25"** ☐

SECTION 3: 8N, 8P, All 10 – 16" PUMPS

Initial Below

1. The impeller is properly torqued to 1000 ft/lb ________
2. The External Tool is properly torqued to 250 ft/lb. ________
3. Impeller/Cutter bar gap to close vane measures __________ inch. ________
   8N, All 10-16": 0.015-0.025" 8P: 0.010-0.020"
4. Pump Out Vane/Back Cutter gap to close vane: _________inch. **0.030-0.050"** ________
5. Clearance between tip of cutter bar finger and external tool tooth ________
   Measurements for 2 locations: a)_____________ inch b)____________ inch
   8N & 8P: 0.025-0.040", All 10": 0.035-0.050" All 12:&16": 0.040-0.060"

JOB ORDER# ________________________

Initial Below

6. Clearance between tips of **external tool** and **anvil**: _____________ inch ________
   8N, 8P: 0.025-0.040" All 10": 0.035-0.050" All 12" &16": 0.040-0.060"
7. Axial clearance between tips of external tool wings and cutter bar fingers ________
8. Cutter Bar inlet diameter is correct per below ________
   8N – 9.0" ☐ **8P – 9.75"** ☐ **10R/10W – 11.0"** ☐ **12W – 13.5"** ☐ **16W – 16.0"** ☐

SECTION 4: 8K PUMPS

Initial Below

1. The Impeller is properly torqued to 500 ft/lb. ________
2. The Cutter Nose is properly torqued to 40 ft/lb. ________
3. The External Tool (if required) is properly torqued to 125 ft/lb. ________
4. Impeller/Cutter bar gap to close vane measures _________ inch. **0.015-0.025"** ________
5. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.005-0.050"** ________
6. Clearance between tip of cutter bar finger and cutter nose **0.030-0.040"** ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
7. Axial clearance between external tool wings and cutter bar fingers _________inch ________
   0.015-0.025"
8. Clearance between tips of external tool and anvil: _____________ inch **0.020–0.045"** ________
9. Cutter Bar inlet diameter is correct **8K – 8.75"** ☐ ________

SECTION 5: PAINTING STATION

Initial Below

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Paint type:________________________ Required paint thickness:_________mils ________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of pedestal beneath the Vaughan nameplate _____MWFT
b) adjacent to nameplate _____MWFT
c) underside of elbow _____MWFT
d) side of bearing housing adjacent to oil hose _____MWFT
e) casing discharge throat on side nearest bearing housing _____MWFT
Initials _____ Date _____

DRY
a) underside of pedestal beneath the Vaughan nameplate _____MDFT
b) adjacent to nameplate _____MDFT
c) underside of elbow _____MDFT
d) side of bearing housing adjacent to oil hose _____MDFT
e) casing discharge throat on side nearest bearing housing _____MDFT
Initials _____ Date _____

JOB ORDER# ____________________________

SECTION 6: FINAL ASSEMBLY STATION

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
3. Pump and drive assembly match drawing referenced above ________
4. If a special pedestal base is required does it match the drawing? Yes ☐ No ☐ ________
5. Verify pump shaft turns over by hand. ________

Motor: Manufacturer ________

Spec/Model/Catalog Number ________________________ Speed__________ RPM

Serial No. ______________________________ Frame Size ____________

6. Motor greased with ___________________________ Not Greaseable ☐ ________
   Note: Check motor manual for grease type when not listed on nameplate.
7. Verify all guards are in place. ________
8. Companion flanges (with gasket) supplied with export pumps? ________
9. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
10. If required the Rotamix trademark has been installed on base of pump. ________
11. Decals are installed on the pump per the decal placement drawing. ________

Vibration Measurements ________

Axial A _______________________

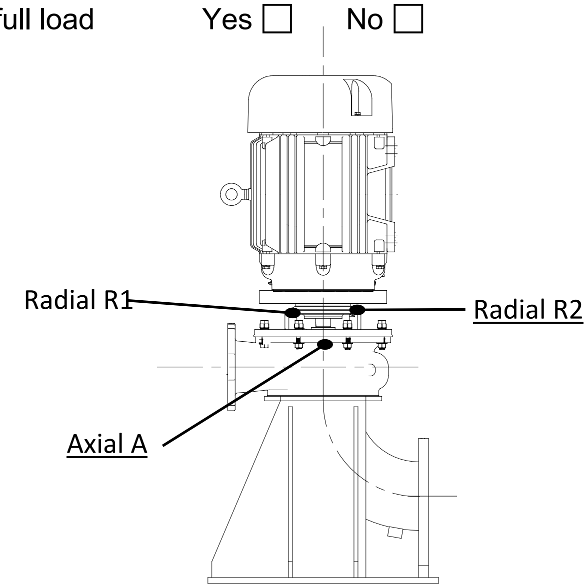
Radial R1 _____________________

Radial R2 _____________________

Motor amps measurement____________ ________

Nameplate full load amps ___________ ________

Reading is below half of full load Yes ☐ No ☐ ________



JOB ORDER# ____________________________

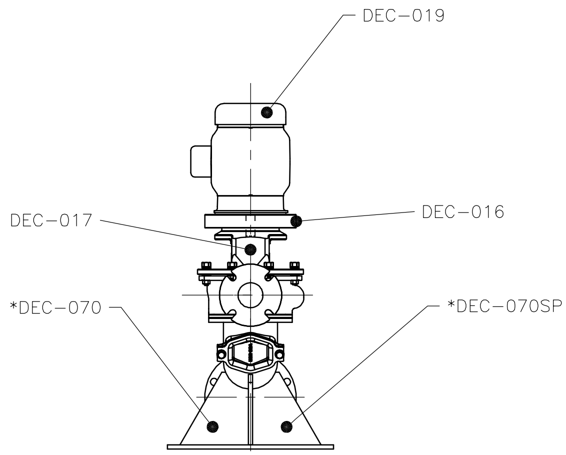
SECTION 7: FINAL SHIPPING STATION

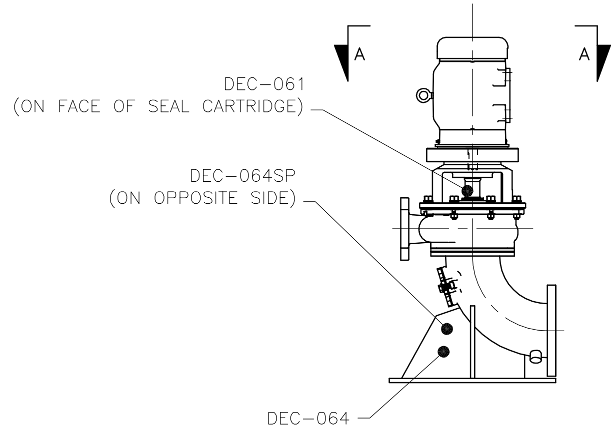
Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#____ ________
3. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
4. Decals are installed on the pump per the decal placement drawing. ________
5. IO&M Manual is attached to the pump at the time of shipping ________
6. QA Sheet is complete:

____

(Shipping Clerk) (Date)

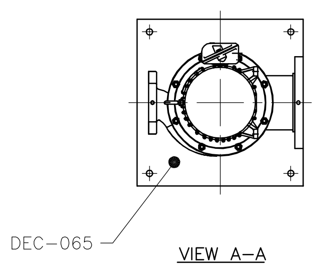




DOMESTIC PUMPS

* DEC-070 NOT USED IF PUMP HAS NO CLEANOUT

PART #	DESCRIPTION
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC-070	4" X 6" WARNING (ENGLISH)
*DEC-070SP	4" X 6" WARNING (SPANISH)



VIEW A-A

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

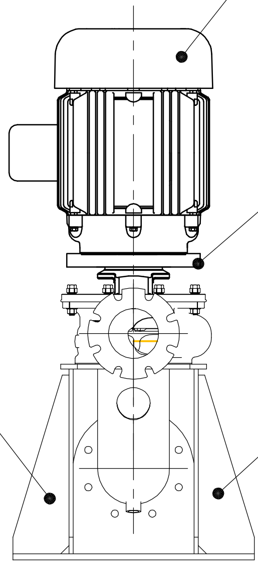
VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

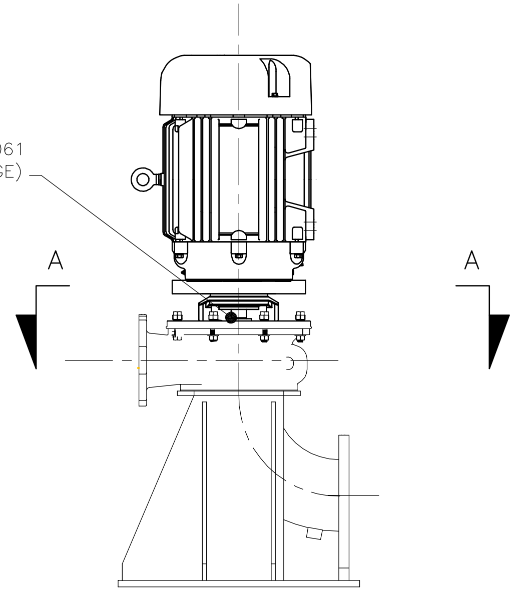
TITLE: CLOSE COUPLED PEDESTAL DECAL PLACEMENT STANDARD BASE ELBOW

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX ± 1/16"
DIMENSIONS, MACHINING: X.XXX ± 0.020"
DIMENSIONS, MACHINING: X.XXXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK	YCC	KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	6/19/17	119229	0

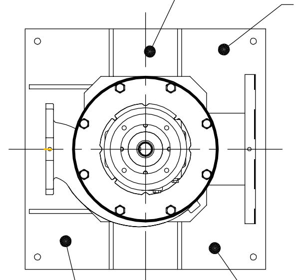




DOMESTIC PUMPS

* DEC-070 NOT USED IF PUMP HAS NO CLEANOUT

PART #	DESCRIPTION
DEC-016	1.25" X 3.5" QUALITY ASSURED
DEC-017	2" X 4" QUALITY CRAFTED IN USA
DEC-019	.5 X 5.5" CLOCKWISE ROTATION
DEC-061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC-064	3.5" X 5" WARNING (ENGLISH)
DEC-064SP	3.5" X 5" WARNING (SPANISH)
DEC-065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC-070	4" X 6" WARNING (ENGLISH)
*DEC-070SP	4" X 6" WARNING (SPANISH)



VIEW A-A

FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC-065 WITH DEC-037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC-065CAN	CANADA
DEC-037CZ	CZECHOSLOVAKIA
DEC-037DK	DENMARK
DEC-037NL	HOLLAND
DEC-037FI	FINLAND
DEC-037FR	FRANCE
DEC-037DE	GERMANY
DEC-037GR	GREECE
DEC-037IT	ITALY
DEC-037NOR	NORWAY
DEC-037PT	PORTUGAL
DEC-037ES	SPAIN
DEC-037SE	SWEDEN
DEC-037UAE	UNITED ARAB EMIRATES

VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

TITLE: CLOSE COUPLED PEDESTAL DECAL PLACEMENT BOX STYLE BASE ELBOW

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: XXX ± 1/16"
DIMENSIONS, MACHINING: XXX ± 0.020"
DIMENSIONS, MACHINING: XXXX ± 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"

BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK	YCC	KCW	1"=1'	

JOB ORDER NUMBER:	DATE:	DRAWING NUMBER:	REVISION NUMBER:
	6/16/17	119190	0

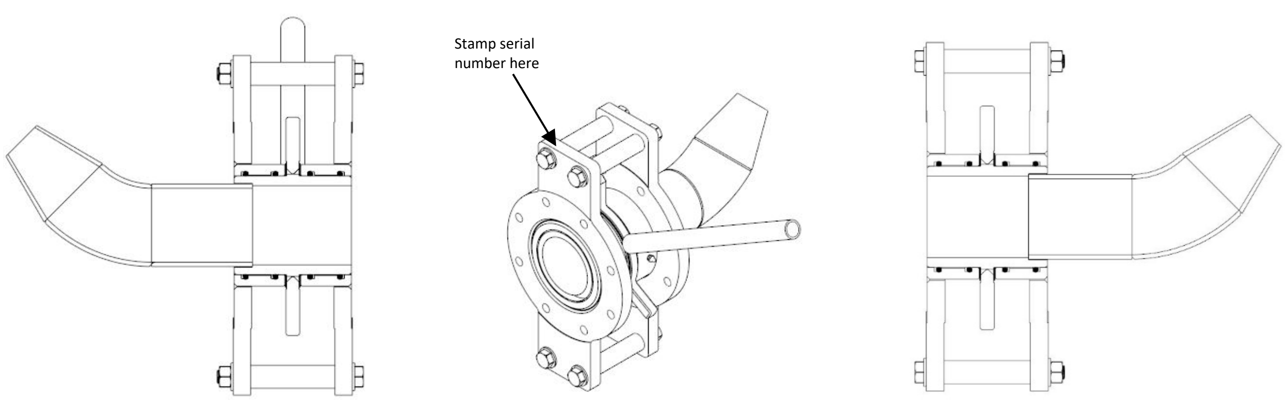
EXTERNALLY MOUNTED STAINLESS MIXING NOZZLE
QUALITY ASSURANCE VERIFICATION

JOB ORDER# ________________________

MODEL# ____________________________ **DRAWING#** ______________________

1. Nozzle: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________(Init)
2. Assembly has been greased with white Viper lube through grease zerks ________(Init)
3. Flange spacers have been included with the nozzles (4 per nozzle) ________(Init)
4. Thrust washers (2) have been installed ________(Init)
5. Operating handle included with nozzle assembly ________(Init)
6. Pointer on spider wheel is aligned with nozzle ________(Init)
7. Bolts have been properly torqued to 50 ft/lb ________(Init)
8. EMSA has been hydro tested for one minute at 100 PSI ________(init)
9. EMSA turns freely with handle while under pressure (100 PSI) ________(init)
10. Decals are installed and serial number is stamped ________(init)

EMSA ADJUSTABLE NOZZLE ASSEMBLY



FINAL SHIPPING STATION

1. Nozzle matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ (Init)
2. Decals are installed and serial number is stamped ________(init)
3. Nozzle matches drawing. Engineer only: ________(Init)
4. QA Sheet is complete: ____
   (Shipping Clerk) (Date)

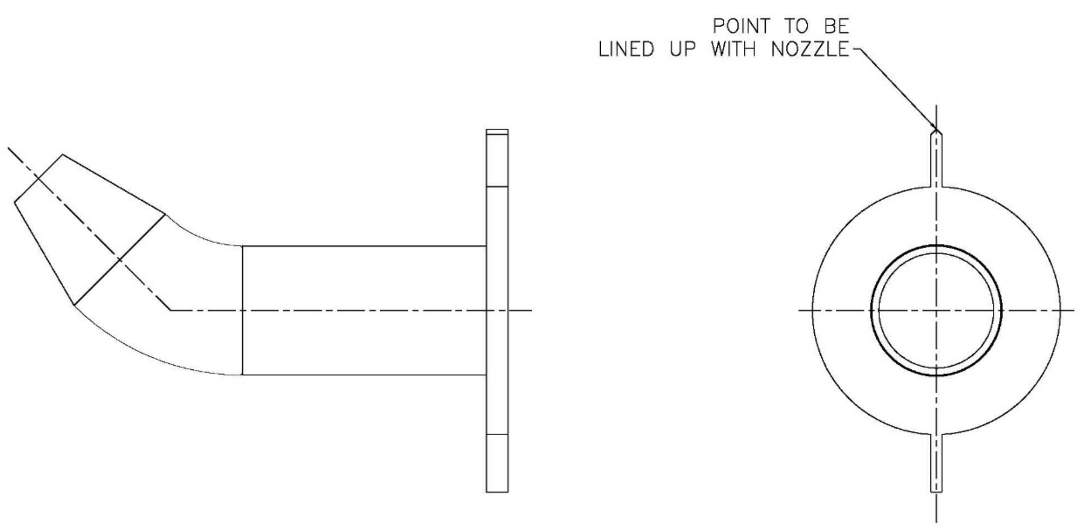
EXTERNAL MOUNT MIXING NOZZLE QUALITY ASSURANCE VERIFICATION

JOB ORDER# ________________________

MODEL# ____________________________ **DRAWING#** ______________________

1. Nozzle matches: Job Order ☐ BOM ☐ ________(Init)
2. Nozzle matches: Drawing ☐ Drawing#_________Rev#____ ________(Init)
3. Nozzle diameter matches order ________(Init)
4. Point lined up with nozzle (as pictured below) ________(Init)

EXTERNAL MOUNT MIXING NOZZLE ASSEMBLY



FINAL SHIPPING STATION

1. Nozzle matches: Job Order ☐ BOM ☐ ________(Init)
2. Nozzle matches: Drawing ☐ Drawing#_________ Rev#____ ________(Init)
3. Nozzle matches drawing. Engineer only: ________(Init)
4. QA Sheet is complete:

____

(Shipping Clerk) (Date)

CLOSE COUPLED HORIZONTAL CHOPPER PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ **SALES ORDER #** ____________________

MODEL #_____________________________ **DRAWING #**________________________

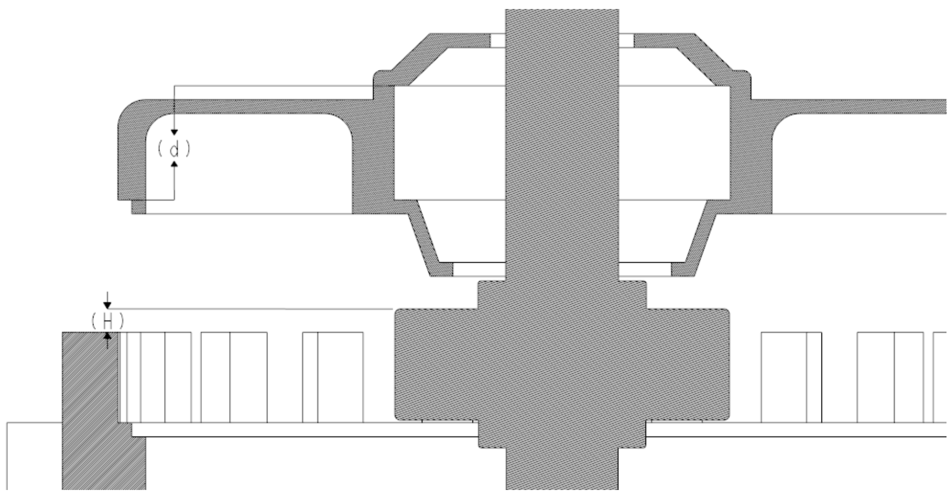
SECTION 1: PUMP ASSEMBLY STATION

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
2. Prior to motor disassembly record motor no load amps_____________ ________

Shaft Thermal Growth Allowances

182-184TC	0.030”	324-326TC	0.039”
213-215TC	0.030”	364-365TC	0.041”
254-256TC	0.032”	404-405TC	0.046”
284-286TC	0.035”	444-449TC	0.068”



Original Bore Depth (**d**) from outer shoulder: ______ . ______ in.

− Shaft Thermal Expansion Allowance: 0.______ in.

= Useable Bore Depth (**D**): ______ . ______ in.

Bearing Height (**H**): ______ . ______ in.

If D exceeds H, OK to proceed and skip step 4. If not, machine bore depth D deeper as needed and fill out step 4. ________

3. Bore Depth (**d**) from outer shoulder after machining: ______ . ______ in.

− Shaft Thermal Expansion Allowance: 0.______ in.

= Useable Bore Depth (**D**): ______ . ______ in.

Bearing Height (**H**): ______ . ______ in.

If D exceeds H, OK to proceed. ________

4. Wave spring has been removed and discarded? Yes ☐ No ☐ ________

JOB ORDER# ____________________________

Initial Below

5. Radial bearing has fan end shield/seal removed. ________
6. Thrust bearing has drive end shield/seal removed. ________
7. Grease port paths are free of obstructions and prefilled with grease. (Pump grease in Zerk until it appears at shaft.) ________
8. Bearings are prepacked with grease from manufacturer Yes ☐ No ☐ ________

   If no, bearings have been hand packed with grease Yes ☐ ________
9. Thrust bearing motor grease zerk has been removed and plugged. ________
10. Impeller has _____ blades and measures _________ inches ________
11. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ ________
12. Impeller has been balanced. ________
13. Shaft run out at impeller: _____________ ________
14. All nuts and bolts tightened with lock washers. ________
15. Discharge orientation matches Job Order ☐ BOM ☐ Drawing ☐ ________

SECTION 2: All 3-6" & 8L PUMPS

Initial Below

1. The Cutter Nut ☐ or External Tool ☐ is properly torqued to 125 ft/lb. ________
2. Impeller/Cutter bar gap to close vane measures __________ inch. **0.015-0.025"** ________
3. Pump Out Vane/Back Cutter gap to close vane: _________inch. **0.005-0.050"** ________
4. Clearance between tip of cutter bar finger and hub tooth. **0.010-0.030"** ________

   Measurements for 2 locations: a)_____________ inch b)____________ inch
5. Clearance between tips of external tool and anvil. **0.010–0.030"** ________
6. Axial clearance between external tool wings and cutter bar fingers __________inch 0.015 – 0.025" ________
7. Cutter Bar inlet diameter is correct per below ________

   3F/3G/3P – 4.25" ☐ **3L/3M/3W – 4.63"** ☐ **3V – 4.0"** ☐ **4K/4L – 5.25"** ☐
   4P/4S – 6.5" ☐ **4R/4T/6U/6W/6X – 7.4"** ☐ **4V – 8.5"** ☐ **8L – 9.25"** ☐

SECTION 3: 8N, 8P, All 10 – 16" PUMPS

Initial Below

1. The impeller is properly torqued to 250 ft/lb _______
2. The External Tool is properly torqued to 250 ft/lb. ________
3. Impeller/Cutter bar gap to close vane measures __________ inch. ________
   8N, All 10-16": 0.015-0.025"   8P: 0.010-0.020"
4. Pump Out Vane/Back Cutter gap to close vane: _________inch. **0.030-0.050"** ________
5. Clearance between tip of cutter bar finger and external tool tooth ________

   Measurements for 2 locations: a)_____________ inch b)____________ inch

   8N & 8P: 0.025-0.040", All 10": 0.035-0.050" All 12:&16": 0.040-0.060"

JOB ORDER# ____________________________

Initial Below

6. Clearance between tips of **external tool** and **anvil**: _______________ inch ________
   8N, 8P: 0.025-0.040” All 10”: 0.035-0.050” All 12” &16”: 0.040-0.060”
7. Axial clearance between tips of external tool wings and cutter bar fingers ________
8. Cutter Bar inlet diameter is correct per below ________
   8N – 9.0” ☐ **8P – 9.75”** ☐ **10R/10W – 11.0”** ☐ **12W – 13.5”** ☐ **16W – 16.0”** ☐

SECTION 4: 8K PUMPS

Initial Below

1. The Impeller is properly torqued to 250 ft/lb. ________
2. The Cutter Nose is properly torqued to 40 ft/lb. ________
3. The External Tool (if required) is properly torqued to 125 ft/lb. ________
4. Impeller/Cutter bar gap to close vane measures __________ inch. **0.015-0.025”** ________
5. Pump Out Vane/Back Cutter gap to close vane: _________inch. **0.005-0.050”** ________
6. Clearance between tip of cutter bar finger and cutter nose **0.030-0.040”** ________
   Measurements for 2 locations: a)_____________ inch b)____________ inch
7. Axial clearance between external tool wings and cutter bar fingers __________inch ________
   0.015-0.025”
8. Clearance between tips of external tool and anvil: ________________ inch **0.020–0.045”** ________
9. Cutter Bar inlet diameter is correct **8K – 8.75”** ☐ ________

SECTION 5: PAINTING STATION

Initial Below

1. Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Paint type:__________________________ Required paint thickness:__________mils ________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) underside of baseplate in the corner nearest the Vaughan nameplate ______MWFT
b) top of baseplate between casing and bearing housing ______MWFT
c) side of suction manifold ______MWFT
d) side of bearing housing adjacent to oil site glass ______MWFT
e) casing discharge throat on side nearest bearing housing ______MWFT
Initials ______ Date ______

DRY
a) underside of baseplate in the corner nearest the Vaughan nameplate ______MDFT
b) top of baseplate between casing and bearing housing ______MDFT
c) side of suction manifold ______MDFT
d) side of bearing housing adjacent to oil site glass ______MDFT
e) casing discharge throat on side nearest bearing housing ______MDFT
Initials ______ Date ______

JOB ORDER# ____________________________

SECTION 6: FINAL ASSEMBLY STATION

Initial Below

1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
3. Pump and drive assembly match drawing referenced above ________
4. Verify pump shaft turns over by hand. ________

Motor: Manufacturer_______________________________

Spec/Model/Catalog Number _________________________ Speed__________ RPM

Serial No. _______________________________ Frame Size ____________

5. Motor greased with ____________________________ Not Greaseable ☐ ________
   Note: Check motor manual for grease type when not listed on nameplate.
6. Verify all guards are in place. ________
7. Companion flanges (with gasket) supplied with export pumps? ________
8. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
9. If required the Rotamix trademark has been installed on base of pump. ________
10. Decals are installed on the pump per the decal placement drawing. ________

Vibration Measurements ________

Axial A ________________________

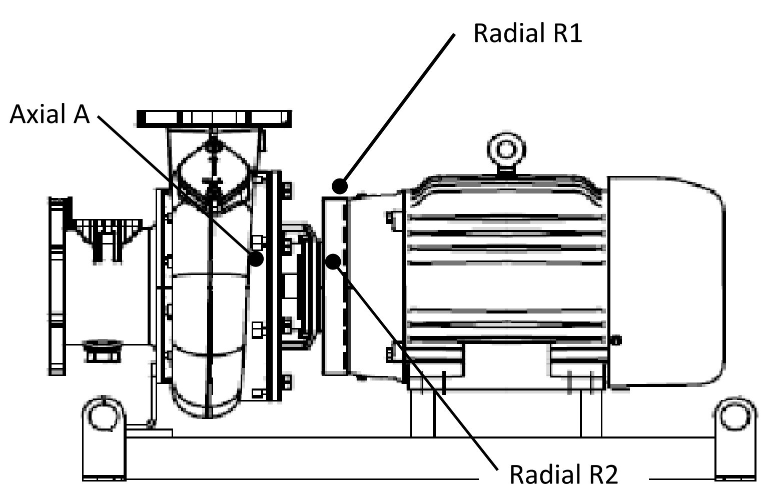
Radial R1 ______________________

Radial R2 ______________________

Motor amps measurement____________ ________

Nameplate full load amps ___________ ________

Reading is below half of full load Yes ☐ No ☐ ________



JOB ORDER# ____________________________

SECTION 7: FINAL SHIPPING STATION

Initial Below

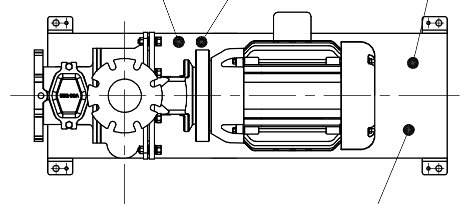
1. Motor or drive matches Job Order ☐ BOM ☐ Drawing ☐ ________
2. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#__________Rev#____ ________
3. Pump nameplate matches the Job Order ☐ BOM ☐ Drawing ☐ ________
4. Decals are installed on the pump per the decal placement drawing. ________
5. IO&M Manual is attached to the pump at the time of shipping. ________
6. QA Sheet is complete:

   ____

   (Shipping Clerk) (Date)

DOMESTIC PUMPS

PART #	DESCRIPTION
DEC–016	1.25" X 3.5" QUALITY ASSURED
DEC–017	2" X 4" QUALITY CRAFTED IN USA
DEC–019	.5 X 5.5" CLOCKWISE ROTATION
DEC–032	NAME PLATE
DEC–061	CAUTION, VAUGHAN CARTRIDGE SEAL
DEC–064	3.5" X 5" WARNING (ENGLISH)
DEC–064SP	3.5" X 5" WARNING (SPANISH)
DEC–065	6.5" X 7.5" PIT DANGER (ENGLISH/SPANISH)
*DEC–067	2" X 4" DANGER (ENGLISH)
*DEC–067SP	2" X 4" DANGER (SPANISH)

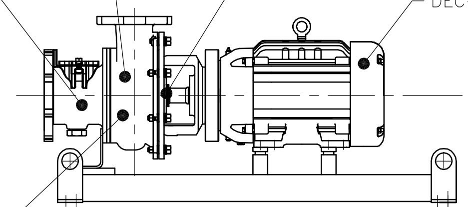


FOREIGN PUMPS

DO NOT USE SPANISH DECALS

REPLACE DEC–065 WITH DEC–037 OR AS LISTED BELOW

PART #	DESCRIPTION
DEC–065CAN	CANADA
DEC–037CZ	CZECHOSLOVAKIA
DEC–037DK	DENMARK
DEC–037NL	HOLLAND
DEC–037FI	FINLAND
DEC–037FR	FRANCE
DEC–037DE	GERMANY
DEC–037GR	GREECE
DEC–037IT	ITALY
DEC–037NOR	NORWAY
DEC–037PT	PORTUGAL
DEC–037ES	SPAIN
DEC–037SE	SWEDEN
DEC–037UAE	UNITED ARAB EMIRATES



VAUGHAN CO., INC.
364 MONTE--ELMA ROAD
MONTESANO, WA 98563
PHONE: (360) 249--4042
FAX: (360) 249--6155

UNLESS OTHERWISE NOTED, TOLERANCES ARE TO BE:
SURFACE FINISH: 125 MICROINCHES
DIMENSIONS, WELDING: X.XX = 1/16"
DIMENSIONS, MACHINING: X.XXX = 0.020"
DIMENSIONS, MACHINING: X.XXXX = 0.010"
GEOMETRIC TOLERANCE, RUNOUT ON CYLINDERS: 0.010"
GEOMETRIC TOLERANCE, PARALLELISM: 0.005"
BREAK ALL MACHINE EDGES 0.005" R UNLESS NOTED OTHERWISE

TITLE: CCHE DECAL PLACEMENT

CHECKED:	ENGINEER:	DRAWN BY:	SCALE:	PART NUMBER:
JK		KCW	1"=1'	
JOB ORDER NUMBER:		DATE: 5/1/18	DRAWING NUMBER: 119816	REVISION NUMBER: 0

PORTABLE CHOPPER PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ **SALES ORDER #** ____________________

MODEL #_____________________________ **DRAWING #**________________________

SECTION 1: PUMP ASSEMBLY STATION:

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____
2. Impeller has ____ blades and measures ______ inches ________
3. Impeller diameter matches Job Order ☐ BOM ☐ Drawing ☐ ________
4. Impeller has been balanced ________
5. If pump is built with Tapered Roller Bearings, shaft endplay is ________inch ________
   Small CBH: 0.001"–0.004" **Large CBH: 0.001"–0.006"**
6. Shaft run out at impeller: ____________ **(0.004" max**.) ________
7. All nuts and bolts tightened with lock washers. ________
8. Discharge orientation is 9:00 when viewed from suction. ________
9. 3/8" NPT vent port is present in casing and plugged. Yes ☐ No ☐ ________
10. 3/8" NPT drain port is present in casing and plugged. Yes ☐ No ☐ ________

SECTION 2: 3L, 3P, 4K, 4S, 4V, 6W, 8L PUMPS

Initial Below

1. The Cutter Nut is properly torqued to 125 ft/lb. ________
2. Impeller/Cutter Bar gap to close vane measures _________ inch **0.015"–0.025"** ________
3. Pump Out Vane/Back Cutter gap to close vane: ________inch. **0.005"–0.050"** ________
4. Clearance between tip of cutter bar finger and hub tooth. **0.010"–0.030"** ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
5. Cutter Bar inlet diameter is correct per below ________
   3P: 4.25" ☐ **3L: 4.63"** ☐ **3V: 4.0"** ☐ **4K: 5.25"** ☐ **4S: 6.50"** ☐
   6W: 7.40" ☐ **4V: 8.50"** ☐ **8L – 9.25"** ☐

SECTION 3: 8P, 10R, 12W PUMPS

Initial Below

1. The impeller is properly torqued to 1000 ft/lb. ________
2. The External Tool is properly torqued to 250 ft/lb. ________
3. Impeller/Cutter Bar gap to close vane measures ___________ inch ________
   8P: 0.010"–0.020" **10R & 12W: 0.015"–0.025"**
4. Pump Out Vane/Back Cutter gap to close vane: ___________inch. **0.030"–0.050"** ________
5. Clearance between tip of cutter bar finger and external tool tooth ________
   Measurements for 2 locations: a)____________ inch b)___________ inch
   8P: 0.025"–0.040" **10R: 0.035"–0.050"** **12W: 0.040"–0.060"**
6. Clearance between tips of external tool and anvil: ______________ inch ________
   8P: 0.025"–0.040" **10R: 0.035"–0.050"** **12W: 0.040"–0.060"**

JOB ORDER# ___________________________

Initial Below

7. Axial clearance between external tool wings and cutter bar fingers _________inch **0.015" – 0.025"** ________
8. Cutter Bar inlet diameter is correct per below ________
8P – 9.75" ☐ **10R – 11.0"** ☐ **12W – 13.5"** ☐

SECTION 4: PAINTING STATION

Initial Below

1. Paint applied matches: Job Order ☐ BOM ☐ Drawing ☐ ________
2. Paint type:________________________ Required paint thickness:_________mils ________

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET
a) bearing housing, above upper plug _____MWFT
b) top of casing, opposite of discharge flange _____MWFT
c) bottom of casing, opposite of discharge flange _____MWFT
d) top of casing at discharge throat _____MWFT
e) casing discharge throat on side nearest bearing housing _____MWFT
Initials _____ Date _____

DRY
a) bearing housing, above upper plug _____MWFT
b) top of casing, opposite of discharge flange _____MWFT
c) bottom of casing, opposite of discharge flange _____MWFT
d) top of casing at discharge throat _____MWFT
e) casing discharge throat on side nearest bearing housing _____MWFT
Initials _____ Date _____

SECTION 5: FINAL ASSEMBLY STATION

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
2. Bell housing matches the Job Order ☐ BOM ☐ Drawing ☐ ________
3. Bell housing is installed with belt cutout at 12:00 ________
4. Adapter flange installed on suction side ________
5. Verify pump shaft turns over by hand. ________
6. Bearing housing oil level is half way up sight glass. ________
7. Bearing housing vent plug is installed ________
8. All guards are in place. ________

JOB ORDER# ____________________________

9. Nameplate is installed on top of the pump with pump model and serial number opposite of lifting eye. ________

10. Seal oil reservoir is installed ________

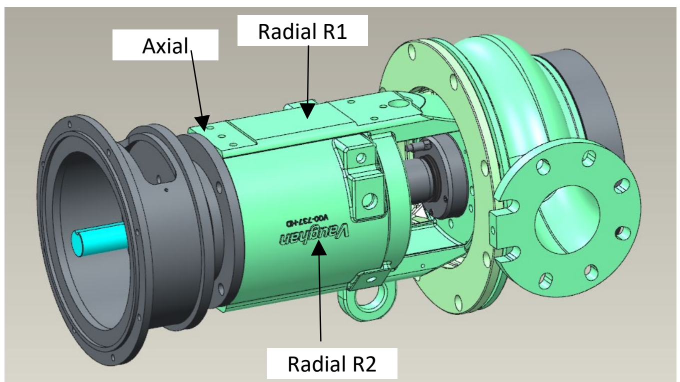
11. Vaughan seal and oil reservoir are filled to the proper level ________

Vibration Measurements

Axial A _________________

Radial R1 _______________

Radial R2 _______________



SECTION 6: FINAL SHIPPING STATION

Initial Below

1. Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ ________
2. Bell housing matches the Job Order ☐ BOM ☐ Drawing ☐ ________
3. Nameplate is installed on top of the pump with pump model and serial number opposite of lifting eye. ________
4. The IO&M Manual is attached to the pump at time of shipping. ________
5. QA Sheet is complete:

____

(Shipping Clerk) (Date)

PORTABLE SCREW PUMP QUALITY ASSURANCE VERIFICATION

JOB ORDER # ________________________ SALES ORDER # ____________________

MODEL #____________________________ DRAWING # ________________________

SECTION 1: PUMP ASSEMBLY STATION

Initial Below

1) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#________Rev#____ ______

2) Impeller model matches Job Order ☐ BOM ☐ Drawing ☐ ______

3) Impeller has been balanced ______

4) Shaft endplay is measured at ________ ______
(Small CBH: 0.001” – 0.004”) **(Large CBH: 0.001” – 0.006”)**

5) Shaft run out at impeller __________ ______

6) Impeller bolt is properly torqued with Loctite 242. ______
3DS, 4DH: 30# ☐ **6EM, 6ES: 40#** ☐ **8F, 10D: 80#** ☐ **12D: 100#** ☐

7) Hub Cutter/Back Cutter Clearance: ________ inch **(0.010” - 0.020" acceptable)** ______

8) Impeller/Suction Cone gap measures _________ inch **(0.015” - 0.025” acceptable)** ______

9) All nuts and bolts tightened with lock washers ______

10) Discharge orientation is 9:00 when viewed from discharge ______

11) 3/8” NPT vent port is present and plugged in casing Yes ☐ No ☐ ______

12) 3/8” NPT drain port is present and plugged in casing Yes ☐ No ☐ ______

SECTION 2: PAINTING STATION

Initial Below

1) Paint applied matches Job Order ☐ BOM ☐ Drawing ☐ ______

2) Paint type:________________________ Required paint thickness:_________mils ______

Dry Film Thickness						
Tnemec 431 Premium	Tnemec 431 Standard	Tnemec 27WB Premium	Tnemec 27WB Standard	Tnemec 1220 Premium	Tnemec 1220 Standard	Tnemec Omnithane Series 1 Primer
Min 30 mils	Min 10 mils	Min 10 mils	Min 5 mils	Min 10 mils	Min 4 mils	Min 2.5 mils

WET

a) bearing housing, above upper plug _____MWFT
b) top of casing, opposite of discharge flange _____MWFT
c) bottom of casing, opposite of discharge flange _____MWFT
d) top of casing at the discharge throat _____MWFT
e) bottom of cone _____MWFT
f) elbow, under flange _____MWFT

Initials _____ Date _____

JOB ORDER# ____________________________

DRY

a) bearing housing, above upper plug	_____MDFT
b) top of casing, opposite of discharge flange	_____MDFT
c) bottom of casing, opposite of discharge flange	_____MDFT
d) top of casing at the discharge throat	_____MDFT
e) bottom of cone	_____MDFT
f) elbow, under flange	_____MDFT

Initials _____ Date _____

SECTION 3: FINAL ASSEMBLY STATION

Initial Below

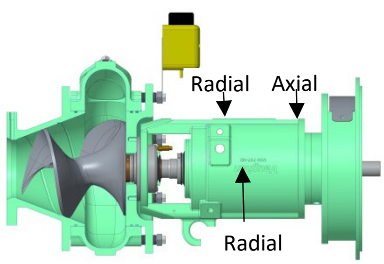
1) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____
2) Bell housing matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ _____
3) Bell housing is installed with belt cutout at 12:00 _____
4) Verify pump shaft turns over by hand. _____
5) Bearing housing oil level is half way up sight glass. _____
6) Bearing housing vent plug and cap are installed. _____
7) All guards are in place. _____
8) Nameplate with pump model and serial number installed on top of the pump opposite of picking eye. _____
9) Seal oil reservoir is installed. _____
10) Vaughan seal and oil reservoir are filled to the proper level. _____

Vibration Measurements

Axial A _________________

Radial R1 _______________

Radial R2 _______________



SECTION 4: FINAL SHIPPING STATION

Initial Below

1) Pump matches: Job Order ☐ BOM ☐ Drawing ☐ Drawing#_________Rev#____ _____
2) Bell housing matches the motor shown on Job Order ☐ BOM ☐ Drawing ☐ _____
3) Nameplate with pump model and serial number installed on top of the pump opposite of picking eye. _____

JOB ORDER# ____________________________

Initial Below

4) The IO&M Manual is attached to the pump at time of shipping. ______

5) QA Sheet is complete:

____

(Shipping Clerk) (Date)

SECTION 11313F

SUBMERSIBLE CHOPPER CENTRIFUGAL PUMPS

PART 1 GENERAL

1.01 SUMMARY

A. Section includes: Requirements for provision of pump systems including submersible non-clog motor-driven chopper pumps for service as scheduled.

B. Tag numbers: As specified in Pump Schedule.

1.02 REFERENCES

A. American Bearing Manufacturers Association (ABMA):
   1. 9 - Load Ratings and Fatigue Life for Ball Bearings.
   2. 11 - Load Ratings and Fatigue Life for Roller Bearings.

B. ASTM International (ASTM):
   1. A48 - Standard Specification for Gray Iron Castings.
   2. A108 - Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished.
   3. A148 - Specification for Steel Castings, High Strength, for Structural Purposes.
   4. A276 - Standard Specification for Stainless Bars and Shapes.
   5. A536 - Standard Specification for Ductile Iron Castings.
   6. A582 - Standard Specification for Free Machining Stainless Steel Bars.
   7. F593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
   8. F594 - Standard Specification for Stainless Steel Nuts.

C. CSA International (CSA).

D. FM Global (FM).

E. Hydraulic Institute (HI):
   1. 9.1-9.5 - Pumps - General Guidelines.
   2. 11.6 - Rotodynamic Submersible Pump Test.
   3. 14.1-14.2 - Rotodynamic Pumps for Nomenclature and Definitions.
   4. 14.3 - Rotodynamic Pumps for Design and Application.
   5. 14.4 - Rotodynamic Pumps for Installation Operation and Maintenance.
   6. 14.6 - Rotodynamic Pumps for Hydraulic Performance Acceptance Tests.

F. Insulated Cable Engineer’s Association (ICEA).

G. National Electrical Code (NEC).

H. Underwriters Laboratories, Inc. (UL).

1.03 TERMINOLOGY

A. The words and terms listed below are not defined terms that require initial capital letters, but, when used in this Section have the indicated meaning.
   1. Allowable Operating Region (AOR): The region over which the service life of the pump is not seriously compromised by hydraulic loads, vibration, or flow separation where the pump's vibration, noise, and cavitation are within acceptable limits.
   2. Preferred Operating Region (POR): The region over which the service life of the pump will not be significantly affected by hydraulic loads, vibration, or flow separation where the pump's vibration, noise, and cavitation are within acceptable limits.
   3. Pump head (total dynamic head, TDH), flow capacity, pump efficiency, net positive suction head available (NPSHa), and net positive suction head required (NPSHr): As defined in HI 9.1-9.5, 11.6, 14.1-14.2, 14.3 14.6 and as modified in this Section.
   4. Suction head: Gauge pressure available at pump intake flange or bell in feet of fluid above atmospheric; average when using multiple suction pressure taps, regardless of variation in individual taps.
   5. Tolerances: In accordance with HI 9.1-9.5, 11.6, 14.1-14.2, 14.3, 14.4, and 14.6, unless specified more restrictively.

1.04 DELEGATED DESIGN

A. As specified in Section 01357 - Delegated Design Procedures.

B. Anchoring and bracing.

C. Rotordynamic analysis.

1.05 SUBMITTALS

A. Submit as specified in Section 01330 - Submittal Procedures.

B. Delegated Design Submittals:
   1. Provide project-specific calculations based on support conditions and requirements to resist loads specified in Section 01850 - Design Criteria.
   2. Anchoring and bracing:
      a. Design of equipment supports and support anchoring to structures.
   3. Rotordynamic analysis:
      a. Level as scheduled.

C. Commissioning Submittals:
   1. For commissioning submittals listed below and specified in this Section and as specified in Section 01756 - Commissioning:
      a. Manufacturer's representative qualifications.
      b. Certificates:
         1) Requirements as specified in this Section.
      c. Test Plans:
         1) Test requirements as specified in this Section.

d. Test Reports.
e. Manufacturer's representatives field notes and data.
f. Owner training.

D. As specified in Section 01600 - Product Requirements.
1. Product data:
   a. Certifications:
      1) Certification of Rockwell C Hardness for cutter bars and impeller.
2. Shop drawings.
3. Calculations:
   a. Torsional analysis.

E. Vendor operation and maintenance manuals: As specified in Section 01782 - Operation and Maintenance Manuals.

1.06 WARRANTY

A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Pump: One of the following or equal:
1. Vaughan Co., Inc., S series.
2. WEMCO Pumps, Model CF.

2.02 DESIGN AND PERFORMANCE CRITERIA

A. Submersible chopper centrifugal pumps with components: Submersible pumps, motors, drivers, bearings, seals, supports, self-aligning discharge connection, base elbows, guide rails and lifting devices, electrical devices internal to pump housing, submersible cabling for power and control conductors, and other items as required for a complete and operational system.

B. Chopping/maceration of materials shall be accomplished by the action of the cupped and sharpened leading edges of the impeller blades moving across the cutter box at the intake opening.

C. Pump shall be capable of continuous operation while pumping primary or secondary scum containing up to 6 percent solids with heavy rags, plastics, grease, hair, grit, and other stringy fibrous material.

D. Design requirements:
1. Pump performance characteristics:
   a. As specified in the Pump Schedule.
   b. Rotordynamic analysis level: As scheduled and as specified in Section 15050 - Common Work Results for Mechanical Equipment.
      1) Vibration analysis expert: Provide when scheduled.
   c. All required conditions (flow/head) shall be within the pump manufacturer's Allowable Operating Range (AOR).

d. Performance tolerances shall be the same as the test tolerances specified in Section 15958 - Mechanical Equipment Testing.

2. Motor characteristics: As specified in the Pump Schedule.

E. Product requirements as specified in Section 01600 - Product Requirements and Section 15050 - Common Work Results for Mechanical Equipment.

2.03 MATERIALS

A. General: When materials are referenced in this Section or on the pump schedule, the compositions shall be the UNS Alloys, Types, or Grades in this article unless specified or scheduled otherwise.

B. Cast Iron: ASTM A48, Class 35 B minimum.

C. Cast Alloy Steel: ASTM A148, Grade 90-60, case-hardened to minimum Rockwell C 60.

D. Ductile Iron: ASTM A536.

E. Steel: ASTM A108, Grade or UNS Alloy as specified or scheduled.

F. Stainless steel: ASTM A276 or ASTM A582, Type or UNS Alloy as specified or scheduled.

G. Fasteners: Stainless steel, ASTM F593 or ASTM F594, type or grade as specified.

2.04 GENERAL PUMP CONSTRUCTION

A. Type: Industrial heavy-duty submersible chopper centrifugal pumps.

B. Other requirements:

1. Vibration: As specified in Section 15958 - Mechanical Equipment Testing.

C. All fasteners shall be Type 316 stainless steel.

2.05 PUMP AND MOTOR CASING

A. Type:

1. 2-piece construction with pump and motor casing bolted together. Seal faces shall be machined and fitted with Nitrile or Buna-N rubber O-ring seals. Bolting shall be Type 304 stainless steel.
2. Pump casing shall be of volute design, spiraling outward to a 125-pound flanged centerline discharge.
3. Back pullout adapter plate shall allow removal of pump components from above the casing, and allow external adjustment of impeller-to-cutter bar clearance.

B. Material: Cast Iron or Ductile Iron.

C. Watertightness: Able to run submerged up to 50 feet and to run dry.

D. Construction: Of sufficient strength, weight, and thickness to provide accurate alignment and watertightness.

E. Motor Cooling: Adequate for continuously submerged operation or unsubmerged operation for a minimum of 15 minutes as scheduled.

F. Design working pressure: Minimum 1.10 times maximum shutoff total dynamic head with maximum diameter impeller at maximum operating speed plus maximum suction static head; suitable for submergence in up to 65 feet of water.

G. O-ring seals: Capable of sealing mated surfaces (major components) watertight; with the following features:
   1. Machined surfaces and grooves.
   2. O-ring contact on four surfaces and O-ring compression on two surfaces.
   3. Does not require specific fastener torque or tension to obtain watertight joint.
   4. Does not require secondary sealing compounds, gasket, grease, or other devices.

H. Testing: Perform 5-minute hydrostatic test of pump casing at minimum 1.5 times Design Working Pressure.

I. Discharge interface:
   1. Sealing of the pumping unit to the discharge connection shall be accomplished by a machined metal-to-metal watertight contact.
   2. Self-aligning without bolting or having to enter the wet well.

2.06 IMPELLERS

A. Impellers:
   1. Type: Semi-open, chopper-centrifugal impeller with pump-out vanes to reduce seal area pressure and smooth water passages to reduce clogging by rags, stringy, or fibrous materials on impellers or shafting.
   2. Chopping/maceration shall be accomplished by the cupped and sharpened edge of the impeller blade moving across the cutter bar.
   3. Impeller shall be threaded or keyed to the shaft with no axial adjustments and no set screws.
   4. Casting: 1-piece, free of cracks and porosity.
   5. Material: Cast alloy steel heat-treated to a minimum Rockwell C 60 hardness.

B. Cutter bar plate:
   1. Material: Cast alloy steel or T1 alloy steel heat-treated to a minimum Rockwell C 60 hardness.
   2. Design cutter bar plate to prevent blockage of intake and binding of debris at the shaft and impeller vanes.
   3. Cutter bar shall be replaceable.

C. Upper cutter:
   1. Shall be threaded into the back pullout adapter plate above the impeller.
   2. Designed to cut against the pump-out vanes and the impeller hub, reducing and removing stringy materials from the mechanical seal area.
   3. Material: Cast steel heat treated to a minimum Rockwell C 60 hardness.

4. The area behind the impeller shall be protected from fouling by the cutting and expulsion action of either serrated teeth or sharpened vane edges in the rear of the impeller sweeping across the spiral grooves in the casing backplate.

D. Balance: As specified in Section 15050 - Common Work Results for Mechanical Equipment to meet the vibration criteria as specified in Section 15958 - Mechanical Equipment Testing.

2.07 PUMP SHAFTS

A. Material: Heat-treated AISI 4140 or Type 316 stainless steel, or Type 416 stainless steel. Turned, ground, and polished.

B. Features:
   1. Connects motor shaft by a bolt and key way.
   2. Strength: Able to withstand minimum of 1.5 times maximum operating torque and other loads.
   3. Resonant frequency: As specified in Section 15050 - Common Work Results for Mechanical Equipment and Section 15958 - Mechanical Equipment Testing.
   4. Deflection: Maximum 0.002 inch under operating conditions.

2.08 MECHANICAL SEALS

A. Provide dual tandem mechanical seal system with oil for seal lubrication and cooling:
   1. Shaft sealing system shall be capable of withstanding volute pressure up to 1.5 times pump shutoff head.
   2. No seal damages shall result from operating the pumping unit in its liquid environment, from running pump dry, or from reverse pump operation.

B. Oil chamber:
   1. Provide oil chamber for shaft sealing system. Design oil chamber to ensure that air is left in the oil chamber to absorb the expansion of the oil due to temperature variations.
   2. Provide drain and inspection plug, with positive anti-leak seal, easily accessible from the outside.
   3. Oil in oil chamber shall be FDA-approved, paraffin type, colorless, odorless, and non-toxic.

C. Upper seal:
   1. Tungsten carbide or silicon carbide rotating seal and tungsten carbide stationary seal or a carbon rotating seal and ceramic stationary seal as specified in Section 15050 - Common Work Results for Mechanical Equipment.
   2. Submerged in oil chamber located below the stator housing.

D. Lower seal:
   1. Tungsten carbide rotating and stationary seals as specified in Section 15050 - Common Work Results for Mechanical Equipment.

E. Springs and other hardware: Stainless steel, 300 or 400 series.

2.09 BEARINGS

A. Pump shaft shall rotate on a minimum of 2 permanently sealed, grease lubricated bearings:
   1. Upper bearing for radial forces.
   2. Lower bearing for combined axial and radial forces.

B. Bearing type: Anti-friction meeting ABMA standards.

C. Bearing lubrication system shall be sized sufficiently to safely absorb heat energy normally generated in bearing under maximum ambient temperature of 60 degrees Celsius under full load conditions.

D. Bearing life: Minimum L10 life of 100,000 hours at rated design point or 24,000 hours in accordance with ABMA 9 or 11 at bearing design load imposed by pump shutoff with maximum sized impeller at rated speed, whichever provides the longest bearing life in intended service.

2.10 MOTOR AND POWER CABLES

A. Motors: Features as specified and as scheduled:
   1. Squirrel cage induction motor, in an air chamber, moisture resistant, UL or FM listed for Class I, Groups C and D, Division 1 service, whether submerged or unsubmerged.
   2. Material:
      a. Motor enclosure: ASTM type A48, Class 25 or better.
      b. Windings: Copper.
      c. Shaft: Type 416 stainless steel.
      d. Rotor: Cast aluminum or copper.
      e. Insulation: Class F, moisture resistant, able to withstand 40 degrees Celsius ambient temperature plus 105 degrees Celsius temperature rise.
   3. Horsepower:
      a. As scheduled.
         1) Listed motor horsepower is minimum to be supplied. Increase motor horsepower if required to prevent motor overload while operating at any point on supplied pump operating head-flow curve, including runout. However, variable frequency drives, generator, and other electrical equipment are sized for scheduled motor horsepower.
         2) Make structural, mechanical, and electrical changes required to accommodate increased horsepower.
   4. Motor shall be designed for continuous duty handling pumped media of 40 degrees Celsius and capable of a minimum of 15 evenly spaced starts per hour.
   5. Motor shall be capable of continuous operation under load with the motor submerged, and a minimum of 15 minutes of full load operation when partially submerged or exposed, without derating the motor.
   6. Motor cooling system:
      a. Design to provide adequate cooling:
         1) At the minimum operating speed with a variable frequency drive.

2) With motor submerged.
3) With motor dry for a minimum of 15 minutes.
4) Motor shall be designed for continuous duty handling pumped media of 40 degrees Celsius.

7. Motor sealing: Design to withstand 200 psi water pressure at seal locations.
8. When VFD is scheduled, motor shall be capable of continuous inverter duty over the speed range specified.

B. Moisture protection system:
1. Provide dual moisture sensing probes extending into the oil chamber to detect the presence of moisture should the outer seal fails.
2. The moisture protection system shall also detect water in the motor chamber.

C. Thermal protection system:
1. Provide temperature detection in motor windings.
2. The temperature detection system shall reset automatically.

D. Power cables:
1. Submersible to same water depth as motor casing.
2. Type SOOW with Carolprene jacket, approved for extra hard usage in accordance with NEC Article 501-11.
3. Insulation rated for 90 degrees Celsius.
4. Non-wicking fillers.
5. Length: Sufficient to connect to surface junction box (without the need of splices) as indicated on the Drawings or 30 feet, whichever is greater.
6. Sized to in accordance with NEC, ICEA, and CSA specifications.
7. Provide stainless steel cable and stainless steel wire braid sleeve to support power cable from underside of wet well roof slab or access frame.

E. Monitoring or control cables:
1. Provide separate cabling (not in common sheath with power cables) for low-voltage or DC cables for monitoring devices.

F. Cable entry seal and junction chamber:
1. Designed to prevent moisture from wicking through the cable entry.
2. Cable entry seal be epoxy encapsulated.
3. Provide Buna-N power and control cable grommets in addition to the epoxy seal leads.
4. The cable entry seal shall provide strain relief for the cable.
5. The cable entry junction chamber shall be separate from the motor chamber to prevent foreign material from gaining access to the top of the pump.

2.11 SUPPORT BASE AND DISCHARGE ELBOW

A. Provide ductile or cast iron support base and discharge elbow for installation in wet pit.

B. Discharge elbow to mate to pump discharge and transition to discharge piping.

C. The entire weight of the pump/motor shall be borne by the pump discharge elbow.

2.12 GUIDE RAILS AND LIFTING DEVICES

A. General: Provide guide rails and lifting devices suitable for wet pit installation as scheduled in Execution and as indicated on the Drawings.

B. Materials:
   1. Wet pit: Type 304 stainless steel guide rails, lifting cable or chain and wall supports; Type 316 stainless steel anchor bolts and fasteners.

C. Wet pit guide rails:
   1. Type: Dual pipe or dual rail able to accurately guide the pump to mate with the discharge elbow.
   2. Strength: Withstand the greater of a minimum of 1.5 times the maximum imposed operating loads or seismic loads in accordance with building code as specified in Section 01410 - Regulatory Requirements seismic loads.
   3. Intermediate supports: Provide at 10-foot maximum intervals; less as required to provide specified support.

D. Lifting device:
   1. Type: Chain or cable attached to lifting eye on the pump casing.
   2. Length: Able to lower pump from top of wet well to operating position as indicated on the Drawings plus 5 additional feet of length.
   3. Retainer: Provide Type 316 stainless steel locking hook or clasp at top of wet well to securely retain the upper end of the lifting chain or cable during pump operation.
   4. Lifting device shall be sized for combined weight of pump and motor.

2.13 HOISTS

A. As required in the Pump Schedule.

B. Hoist material:
   1. Type 316 stainless steel.

C. Winch material:
   1. Type 316 stainless steel, marine grade.

D. Anchor bolts: As specified in Section 05190 - Mechanical Anchoring to Concrete and Masonry.

E. Mounting: Type 316 stainless steel socket, off side of sump access opening.

F. Winch: Manual adjustable boom with, safety hook, and sufficient lift for pump removal from deepest sump.

G. Hoist and winch safe load rating: Minimum 1.25 times heaviest sump pump furnished or 500-pound minimum with maximum reach, whichever is greater.

H. Anchors:
   1. Flush with internal thread anchor bolt sockets at each sump and socket with allowable working load of 5 times hoist pull out tension load on anchor bolts.

2.14 SPARE PARTS AND SPECIAL TOOLS

A. Spare parts: Provide one of the following for each size or type of pump:
   1. Mechanical seals: Upper and lower mechanical seal set.
   2. Pump bearings: Upper and lower bearing set.
   3. Impeller and cutter bar: 1 set.
   4. Casing seal gaskets or O-rings: 1 set.
   5. Power cable entry seal set: 1 set.

B. Special tools: Deliver 1 set for every furnished pump type and size needed to assemble and disassemble pump system.

PART 3 EXECUTION

3.01 PREPARATION

A. Anchoring and bracing to structures:
   1. Prepare equipment anchor setting template(s) and use to position anchors during construction of supporting structure(s).
   2. Install anchors of type and material indicated on approved anchoring designs.
   3. Install anchors with embedment indicated on approved anchoring designs.

3.02 INSTALLATION

A. Install the equipment in accordance with the accepted installation instructions and anchorage details.

3.03 COMMISSIONING

A. As specified in Section 01756 - Commissioning.

B. Source Testing: As specified in the Pump Schedule. Test as specified in Section 15958 - Mechanical Equipment Testing.

C. Functional Testing: As specified in the Pump Schedule. Test as specified in Section 15958 - Mechanical Equipment Testing.

D. Refer to Section 16950 - Field Electrical Acceptance Tests and Section 17950 - Commissioning for Instrumentation and Controls for additional requirements.

E. Manufacturer services:
   1. Provide certificates:
      a. Manufacturer's Certificate of Source Testing.
      b. Manufacturer's Certificate of Installation and Functionality Compliance.
   2. Manufacturer's Representative onsite requirements: Require manufacturer's representative to perform the following services as described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the Owner to meet the required quality of work. Work to be done in a minimum of 4 trips:

a. Installation assistance: As required:
   1) Advise/observe the Contractor on the installation of the equipment.
   2) Provide additional assistance as required.
b. Installation inspection: 1 trip, 5-day minimum.
c. Functional Testing: 1 trip, 4 days.

3. Training: As defined in Section 01756 - Commissioning and this Section:
   a. Operations: 2-hour class, 2 sessions.
   b. Mechanical Maintenance: 3-hour class, 1 session.
   a. Combined Electrical and I&C Maintenance: 1-hour class, 1 session.
4. Final acceptance checkout: 1 workday (trip may be combined with training).

3.04 PUMP SCHEDULE

Tag Numbers	**03-PMP-1610, 03-PMP-1620**
General Characteristics	
Service	Primary Scum
Quantity	2 (1 duty + 1 standby)
Installation Configuration	Wet Pit
Rotordynamic Analysis Level	None
Vibration Analysis Expert	Not Required
Pump Characteristics	
Speed Control	Fixed
Maximum Pump rpm	1,800 rpm
Shaft Seal Type	As Specified
Rated Design Point (At Maximum Revolutions per Minute)	
Flow, Gallons per Minute	235
Head, Feet	43
Minimum Hydraulic Efficiency, Percent	33
Required Condition 2 (At Maximum Revolutions per Minute):	
Flow, Gallons per Minute	150
Head Range, Feet	46 to 52
Minimum Hydraulic Efficiency, Percent	25
Required Condition 3 (At Maximum Revolutions per Minute):	
Flow, Gallons per Minute	325
Head Range, Feet	30 to 33
Minimum Hydraulic Efficiency, Percent	35
Required Condition 4 (At Maximum Revolutions per Minute):	

Tag Numbers	03-PMP-1610, 03-PMP-1620
Flow, Gallons per Minute	400
Head Range, Feet	12 to 27
Minimum Hydraulic Efficiency, Percent	20
Other Conditions	
Minimum NPSHa at Every Specified	N/A
Minimum Discharge Size, Inches	3
Maximum Shutoff Head, Feet	70
Driver Characteristics	
Driver Type	Motor
Variable Frequency Drive	N/A
Minimum Submergence	Exposed
Maximum Driver rpm	1,800
Minimum Driver Horsepower	7.5
Voltage/Phases/Hertz	460/3/60
Service Factor	1.15
NEMA, Design Type	B
Accessories	
Hoist	Required
Source Testing	
Test Witnessing	Not Witnessed
Performance Test Level	1
Vibration Test Level	None
Noise Test Level	None
Functional Testing	
Performance Test Level	1
Vibration Test Level	None
Noise Test Level	None

END OF SECTION